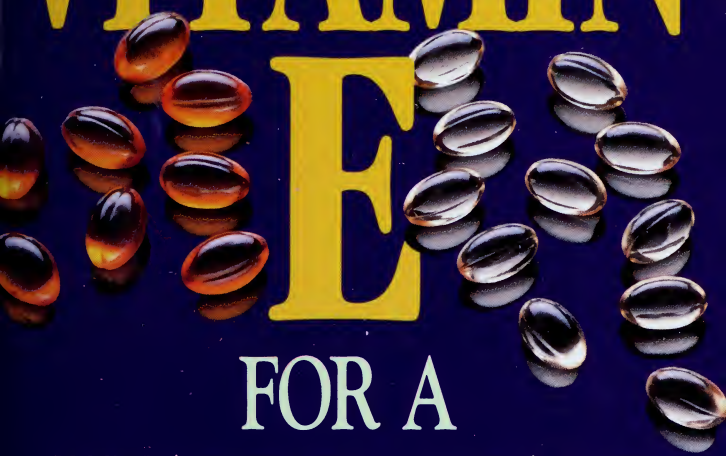


"VITAMIN E GREATLY REDUCES RISK OF HEART DISEASE."
—front page, *The New York Times*

VITAMIN E



FOR A
HEALTHY HEART
AND A LONGER LIFE

Herbert Bailey

Best-selling Health Sciences Author
Foreword by Bill Ingram, editor of Medical Tribune

THE VALIDATION OF VITAMIN E

“It’s very exciting [results of Harvard Surveys] because things like Vitamin E are not all that expensive compared to the drugs that are used to treat cholesterol.”

—*Dr. John LaRosa, designated spokesman,
American Heart Association*

“... and for good measure, I take 200 units of Vitamin E a day.”

—*Eric Rimm, Ph.D., Epidemiologist, Head of
Vitamin E Survery [Men’s] Harvard School
of Public Health*

“Your book was the spark that lit the Vitamin E bonfire.”

—*Dr. P. L. Harris, former Director, Depart-
ment of Nutrition, the Food and Drug
Administration*

Vitamin E “may reasonably be expected to add five to ten or more years of healthy, productive life to the life span of the average person.”

—*Denham Harman, M.D., Ph.D., Director of Medical Research, University of Nebraska Medical Center*

“A study conducted at the National Institutes of Health of individuals taking 800 units over an average of three years found no evidence of toxicity. . . . There is good reason to believe that cardiologists could use Vitamin E to replace aspirin.”

—*Dr. Max K. Horwitt, Professor Emeritus, St. Louis University of Medicine*

WHAT THIS BOOK CAN DO FOR YOU

- 1) Establish why you need the all-important Vitamin E.
- 2) Why you need the other antioxidants, Vitamin C and Beta Carotene (Vitamin A).
- 3) Why you need other vitamin/mineral supplements.
- 4) The recommended dosage for “normal” healthy people and for those who need varying amounts because of deficiency or disease.
- 5) Learn the fascinating story of how Vitamin E was developed and how it probably will help save your life—and prolong it in good health.
- 6) All this backed by the most prestigious of the *advanced* researchers—whom you will read about in this book.

Herbert Bailey

VITAMIN E

*For a Healthy Heart and
a Longer Life*

Herbert Bailey

With a Foreword by Bill Ingram

Carroll & Graf Publishers, Inc.
New York

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EVAN V. SHUTE, F.R.C.S.(C), F.R.S.M.
*Founder, Shute Foundation for Medical Research,
London, Ontario, Canada
(1906-1978)*

This book is dedicated to the memory of Dr. Evan V. Shute, pioneer in Vitamin E research and clinical application during nearly a half-century of ridicule and nonacceptance by most of the medical fraternity.

He lived to see his dream coming true.

He was a friend of mine and of all mankind. We shall miss this giant of true scientific endeavor.



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Acknowledgments

Of course my acknowledgments go first to the researchers and clinicians who made this book possible and with whom I have developed many friendships. These will be noted throughout the book. Unfortunately, several have been lost either through death or retirement. They are gone, but they are still remembered most gratefully.

Then there comes Miss Barbara, my wife, who has criticized, made suggestions (many of which we followed), and integrated parts of my other two Vitamin E books into this current volume, despite many adversities which occurred during its preparation. And *truly* "without whom. . . ."

Next is my old friend and colleague, Neal Thorpe. Neal is the vice-president of Man's Frontiers, an organization I founded many years ago and which I have not been able to keep active due to lack of funds. It seems that the grantors of the necessary funds, following true-to-style of the various agencies described in this book, would rather pour their largess down the same old worn-out drains instead of investigating the new thrilling approaches.

Same sad old story, as my readers know by now. Yet somehow mankind will probably "muddle" through. We hope so.

Another of our lieutenants who still survives is Harold Salkin, who assisted in several of our previous books and articles, as described in the main text. Harold, a former staffer with the late General Douglas MacArthur, is the secretary of Man's Frontiers.

Another of my researchers who deserves special credit is Thomas Hackney, who through many years

has shown his versatile abilities to ferret out little-known facts. Then there is Mike Luckman, top public relations counselor, who arranged for our second series of meetings with Muhammad Ali in New York City.

My appreciation also to Grace Bechtold (now deceased), then senior editor of Bantam Books, for her intuition in choosing books which are not only of literary quality but are also those that sell. She helped guide my previous Vitamin E book to successful completion.

My thanks also to my agent, Julian Bach, who conducted the necessary negotiations with his usual knowledgeable, friendly aplomb. Carolyn Krupp is his very able assistant.

And finally, my appreciation to Herman Graf who, with a single phone call to me, insured the publication of this book. It was Herman who many years ago, successfully sold my *Vitamin E: Your Key To A Healthy Heart*.

So it was simple: Herman believed in me and I believed in Herman.

Now we trust that *you* once again, believe in *us*.

—Herbert Bailey

Sandy Hook, Connecticut

Researchers' Statements

EVAN V. SHUTE, F.R.C.S.(C), F.R.S.M. Founder, Shute Foundation for Medical Research, London, Ontario, Canada

“You have been the most effective force in bringing the facts about Vitamin E before the public, more than any man I know. Any scientific discovery is meaningless till it is known, discussed, and tried by all of us. First, thanks largely to science writers like Herbert Bailey, we have to be given the evidence. *Here it is!*”

ALBERT SCHATZ, Ph.D., F.R.S.H., Microbiologist and Biochemist. Discoverer of Streptomycin.

“In my many years as a researcher, I have not encountered a more promising, scientifically justifiable substance than Vitamin E for the treatment of many of mankind's worst illnesses. The foremost, of course, is our Number One Killer, heart disease and related cardiovascular problems.

“I am proud that my good friend and fellow worker for the truth, Herbert Bailey, the author of this book, has been primarily responsible for attracting public—and medical—attention to the long-neglected values of Vitamin E. As will be noted in the book, my colleagues in university and research circles from South America to the former Soviet Union have corroborated the worth of alpha tocopherol in human therapy, both preventive and curative.”

RICHARD A. PASSWATER, Ph.D. Renowned Researcher, Author, and Lecturer.

“After receiving your letter, I pulled your *Vitamin E* book off of the shelf to refamiliarize myself with it. It has been several years since I have had the time to read your book, but the dogears and underlinings quickly reminded me what a friend the book was. Your views in Chapter 5 are stated elegantly and were a decade ahead of most researchers. I am pleased that you are going to update.”

SAMUEL AYRES, M.D., Diplomate American Board of Dermatology

“Your book was the beginning of my interest in Vitamin E.”

ALTON OCHSNER, M.D. Ochsner Clinic, New Orleans, Louisiana

“It [Vitamin E] has been very efficacious.”

LADY PHYLLIS D. CILENTO, M.B. B.S., Brisbane, Australia.

“I am glad that you are putting out a further edition of this book, as I had purchased it some time ago and consider that it is a most necessary exposé of the active prejudice against Vitamin E. Since writing it much water has flowed under the bridges and the evidence for its use in many fields, especially coronary heart diseases, is even stronger than when you wrote it in 1964.

“Although I have recently retired from general practice, I have had a lot of experience of the tremendous benefits accruing to many people by taking Vitamin E in suitable dosage. My husband [also a physician] has overcome both venous and arterial thromboses in the leg with 800 to 900 milligrams of Vitamin E daily, and his specialist considers the reestablishment of circulation a ‘miracle.’ I myself take 400 to 600 milligrams

a day and, at seventy-eight, feel well and full of energy and just as capable of a good day's work as I was twenty years ago."

EDWARD R. HARVEY, M.D., LL.B., Clinician and Noted Vitamin Researcher, Seymour, Connecticut.

"I have used Vitamin E therapy for my patients since reading Herbert Bailey's book on the subject several years ago. In my experience Vitamin E has proved to be of great benefit, particularly in degenerative diseases and in ameliorating the symptoms of aging.

"I have found that adequate dosages of alpha tocopherol [Vitamin E] have been effective when other treatments were of little use.

"We must thank Mr. Bailey for being the first writer to state the case for Vitamin E so clearly and convincingly."



Foreword

Medical Tribune reported last November [1992] that in two landmark Harvard University studies presented at the American Heart Association meeting, 127,000 men and women who took vitamin E at 100 IU for at least two years lowered their heart disease risk by about 40 percent.

I asked our 150,000 doctor-readers a simple question: "Do you believe in vitamin E?"

There was an avalanche of response: "yes"—at dosages of up to 1000 I.U.

Here were the best informed—and most conservative—judges of nutritional supplementation in this country saying that they know what vitamin E can do because they are taking it themselves and recommending it to their patients.

They learned about a number of studies, starting thirty years ago, testifying to the same effect the Harvard team had nailed down. Most important, they had observed cardiovascular improvement in their own heart patients—and in themselves.

Some of the doctors related the experience of *patients coming to them* as far back as the early 1960s saying they had read about the antioxidant properties of vitamin E. These revelations came in books written by Herbert Bailey.

Bailey put it in print over and over: Vitamin E confers cardiovascular benefits.

Yet FDA last year declared that the vitamin E recommended daily allowance should be slashed from 16 I.U. to 8 I.U.!

Vitamin experts recently made a plea before the National Academy of Sciences that the Vitamin E RDA

can no longer be set at a level to merely prevent deficiency when at six times that dose, it demonstrably *prevents disease*.

Herbert Bailey was on to this paradigm of modern nutrition decades ago. Researchers have now verified that the vitamin indeed prevents oxidation of low density lipoprotein on the wall of coronary arteries and thus prevents atheroma, the primary lesion of atherosclerosis.

Tragically, the estimated 10 million Americans who now take Vitamin E are in the middle and upper socioeconomic strata. The poor and badly nourished, who need the vitamin the most, are ignorant of the nickel-a-day therapy or simply shun it.

Herbert Bailey has told the Vitamin E story to millions of readers. It's imperative that he tell it all over again and update it for the sake of millions more who will otherwise be deprived of lifesaving nutritional support.

Bill Ingram
Editor
Medical Tribune

Preface

“Formulation of human diets, adequate in essential nutrients but designed to minimize random free-radical reactions in the body [Vitamin E], may reasonably be expected to add five to ten or more years of healthy, productive life to the life span of the average person.”

DENHAM HARMAN, M.D., PH.D.
*Director of Medical Research,
University of Nebraska Medical Center,
Omaha, Nebraska*

(Man's median life span has now reached about seventy-five, only five years above that of all recorded history. The biblical version is three score years and ten . . . written around 5,000 years ago.)

Would you like more authorities' findings? You will read about them later on in the book or see them cited in the Appendices—Jeffrey Bland, Lester Packer, Alton Ochsner, the Shute brothers, Richard Passwater, Lawrence Machlin, M. K. Horwitt, W. A. Pryor, Emanuel Cheraskin, Samuel Ayres, et al., and many others. I could name other researchers by the score, including hundreds of foreign researchers, who agree with Dr. Harman that antioxidants such as Vitamin E, along with other nutrients, will prolong man's life span many years longer than now, in full vigor.

And almost all the famous American authorities who opposed the innumerable favorable findings on Vitamin E—as reported in our first book in 1964, *Vitamin E: Your Key to a Healthy Heart*—have changed their opinions.

Now the pendulum is swinging in the other direction since all serious researchers are finding that Vitamin E does, indeed, perform for the human being almost precisely what we reported twenty-eight years ago. Some are going far beyond the conservative Harman and predicting that man should be able to live to about 125 years in good condition. Some others are going so far as to predict that with technological developments man can become immortal! As if man in his current state of development would desire equal rating with the Author of the Big Bang!

INVICTUS

*Out of the night that covers me
Black as the Pit from pole to pole
I thank whatever gods may be
For my unconquerable soul.*

*In the fell clutch of circumstance,
I have not winced nor cried aloud;
Under the bludgeonings of chance
My head is bloody, but unbowed.*

*It matters not how strait the gate,
How charged with punishments the scroll,
I am the master of my fate;
I am the captain of my soul.*

WILLIAM ERNEST HENLEY (1849–1903)

The first of these is the fact that the
 year 1875 was a year of unusual
 dryness in the West. The second
 fact is that the year 1875 was a
 year of unusual high water in the
 Mississippi River.

The third fact is that the year 1875
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The sixteenth fact is that the year 1875
 was a year of unusual high water in the
 Mississippi River.

The seventeenth fact is that the year 1875
 was a year of unusual high water in the
 Mississippi River.

CHAPTER 1

Escape Your Heart Attack— and More

The week beginning May 20, 1993, was a week to remember for millions of heart-disease sufferers—and millions more who feared the most deadly disease of the second half of the twentieth century. During that week the news flashed—carried by every form of communication—TV, radio, newspaper, magazines—that a single vitamin, Vitamin E, in heavy doses, greatly lowered the awesome risk of heart disease.

That reassuring news was afforded by surveying more than 120,000 female nurses and male health professionals over a four- to ten-year period by two medical teams sponsored by the Harvard School of Public Health and Brigham and Women's Hospitals, commonly referred to as the "Harvard Surveys."

Those who took supplements of Vitamin E—more than 100 International Units (I.U.)—suffered about 40 percent less of heart disease than those who received no Vitamin E—except the tiny amounts in their diet, approximately 10 I.U.

There was unanimous agreement among medical authorities with the startling, favorable news.* The American Heart Association spokesman concurred with the others: "It's very exciting because . . . Vitamin E [is] not all that expensive compared to drugs that are used to treat cholesterol."

As pointed out by Bill Ingram, editor of *Medical Tribune*, "The Physician's Newspaper" in his Fore-

*See Appendix B for other researchers' observations.

word to this book, a Vitamin E supplement can be had for five cents a day. The drugs cost *many dollars*—as any user knows.

I began receiving calls from people all over the country congratulating me on the wonderful news, although I was not mentioned in the reports. (Of course, I did not *expect* to be mentioned because the news came from the Harvard medical researchers and was published in the *New England Journal of Medicine*.)

The callers were readers of my two books on Vitamin E. The first, *Vitamin E: Your Key to a Healthy Heart*, published in 1964, reached millions of people interested in avoiding a much-dreaded heart attack. Each book advocated the use of Vitamin E both as a preventive measure and a treatment of heart disease. According to the media, the *New York Times* among many others, *Your Key to a Healthy Heart* was mainly responsible for the Vitamin E “boom” beginning in that period and continuing to the present. (Sales of Vitamin E were \$400,000,000 in 1992.)

Both books, listing unimpeachable medical sources, induced the public to take Vitamin E. These books also persuaded prominent medical doctors and researchers to begin further research on Vitamin E.

I told my callers, and later wrote in newspaper articles, that I was indeed gratified that universal acceptance had finally come for Vitamin E—after 30 or more years of nonacceptance by most of the medical fraternity.

Then I phoned my supportive medical friends and told them I would update my books to include much more new information. Among the many callers, I talked with Bill Ingram, my old editor-friend who had been reporting the tremendous surge of Vitamin E doctor-and-patient users.* Bill said: “Well, Herb, our campaigning has really paid off.” Indeed it has, Bill.

My new book would not only confirm the past and

*See Appendix B for *Medical Tribune* results of doctors’ surveys.

Escape Your Heart Attack—and More 27

present values of Vitamin E in preventing and treating heart disease, but we would go far beyond—into the various other diseases for which Vitamin E was not considered by conventional medicine to be of any use.

Yet Vitamin E had *already* proved its value in many other diseases, among them diabetes, cancer, kidney, and skin diseases. Also, Vitamin E is a remarkable antiaging agent.

However, examine Dr. Richard Passwater's survey on 18,000 people, which came nearly *twenty years* before the Harvard Studies. As stated previously, Dr. Passwater is a renowned researcher and author. His survey begins on page 82 and his article showing the comparison with the Harvard Surveys is in Appendix B.

Dr. Passwater's survey exceeded by far the results of the Harvard Surveys. Dr. Passwater and an independent statistician found an 80 percent reduction in heart disease by Vitamin E instead of 40 percent, which was double that of the Harvard Surveys.

Further, his studies of those in the fifty to ninety-eight age bracket, showed a far greater *range* of cardiovascular conditions benefited from Vitamin E treatment than did the Harvard Surveys.

Now, we will begin with Muhammad Ali and how many years ago Vitamin E helped him regain his heavyweight title. Then we will proceed to how my own heart attack led to my writing these books.

CHAPTER 2

Vitamin E: The Real Secret Weapon of Muhammad Ali

Muhammad Ali, heavyweight boxing champion of the world, reached into his hip pocket and pulled out a big bottle of capsules. He turned to me, waving the bottle for all in the crowded dressing room to see, then with a triumphant grin as only he can effect, exclaimed exultantly:

“Now, does *that* satisfy you, Mister Writer? I take these Vitamin E pills every day of my life! I told you several times before how much this Vitamin E has done for me! It gave me the extra energy to whip Norton and Frazier and Foreman—and the others!”

Ali then signed a statement attesting to these facts. He had the counsel of his close personal adviser and trainer, Bundini Brown.

This scene took place at Ali's training compound near Deer Lake, Pennsylvania, about a month before his last fight with Joe Frazier, which he won by a technical knockout.

The past three years had been an unbroken series of victories for the ex-champ. In his successful comeback, he won over two formidable opponents who previously had defeated him—Ken Norton and Joe Frazier. At last came the ultimate test: the mighty George Foreman, who had blasted every challenger with contemptuous ease. Yet, as most know—even those who make their abode in remote caves—Ali knocked out Foreman in Zaire, Africa, and went on to become the most recognized human being on the planet. His fame spread not

just because of his successful comeback (which no other heavyweight champion had done after a three-year layoff) but because he contributed both his time and a goodly part of his winnings to help people of the Third World.

Three years earlier, I had visited Muhammad Ali in his training quarters, along with my friends and associates Harold Salkin and Timothy Beckley. There were many others in this strange outré caravan proceeding to see a man who was as multifaceted as any of the ones who sought him. Included were Uri Geller, the controversial mystic, who apparently performs miracles by bending spoons, metals, and other materials by the power of his mind; Dr. Joseph Gross, a well-known New York psychiatrist, and Robert Marks and his lovely wife Linda, breeders of thoroughbred horses.

After observing Ali for some time, I came to respect him, since I saw he was really sincere and extremely intelligent, in spite of the bombastic attitude he assumes in public and around strangers.

I had not come to sell Ali on my book or Vitamin E; I came along at the last minute because we were going to watch films of flying saucers and see Uri Geller's "magic." (Ali is interested in "far-out" subjects.)

Also, I was curious about the man who was the talk of the world. Whether or not you liked him or liked boxing, his charisma was so overpowering that it would have been an insensitive writer, indeed, not to want to find out what kind of man this was who could get almost everyone excited, intellectuals and common folk alike.

At this point, I will not go into detail as to why Ali and his (then) wife Belinda were sold on my book and on my advice that he should take 2,000 units of Vitamin E a day and then, twenty-four hours before each fight, 4,000 units. (I scaled down these figures from doses administered to race horses whose winnings in-

creased 100 percent on a Vitamin E regimen—as described later.)*

What followed is a matter of record: Ali did take the suggested 2,000 units a day and 4,000 before his fight with Foreman.

Now back to the opening scene at Deer Park:

I had challenged Ali by asking, “Are you *really* taking Vitamin E as you told us repeatedly, or is this just another one of your famous rope-a-dope tricks?”

I knew the answer since Harold and I had heard it from him several times during the years when he was fighting to regain the championship and after he KO’d Foreman.

But I wanted to hear it again in front of his assembled selected group of managers, trainers, public-relations men, and various worshippers—the latter having managed to con their way through the sentries on duty at the door marked PRIVATE.

He gave me the answers I knew he would. But he did it in a way which is typical of a very complex, honest, yet subtle man. He pulled the bottle of Vitamin E capsules out of his pocket, in front of everybody. I had not expected so dramatic a move. Especially since he had kept the fact of taking massive doses of Vitamin E a trade secret. Who would want opponents taking it, too?

(Confirmation of all the foregoing is in Appendix C in the form of recorded telephone conversations between Muhammad Ali and Harold Salkin, and me.) In these conversations, Ali enthusiastically declares that Vitamin E gave him the extra stamina necessary to carry on in the final rounds of most of his fights, and that Vitamin E had given him the added power for a knockout punch (against Foreman) which he never had before.

*Ali found out what race horses like Northern Dancer (and all the Taylor Farms horses) had experienced, that heavy doses of Vitamin E was like having an invisible oxygen tank.

While this book is about Vitamin E and the exciting new discoveries made after my first book was published in 1964, Muhammad Ali's experience with Vitamin E deserves a prominent place. His story represents millions of persons all over the world who have had the same favorable results, in one way or another, with this magic, versatile vitamin. They form the hitherto-unknown cadre of persons who secretly smile at the official pronouncements saying Vitamin E is of no value in heavy doses. Like Muhammad Ali, they are concerned only with provable results. When they see these results, they know the doctors who tell them the results are "coincidental" are either guilty of dissimulation or are ignorant of the facts.

Then they seek other doctors who may be more interested in actual results than in parroting the official line of the Medical Establishment. The establishment, throughout medical history, has fought the new and the unorthodox.

This is why I think that Ali's story deserves top billing. He is only one of millions, but he can speak with vast influence for those millions, which to his everlasting credit, he is now doing.

In subsequent chapters we shall explore the incontrovertible proof amassed in thousands of clinical and laboratory tests throughout the world, based not merely on testimonials (valuable as testimonials are when confirmed by other scientific tests), but on controlled experiments. We shall see just how Vitamin E works in the body to effect the miraculous results Muhammad Ali obtained.

We repeat: Most of the evidence will be documented by controlled tests on both humans and animals; also confirmed in the laboratory on cells. It will be an exciting journey, for it affects you and me—and everyone else now alive.

CHAPTER 3

My Personal Experience

Readers of my first book on Vitamin E will probably recall my experience with a massive heart attack, and therefore my interest in Vitamin E, but I think the episode is well worth repeating. This is especially true since according to the *New York Times*, the *Washington Post*, many other newspapers, and various other media (magazines, radio, and TV), the book was credited with starting the Vitamin E boom. It touched off at least ten other Vitamin E books and countless magazine articles. Shortly afterward Vitamin E was the “in” vitamin, stimulating much more serious research than it had previously received.*

Now we return to the account of my heart attack, which apparently triggered the explosion of Vitamin E into public and scientific consciousness.

My heart attack occurred on one of the finest days of my life. I had just returned from a leisurely, month-long trip and, feeling full of energy, I immediately set to wrestling huge logs up from the beach. I spent the whole day at this violent activity even though the logs were really too heavy for anyone in his right mind to tackle. But I was feeling the need for exercise, which had long been frustrated, and so by the end of the day, working continuously, I had got the logs stacked neatly by our house. But when I came in for dinner, I collapsed.

I was suffering severe pains around the heart and I couldn't move. I seemed to be suffocating, but I could

*See Appendix D for media reactions.

not breathe fast enough to overcome the awful sensation. I felt that I was going to die—together a most horrible, helpless, panicky feeling. Our excellent general practitioner and good friend, Dr. Edward Harvey, was called in, and he immediately prescribed digitalis and nitroglycerine—plus bed rest for three weeks. I was flabbergasted. I was working on an important book at the time and really needed every spare moment for hard labor. But Dr. Harvey warned me that I had had a severe heart block and near failure due to congestion and that I required complete rest and no work at all. (A heart block means the pacemaker in the heart isn't transmitting properly; therefore, the necessary rhythm of the heartbeat is disrupted.) I asked him about massive doses of Vitamin E which I had read about briefly somewhere; he argued that he knew of no favorable literature on the subject for humans, just on rabbits and rats. We haggled a bit over the pros and cons of organized medicine, then he left me to my rest, digitalis, and nitroglycerin.

While trying to follow his orders, I received the news via a courageous radio nutrition commentator, Carlton Fredericks, Ph.D., and a courageous publisher, J. I. Rodale, the late editor of *Prevention Magazine*, that Vitamin E was in fact the best therapy for heart ailments of almost all types. I was informed that much medical literature on the subject was actually available; that, indeed, a book had been published by the pioneers of Vitamin E therapy, substantiating in great detail the proof of Vitamin E's efficacy in the treatment of heart and cardiovascular diseases.

Thus, my own heart attack was responsible for introducing me to the relatively unknown remedy of Vitamin E for heart disease. I obtained the book, as well as some other literature, which was not an easy task. Also, the book was written for doctors. Its title was: *Alpha Tocopherol [Vitamin E] in Cardiovascular Disease*. It was compiled by Dr. Evan V. Shute and Dr.

W. E. Shute of London, Ontario, Canada, but it included evidence from all over the civilized world. The senior author, Dr. Evan V. Shute (now deceased, though not from heart disease), was a Fellow of the Royal College of Surgeons (Canadian), and each of the contributors in the clinical tests was a well-recognized M.D. There was one exception: N. M. Lambert, of Dublin, Ireland, who was then and is now a Fellow of the Royal College of Veterinary Surgeons. It was he who did the first favorable work with Vitamin E on dogs and cats with heart disease.

Words cannot describe my reaction when I read the book. Although I was acutely aware of the adverse medical attitude toward any proposed cancer remedies by the medical "authorities," I was not prepared to admit that the suppression was also manifest in heart disease.

And yet there it was: full and complete evidence on more than 10,000 heart-disease victims which the nonprofit Shute Foundation for Medical Research had treated in Canada. Many more thousands had been treated with beneficial results in other countries. (At this writing more than 80,000 persons throughout the world have been saved or benefited by Vitamin E therapy at the Shute Institute.) And this overwhelming evidence had been stifled, ignored or, even worse, denied, in the medical press and, therefore, in the popular press as well. These reports and references are available, however, in good medical libraries.

I found out from Dr. Shute's book that I should probably take 600 international units of Vitamin E a day at the beginning. I did.

On the third day of Vitamin E therapy, I felt a surge of energy, such a new awareness of physical well-being that I said to my wife: "I'll be damned if I'll stay in bed any longer." Nor did I. I got up and have never been confined to bed again because of heart trouble. And that is why years later, my digitalis and nitroglyc-

erin tablets forgotten, I am able to take up the writing of this book. I am able to work at my writing all through the night, to saw a few logs as the dawn comes up, and then to take a spin on my bicycle around the countryside without feeling any aftereffects except the exhilaration which comes from stimulating exercise.* I know Vitamin E therapy is effective against almost all forms of cardiovascular ailments, unbelievable as it may seem. I want to pass on what I have learned so that others may benefit.

I know it will be difficult to convince everyone, especially the die-hard doctors, of the incredible value of Vitamin E, but I can try. The difficulty is that the clinical physicians who treat you, also treat many other patients a day and don't have enough time to read what is happening in the world of medical research.

When I was still in the skeptical stage about the value of Vitamin E, I made a personal determination for myself. A fourteen-year-old cat, named Old Tom, which lived next door to me, used to be the terror of the neighborhood. Wherever he went, the other cats moved aside. Females which were in heat, of course, were excepted. In fact, Tom forced many a powerful male cat to move from the neighborhood because he established precedence over the females (and males) and would not tolerate any other males which might seek to share *affaires d'amour* with his many lady loves.

In battle with other males, Tom had no match. In fact, even dogs were afraid of him. Some did test his fighting prowess at first, but as he was not only powerful but ingenious, he gained a most respectable reputation, even among dogs. Thus, he ruled the neighborhood for twelve years, which brings us to the point of this passage. Old Tom suddenly began to fail. It was almost unbelievable at the beginning. First, his sexual impulses

*Now, thirty years later, I omit the log-sawing!

diminished to the point where he could look at a female cat in heat without displaying the slightest reaction. Then all the male cats in the neighborhood which he had intimidated and/or beaten (including two of ours) began to take certain liberties. They would jump in his yard and some would even steal his food, to which atrocious actions he offered not the slightest objection. He would just sit in the sun and ignore everything.

However, since his owner had to be away from home frequently, Old Tom did not receive his food as regularly as he would have wished. He began frequenting our house, and my wife could not resist feeding him. But she found that his whole attitude had changed. Before, he had been a bristling, belligerent animal, hostile to almost anyone and everything. Now he was a whining, crawling beggar. He did not show any hostility when our cats would hiss and growl at him. He was completely subdued by life, or if we wish to put it bluntly, he was subdued and cowed by old age. There was nothing left for him to do but die or beg.

To put Vitamin E to another test, I added 50 I.U. daily to Old Tom's diet. The results would amaze someone who knew nothing about Vitamin E's action. Within a week, Old Tom began to take an extremely renewed interest in life. He savagely mauled the various cats who thought he was done for (including one of ours); he pursued the female cats who were in heat (there were several), but most significant of all to me, his pride came back. He stopped begging food from us. Prior to Vitamin E in massive doses, he would look at us and immediately his mouth would fly open in an automatic wail. The cat had no pride! Now, under Vitamin E, his pride had returned. However, since he had mauled one of our cats and was attempting to drive them away as is the custom of the species, we decided not to give Old Tom any more Vitamin E.

The effect, after about two weeks, was almost as miraculous as the previous experiment. Old Tom re-

turned to his old senile patterns and became on good terms with our cats and all cats without sex, without any drive except for food and perhaps some petting. But he wailed at us day and night, helpless, and without pride, waiting for death without knowing it.

Let us hope we never have to encounter death in that fashion; I do not believe we will if Vitamin E and other nutrients are utilized in the proper fashion. There is no question but that every animal and every human has to die; yet life might be preserved, more abundantly, in full vigor, for much longer had we the wisdom to apply the findings already known about Vitamin E.

This section is almost self-evident, since I am still among those who call themselves living. I haven't had another heart attack, thanks to that miracle "drug" Vitamin E. That, along with other vitamins and nutrients, has made me supremely confident—has rendered me free of fear, a most important factor to anyone who has ever suffered a massive heart seizure.

Although I am thirty-three years older, my life-style remains basically the same, except that I have increased my interests and activities considerably. My energy output both physically and mentally has kept pace with my increased activities. Friends tell me I look to be in better condition than I was thirty years ago, before I "discovered" that Vitamin E and various other vitamins and nutrients, when taken in mega dosages, do indeed halt, and often reverse—for a time at least—the aging process.

Of necessity, I live what is generally termed a tension-ridden life, because I believe I must do all I possibly can to help my fellow bits of cognizant protoplasm in our search for various truths.

I exercise at least an hour a day, try to eat correctly when possible, and break some of the present-day rules of so-called good living frequently. Yet several doctor

friends who check me over from time to time attest that my heart is normal; other organs, including the liver and kidneys, are performing their duties quite well. Bones, broken accidentally, have mended in record time, thanks to heavy dosages of A, E, C, the B-complex, calcium, magnesium, choline, inositol, along with other supplemental nutrients.

A relevant confirmation: about three years ago while riding my bicycle, I hit a hidden pothole. I crashed to the pavement, suffering a fractured hip. Examining my X-rays, the surgeon exclaimed: "You know, I'm amazed—you have the bones of a much younger man."

Now there are genetic factors, of course, in human illness and the prevention thereof, but the great researchers know that propensities for illness can be altered or even reversed by the proper external and internal (stress) environments, which include nutritional therapy—which in turn includes usage of the nutrients as therapeutic agents, not just as substances demanded by the body in order to eke out a bare existence.

As any victim of a heart attack can tell you, he or she lives with a constant nagging sensation that the seizure might happen again. And statistics confirm that this horrible fear is well founded, whether due to "infarction" or "spasm" of the coronary arteries. Many victims who are treated with the conventional therapies do indeed have one or more further "episodes" and finally achieve their transfer from this plane of existence via some breakdown of the cardiovascular system. This includes the bypass. But not those who take sufficient amounts of Vitamin E and other substances, and conscientiously stick to the regimen, no matter who tells them (mostly doctors) they are wasting their time and money.

They are the ones who write me—thousands of them—about the amazement of their doctor when he examines them. And sometimes they tell the doctor their prolonged improvement is due to Vitamin E. Then

when he shrugs and says "Impossible!", they go home and slug down some more of that impossible vitamin which they know is saving them from the repetition of that awful clutching agony of a major heart attack. It will not be the first or the last time patients have had to force an effective treatment down their doctors' stubborn gullets.

This time, however, patients who have access to the knowledge that a specific treatment works—in this case Vitamin E alone—can, if absolutely necessary, dispense with the doctors, because they can obtain the remedy without a prescription, a somewhat unusual situation for both doctor and patient. I have great hopes that the patients will win this battle because of the just-mentioned factor: patients are not really dependent on their doctors for either control or prevention in most cases.

There is another element which also makes me optimistic. More and more doctors, especially the younger ones, are breaking the strict bonds with which most doctors are tied. We see doctors' groups being formed in opposition to the rigid stand taken by the American Medical Association. Doctors are becoming increasingly aware that the monolithic, tyrannical structure housed in Chicago has steadily undermined their image as healers in the eyes of the American people. They know this in spite of AMA-oriented TV shows and AMA control (medically, at least) of the large circulation magazines and newspapers. They know that fictional medical heroes are somewhat rare in the "real" world. Their existence lies only in the minds of the Hollywood scripters, who check carefully with the authorities (AMA) to be sure their scripts are medically "sound."

Still another trend is apparent that gives me cause for optimism. Some time after my first book was published, I was invited to speak before the University of Illinois Medical Students' Association. The subject: my

book and Vitamin E. Such an action would have been unthinkable a few years earlier. Yet several of the medical students had read my book and were impressed by the vast amount of evidence it contained—evidence that, as my readers know, has been largely suppressed and ignored by the medical potentates in this country.

My talk was well received, judging from the reaction of the doctors-to-be. I was very much impressed by this new type of medical student: eager to learn, not willing to accept dogmatic statements from anyone, teachers included. Most of all, they exhibited a desire for responsibility, to right injustices, social as well as medical.

So, the trend is there and if it continues, there is every indication there will be a new type of doctor shortly, one who will be a true physician and researcher—which type is in the minority today. I shall eagerly await the change. And I am sure you will, too.

CHAPTER 4

Vitamin E's Attributes: How and Why Vitamin E Works

At this point, however, perhaps we should “encapsule” what Vitamin E is, why it has such an almost unlimited capacity to affect a number of maladies which afflict all of us, from the unborn—not yet “mewling and puking in the nurse’s arms” (as Shakespeare depicts the infant)—to the old-age syndrome: “shrunk shank—sans teeth, sans eyes, sans taste, sans everything.”

1. Vitamin E is an oil-soluble vitamin, like vitamins A, D, and K. It is found in the liver and all other organs, such as the heart. Milk and eggs also contain it. Grains are a source and, to a much lesser degree, some leafy vegetables. Blood also has Vitamin E, depending, of course, upon the level of E in the individual. We will discuss the low level of E in most Americans later.

2. It has been established that Vitamin E is a natural antioxidant and, therefore, an oxygen conservator. (Vitamin C is also to a lesser degree, in addition to possessing many other properties, as does E.) This means that Vitamin E prevents the oxidative processes of polyunsaturated fats, thus stopping the process which turns cells “rancid,” leading to their hasty death. The cells’ death, of course, contributes significantly to our own demise since almost all disease begins at the molecular or cellular level, including the brain, heart, blood, skin, bones, and just about every other constituent of that creature we call human. Recent studies

show also that Vitamin E almost certainly plays a very important role in the “immune” as well as the “auto-immune” processes.

3. Vitamin E is also a vasodilator. This means that it opens up the blood vessels, even the all-important tiny capillaries, which “feed” every cell in the body, including naturally the heart, brain, and all the other vital organs. It is providing more food—in the form of blood—with less strain, particularly if the vessels are clogged with fatty deposits (atherosclerosis) or calcium deposits which are associated with “hardening” of the arteries.

Either or both produce almost the same symptoms and usually result in heart attacks or strokes, in addition to a multitude of other accompanying ailments.

4. Vitamin E is the only known agent to prevent “plaques” from forming and depositing their deadly cargoes of LDL cholesterol inside the arterial walls, thereby blocking the blood flow to the heart. Vitamin E accomplishes this astonishing feat by “oxidizing” the LDL *before* it can even form plaques, thus, of course, preventing most heart attacks and cerebral strokes.

5. Vitamin E protects red blood cells from aging. In classic experiments, several researchers found that not only in every species of animal studied, but in man himself, red blood cells lived longer when extra Vitamin E was supplied.*

Perhaps this finding was initiated by the astronauts, many of whom came back from their celestial journeys suffering from severe anemia caused by destruction of red blood cells. This condition was not caused by lack of proper food, but by lack of sufficient Vitamin E, related to their living in a weightless environment for a long period and by breathing pure oxygen for days with little exertion.

*See Chapter 7, “Researchers Who Find Vitamin E Most Essential,” Hayflick, Packer-Smith, Jeffrey Bland et al.

When they received extra amounts of Vitamin E and calcium and adjusted the oxygen-nitrogen ratio, the anemia vanished promptly. This should have made headlines and been on every newscast, but instead was buried in medical journals. Reason? The Medical Establishment, of course, which didn't want Vitamin E extolled for anything. The late professor David Turner received a NASA award for explaining the cause and cure of the "space" anemia. (See *The Summary*, Appendix A.)

6. Vitamin E is a safe, effective anticoagulant (blood-clot preventive and dissolver). For almost two and one-half decades, the famed Dr. Alton Ochsner, first at Tulane Medical School and then later at his own clinic, routinely administered Vitamin E prior to surgery. The results were amazing. The death rate from clotting dropped phenomenally.*

Throughout the years, Dr. Ochsner reported his findings to various medical journals and they were published. But amazingly enough, despite his reported findings, surgeons continued to use the dangerous drugs heparin and dicumerol, just as if their eminent colleague's findings had not existed! And they still ignore them, continuing to use these same drugs, which must be carefully monitored. They still have the same problems of postoperative clotting, which often leads to fatal thrombosis in the lungs, brain, or heart! (The operation was a success, but the patient died! The same old story.) Nothing short of an atomic blast in the right anatomical place can move the old boys who control the establishment (and pass on their outdated notions to medical students). Only after the blast will the past change to the present and perhaps the future.

Yet a note of hope: As we state elsewhere (pp. 68-109), many prestigious researchers have been forced to change their opinions about the worth of Vitamin E,

*See Dr. Ochsner's personal letter to me, Appendix E.

both in treatment and in prevention of many diseases and conditions affecting the human body.

As far back as 1976 Dr. M. K. Horwitt wrote in the *American Journal of Nutrition*, "There is now evidence to support investigators who claimed that supplementation with Vitamin E affects coagulation."

He goes further: "A recent study at the National Institutes of Health of individuals taking 800 units [of Vitamin E] for an average of three years found no evidence of toxicity. . . . *There is good reason to suppose that cardiologists could use Vitamin E to replace aspirin* . . . [also] Vitamin E could prove useful in decreasing the risk of thrombosis in thousands of women now using oral contraceptives."

Dr. Horwitt was also a pioneer in establishing that the more polyunsaturated fats you use, in spite of all the propaganda over the last twenty-five years, the more you need Vitamin E in direct proportion to the amounts of polyunsaturated fats you consume. Later on, we'll explain this in terms of "free radicals" and other factors (see p. 100), but meanwhile, welcome to the club, Dr. Horwitt. It's always refreshing to receive a scientist of your stature. They are few, but they are growing in numbers and power.

The bandwagon has already started rolling and will not stop for those wishy-washy scientists who are waiting to see which way the establishment will jump.

Well, I can tell you, friends, the establishment has already been punched soft by the scientific innovators and soon the establishment doctors will leap on the bandwagon belatedly, claiming they were "waiting only for absolute scientific proof." They never waited on absolute proof for anything else, sponsoring all kinds of medical fads, then having to retract without ever admitting their mistakes. (Other qualities of Vitamin E can be found elsewhere in the book and the Appendix.)

At this point we are tempted to explain the whys

and wherefores of Vitamin E's anticoagulant action in preventing platelets from clumping together to form a clot, but we would have to explain what platelets are, the prostaglandins, and their complex role in the body. All the technical explanations (and proofs) are in the Appendix so that you, or your doctor, may examine them there, rather than interrupt our main narrative.

Vitamin E's other qualities, such as increased fertility for both male and female, healing of wounds and burns, favorable results with diabetes (in which insulin requirements are reduced, sometimes by one-half), heightened energy, infections, counteracting air pollution—indeed, toxins in all forms—as well as a host of other discoveries will be dealt with in subsequent chapters.

Heightened energy, physical or mental, and endurance under stress is probably one of man's most sought-after goals. Dr. Hans Selye, probably the world's foremost researcher on stress, has emphasized repeatedly in *Stress Without Distress* (New York: Lippincott/Signet) that stress is a normal, even a necessary component of living. Yet it resolves to the questions of how much? what kind? what environment?

As reported in my first book on Vitamin E, Dr. Selye made one of the first experiments on stress with rats, using a "double-blind" technique. He found that rats treated with Vitamin E were able to endure much more stress (of many types) than their untreated litter mates. The text, with pictures, printed in a reputable medical journal was (to my knowledge) never reported in the media. And Dr. Selye never followed up on his original findings on Vitamin E and stress. The reason? I'll lay you one Nobel Prize against one adjusted dollar that you guessed it right—assuming that you are of sound mind and can read properly—that is, between the lines of propaganda releases from the establishment represented by the old boys—although their chronological age may be rather young.

Nevertheless, many other researchers, investigating

stress and Vitamin E, found that rats could swim twice as long when pretreated with Vitamin E than their untreated litter mates—that every animal species tested, including the race horse, increased their performance under stress when treated with Vitamin E. Except *man*? Whose body is essentially animal, yet whose mind, continually evolving, is able to partially comprehend the creation? Hardly.

Many studies on humans which we report elsewhere have shown that Vitamin E *does* increase endurance under stress and therefore produces heightened energy.

Now we will review the history of Vitamin E, because if we “cannot remember the past [we] are condemned to repeat it,” as George Santayana put it.

CHAPTER 5

The Remarkable History of Vitamin E

Vitamin E was discovered in 1922 by researchers H. M. Evans and K. S. Bishop. Unfortunately for human beings, but perhaps fortunately for science in the long run, these researchers picked the rat to prove the existence of the then unnamed substance. Had they chosen another animal, the results of Vitamin E deprivation would not have been the same as they were in the rat. The goat, for instance, apparently manufactures its own Vitamin E (just as the dog manufactures its own Vitamin C). However, in all other animals tested (sixteen thus far) deprivation of Vitamin E caused serious disorders—and finally death.

The original researchers chose the laboratory rat. Like man, the rat is an omnivore, i.e., it eats almost anything, including meat, vegetables, and grains. The rat is even closer to man, nutritionally speaking, than most other species, including our “closest relatives” the apes, which eat vegetables and fruits primarily. Apes will not usually eat meat; neither will the horse or elephant, which may partially explain why these animals in the natural state have to spend almost all their waking time eating or looking for food. They have to supply their need for protein by eating large quantities of vegetative products rather than obtaining it in its concentrated form, which is meat, or animal products like eggs, milk, and cheese.

Vitamin E is absolutely essential for the rat in order to achieve reproduction. Certain other adult animals,

such as chickens, lambs, and cows, if deprived of Vitamin E, *may* reproduce, but their *offspring* never get a chance to reproduce. The reason? Vitamin E-starved chicks die of "crazy chick disease" (encephalomalacia), a disorder of the brain; lambs deprived of Vitamin E die of "stiff-lamb disease" within days or weeks; minks and mink kits die of "yellow fat" disease. Reproduction is thus eliminated in the second generation because these animals die before they can reproduce. If the rat is deprived of Vitamin E, it either cannot reproduce or else the fetus is "resorbed" by the mother and there is no birth at all.

This was a seemingly significant discovery at the time. It still is significant in the light of later research. However, the implication for the human being concerning reproduction was not long in being deduced. Vitamin E became known as the "fertility" vitamin. Then rapidly, because of the peculiar way our society is constituted, Vitamin E achieved the medical status of being ranked along with Voronoff's "monkey glands" as a sexual rejuvenator. It is difficult to believe, in view of the thousands of reports published since then concerning its diverse roles in metabolism, that there are still some medical "authorities" who think of Vitamin E *only* as a "fertility" substance. It does happen that humans, both men and women, are affected beneficially by Vitamin E in the reproductive process, but that is a relatively minor factor—almost a side effect—of the vitamin when compared with its other capacities.

At first, Vitamin E was also thought to be synonymous with wheat germ oil, but although wheat germ oil does contain some Vitamin E, the amount is usually not enough for therapeutic or even preventive dosages for degenerative diseases. But this fact was to be learned much later after many experiments and many scientific arguments. (However, octacosanol, concentrated from wheat germ oil, has additional stamina-building qualities other than that contained in Vitamin

E and may be taken in conjunction with Vitamin E for optimal benefit.)

Thus, there were double anathemas cast upon Vitamin E; namely, (1) it was a "sexual" vitamin and, therefore, could not be tolerated in an ostensibly puritanical society where sex was a dirty word; and (2) the early researchers did not have Vitamin E in sufficient quantities (they were using wheat germ oil). These early experiments were also confined to rats and rabbits for the most part, which hampered progress even more. And no one knew then that the wheat germ oil being used was probably turning rancid and, was, therefore, useless since rancidity not only destroys Vitamin E but has a most significant adverse effect on cellular metabolism.

However, to add an even more devastating roadblock in the path of the researchers, in 1937 Vitamin E was finally synthesized as a chemical laboratory product and more researchers began to use it, but no one was to know until 1962 that the synthetic product would be proved by the strictest test to be only one-half as powerful as the product derived from natural sources. Meanwhile, many researchers were using the synthetic product and claiming poor results.

Ultimately there were two forms of the synthetic Vitamin E, but the one finally accepted possesses only 50 percent of the potency of the naturally produced Vitamin E. This according to several researchers, Drs. Horwitt, Bland, and others.*

Therefore, if you buy the synthetic—which I hope you don't—you will be losing at least 50 percent of what you paid for. All vitamin manufacturers are forced to identify on their labels the natural or synthetic form of Vitamin E. The natural Vitamin E is indicated as *d*-

*Much of this information regarding synthetic Vitamin E was provided to me by Carlson Laboratories, Arlington Heights, Illinois. I am grateful, as I am to all of our other researchers.

alpha tocopherol; the synthetic as *dl*-alpha tocopherol. Read the labels. Need I tell you which to buy?

Most of the early workers were trying to establish Vitamin E's efficacy, or lack of it, in various types of animals and in humans—purely on its relationship to fertility—without having enough (as we now know) of the vitamin to make a valid test even on the reproductive basis.

There were a few pioneers during this period who used Vitamin E for various ailments and who obtained good results, but, regrettably, these findings were never followed up. Therefore, Vitamin E encountered a long dry spell between 1922 and 1945, when Dr. Evan V. Shute and his brother, Wilfred, began seriously to investigate Vitamin E. In the book, *Alpha Tocopherol [Vitamin E] in Cardiovascular Disease* (1954), the Shutes describe how they arrived on the scene by a "series of lucky accidents." They had been intrigued by Vitamin E some years before (in 1936) but had "ignored the clues."

Evan Shute was one of Canada's most respected physicians. He was an obstetrician and gynecologist, but his interests in medicine were far-ranging. His younger brother, Wilfrid, also a physician, was a cardiovascular specialist.

Evan qualified at a very early age as a genius. He entered high school at the age of nine, a feat never surpassed in Canada. He sailed through high school, obtaining the highest standing in Essex County, Ontario, and then won a scholarship to the University of Toronto at the age of fourteen—another record never before achieved in Canada. He received his Bachelor of Arts degree in 1924 and three years later, his medical degree.

In 1933, Dr. Shute was made a diplomate of the American Board of Obstetrics and Gynecology, which was quite an honor for a young physician. However, an even greater accolade was to follow two years later

when he was selected in Canada as a Fellow of the Royal College of Surgeons. Other honors followed in rapid succession, which we will not enumerate here due to lack of space, but we must mention that he authored over 120 medical papers and has lectured at international medical meetings in London, New York, Naples, Venice, Japan, and Amsterdam. He was also listed in Canada's *Directory of Medical Specialists*, *American Men of Science*, and *American Men of Medicine*.

We feel it necessary to acquaint the reader with Dr. Shute's accomplishments and honors in the field of orthodox medicine so that no one could possibly question his medical background. Now we will return to the history of the Shute brothers' discovery.

Together, they began treatment of cardiovascular diseases with Vitamin E as early as 1936-37 but, as they term these studies now, they were "abortive." Preparations of the substance were poor and lack of facilities curtailed their research. Nevertheless, the results obtained were so suggestive that they never forgot them.

In 1945, due to a brilliant type of research which Dr. Evan Shute proposed to a doctoral candidate, Floyd Skelton, the intriguing question of Vitamin E's efficacy in the matter of blood coagulation arose. Purpura (a bad breakage of the smaller blood vessels) was induced in dogs—the animals chosen for the experiment—and then relieved by large doses of Vitamin E. Skelton's research proved so fascinating that the Shutes resolved to try Vitamin E on a human patient. The patient they selected was in the last stages of purpura. His doctors were considering removing his spleen in the hope that somehow the condition could be helped.

Now, removal of the spleen is a most serious operation—for the patient, that is—since the spleen is foremost among those organs and tissues which comprise the body's natural line of defense against infections and diseases of all types. (This natural defense mechanism

is termed the immune system. Among the immune system's many activities is the manufacture of antibodies and defensive white cells.)

An operation, however, was not actually possible since the patient also had severe heart failure. You cannot remove a man's spleen when he is suffering desperately from heart disease. Therefore, researchers Skelton and Shute, calculating on the basis of the dosage used to cure the dogs of purpura, administered 200 IU (International Units) of Vitamin E a day to their despairing patient. This dosage was much higher than had ever been used before in treating human beings. Today we know this dosage is low; it helps to explain why some researchers could not obtain uniform or even satisfactory results.

Even so, the patient's heart trouble quickly disappeared *before* his purpura, though that, too, yielded under the onslaught of the then mysterious substance called Vitamin E.

You can imagine the Shutes' thoughts at the miracle accomplished by the administration of a *mere* vitamin, the dosage of which would appear as an infinitesimal speck in the palm of a man's hand. Reacting as true scientists, they immediately looked about for more patients—only those with heart disease this time—on whom they could again test Vitamin E.

They did not have far to look. Dr. Evan Shute's erstwhile barber was dying from recurrent coronary thrombosis (the disease that kills more than half the men over forty-five). The barber was in the last stages of heart failure, complicated by a type of angina which was extremely painful and which no anodyne could relieve. Now, anyone who has suffered angina knows it is one of the most "painful pains" known to man and also one of the most terrifying. When you know it is your *heart* which is sending out this most urgent call for relief, it is a slightly different matter from suffering from any other pain. You know, for instance,

that the almost unendurable pain experienced in the aftermath of a torn-off leg will eventually disappear and that your life will probably be saved if you are in competent medical hands. But, in this case, you know that if your heart stops functioning, *you* stop functioning also.

There are millions of persons who were and are now enduring exactly the same type of pain from the same type of disease which Dr. Shute's barber was suffering in 1945. Very likely they could have been helped if Dr. Shute's findings had become common knowledge in that year and in succeeding years. Dr. Shute's dying barber, after being treated with Vitamin E for three weeks, *was playing the drums in one of the local theaters*. We may assume that he had returned to his regular profession of cutting hair as well, although this information is not vouchsafed to us by Dr. Shute in his book.

Next, the Shute brothers treated their mother with Vitamin E. She was suffering from severe angina pectoris. Again the vitamin worked; their mother's gripping pains ceased, and she was able to resume her normal activities.

Encouraged by these results (and who wouldn't be?), the Shutes, with the assistance of Dr. Arthur Vogel-sang, a general practitioner and former student of Dr. Evan Shute, began a series of treatments on persons suffering from cardiovascular diseases.

Their results were as phenomenal as those previously described. At the time of compiling their monumental work in 1954, Dr. Shute and others had treated more than 10,000 heart patients—with fantastic, scarcely credible results. Overnight the picture of gloom which had pervaded the whole area of heart disease was changed into the bright, glowing colors of optimism. At least that is what the Shutes thought. And that is, of course, what should have occurred.

But the Shutes had to learn the hard way. One would have expected that the medical profession and the med-

ical journals would have seized upon this radical discovery when it was backed up by the most careful type of scientific evidence. Naturally, this evidence included the living, recovered patients who had been doomed to die. The patients possessed undeniable records of their illness as well as physical proof of their return to healthy life. Almost none of the diagnoses of heart disease had been made initially by the Shutes. These were patients given up as "hopeless" by cardiac specialists.

The Shutes expected that the medical journal to which they sent their report would be proud to be the *first* medical journal to break the news to doctors—and therefore the world—that finally a *real key to heart disease* had been found with indisputable proof. To their amazement the Shutes' report was rejected. The doctors thought there must be some mistake. However, the second medical journal refused to publish also, as did the third. As Dr. Evan Shute puts it in his book in a classic masterpiece of understatement: "To give some slight indication of our problem, we once had resigned ourselves to being unable to publish our work in any medical journal. We sought unsuccessfully to register our manuscripts with the National Research Council of Canada, planning then to retire from this field of endeavor."

However, "the Council felt unable to assist us in this way, so we *perforce* continued our studies." (Italics mine.) The Shutes were forced to continue their research with Vitamin E because they could not abandon their search for the truth in spite of official skepticism and opposition. They eventually founded the nonprofit Shute Foundation for Medical Research and have, since 1954, treated successfully over 80,000 heart disease sufferers. They have collected reports from all over the world, attesting to the value of Vitamin E not only in heart disease but in many other ailments afflicting human beings. (See Chapter 10,

“Vitamin E Benefits Still More Other Ailments,” and Appendix A.)

As Evan Shute relates in his book, heart specialists (cardiologists) condemned his findings at first sight without ever reading the evidence. Similar experiences have occurred to other scientists.

In spite of Dr. Shute's monumental researches in Vitamin E, he was not accorded the recognition he deserves. It was he, more than any other doctor I know, who worked harder, and was, of course, responsible for the Shute Foundation. It may seem almost incredible—except to those who have followed medical history—that a Fellow of the Royal College of Surgeons (Canadian) and the recipient of innumerable honors and awards from the most orthodox of medical groups should not be given a hearing by those groups. Yet this is the case, not unusual among pioneers.

It should be noted that his brother Wilfrid (now deceased), who was active in the earlier work on Vitamin E, was not a member of the Shute Foundation for many years, having a private practice. However, he wrote several books on Vitamin E, relating his experiences.

Unfortunately, Dr. Evan Shute did not enjoy the protection of the king or of any premier, president, or potentate as did Dr. William Harvey, discoverer of the circulation of the blood, who was protected by James I from the wrath of the medical fraternity. The medical men of that time were just as adamant against what they considered quackery as are their counterparts of today. For instance, they *knew*, as only the abysmally ignorant can know, that any doctor who said that the blood circulated throughout the body was guilty of quackery and should forthwith be dismissed from the medical society. It was only the king's influence that saved Harvey from a fate that has been accorded hundreds of his less fortunate confrères.

It is interesting to note that had it not been for a

king's belief in his doctor, all medical science might have been delayed indefinitely, and we might not now be reaping the benefits of knowledge of the fact that the blood does indeed circulate through the body. This basic knowledge alone has led to untold thousands of other vital discoveries. Do you dare think of the thousands of brilliant findings about the human body and mind that did not have the king's favor (or the sanction of a medical Establishment, which passes as the same thing today) and so have been repositied in limbo?

Canadian Inspection—the Shute Institute

As I reported in one of the last editions of my previous book, my first trip to Canada to investigate Vitamin E was a most enlightening experience. It bears repeating since conditions are essentially the same as they were then.

Dr. Evan Shute conducted me through the Shute Institute, where I met several of the patients and many of the staff members. The Institute was what one would expect of a man of Shute's caliber and the procedures were thoroughly scientific in every detail—from the initial examinations, replete with X-rays and other tests, blood examinations, and the electrocardiograph, to the careful workup of the patient's present condition, including a painstaking review of the patient's history and previous treatment.

It was as complete an approach toward diagnosing a patient's ills as one could hope for. Having been a health/science writer for many years and having inspected many clinics and laboratories, I can report that I was extremely impressed by the work being done at the Shute Institute, as well as by the high morale of the

staff members, who were and are dedicated, objective workers seeking the truth about Vitamin E.

Several years ago, I talked with Dr. Evan Shute (among many other conversations over the years of our friendship); he told me the Institute was swamped with patients. Some were having to wait three or more months for appointments. While it is pleasing to learn of Vitamin E's success both clinically and financially, it is disappointing that most doctors, particularly cardiologists, have refused to remove their blindfolds in order to see the value of Vitamin E in heart diseases. Most, though, are now peeking through their blindfolds and are beginning to see the sun shining through the heavy layers of clouds which have hitherto obscured the scientific truths about Vitamin E.

I asked Dr. Shute to consider enlarging the Institute's facilities so it would accommodate the increasing numbers of heart disease sufferers who could find no doctor who would treat them adequately—i.e., with Vitamin E and other therapies.

Dr. Shute promised he would seriously attempt this, although hiring competent heart specialists would not be easy, even in this era of overwhelmingly favorable reports. He did do so, however, and the Institute can now handle more patients.

In one of our last conversations he said: "I am counting on you to save the situation so that doctors will come here, and that even more will use Vitamin E in their clinical practice. Then we will see more and more people saved and we will achieve victory in our lifetime."

I thanked Dr. Shute for his confidence in me and told him I would never rest until I had done everything possible to win a total victory for the truth about Vitamin E.

He replied: "You have been the most effective force in bringing the facts about Vitamin E before the public, more than any man I know. I think we'll win."

I said, "I know we will."

End of conversation with one of the greatest scientists and humanitarians, now in another dimension, but still embattled, I believe. But *we shall win*, Dr. Shute.

I am told by various persons whom I have sent to the Shute Institute that the work done there has maintained its same high quality of scientific integrity throughout the years. The Shute Institute now employs a staff of three full-time doctors as well as the nursing staff. The directorship of the Shute Institute has been assumed by Evan Shute's son, Vere Shute, Ph.D., a biochemist, who assured me the work of the Institute is being maintained in the same spirit of science and humanity that his father would have wanted. Dr. G. E. Desaulniers is head of the medical staff, and Dr. Shute's daughter, Janet Shute, is administrator.

Canadian Inspection—the Taylor Farms

On one of my visits to Canada, Dr. Shute introduced me to another man who had helped to keep alive the truth about Vitamin E during the dark years when it was being ignored or under heavy attack. That man was W. J. (Bill) Gutterson, the former president of Webber Pharmaceuticals, Ltd., of Rexdale, Ontario. Mr. Gutterson became convinced by the Shute brothers' early findings as to Vitamin E's efficacy, and he soon began to supply them (and many other researchers) with Vitamin E in potent form for various research experiments.

Vitamin E is not inexpensive to make since it takes a tankful of the various vegetable oils (primarily soya) from which it is derived to produce a cupful of the dry, pure vitamin. Mr. Gutterson supplied a number of university research projects, including the famous series

of experiments on the E. P. Taylor Farms, which were described in some detail in my first book.

Also, since I wanted to see firsthand the effects of Vitamin E on horse racing and breeding, I accepted Bill Gutterson's offer to drive me to the Taylor Farms. There I met Gil Darlington, the manager of the stud farm and co-author of the *Darlington-Chassels Report*.

Gutterson told me that more than a thousand physicians were ordering Vitamin E from him for their personal use and more than two hundred Canadian hospitals were ordering Vitamin E for treating their staff doctors. Not many of their patients! Yet Gutterson estimated that "medical acceptance of Vitamin E therapy was in the 60 percent range and increasing daily."

Since the first book was published, many persons have asked me if the Taylor Farms are still using Vitamin E. I can personally report that most emphatically they are, and, if anything, they have increased the dosages, in some cases dramatically.

Mr. Darlington drove us around the stud farm and also conducted us through the laboratories the farm maintains. In my opinion, these horses receive much more careful medical attention than do most human beings, even in developed countries such as the United States. Every aspect of their well-being is watched over. For example, the farm raises several hundred head of cattle, which provide the organic fertilizer to produce the superior grass upon which the horses graze. (They are also given special food daily, of course, containing 5,000 or more units of Vitamin E in the form of pure alpha tocopherol.)

Samples of the horses' blood are taken regularly and examined to ascertain if various constituents, such as vitamins and minerals, are present in the proper balance. Also, if signs of any incipient disease should be manifest, the several veterinarians and medical technicians on the staff are quick to pick them up and institute the proper treatment.

The breeding itself follows extremely scientific procedures. (I am not so sure that human beings would appreciate being subjected to this kind of "scientific sex"; the horses, of course, are not aware that they are being handled and observed, not only physically but psychologically as well.)

The instant two horses have been bred, the stallion and the mare are led away to their separate stalls. Meanwhile, a technician, who is trained for the job, has caught some of the stallion's semen in a bottle. He rushes it to the laboratory, where it is examined for viability of the sperm cells. By microscopic examination the veterinarian can tell, with almost unfailing accuracy, whether or not the coitus will result in conception. This development is extremely important, for, in the past, no one could tell for weeks or months whether or not a mare had been impregnated. Thus, in many cases, valuable time was lost to the horse breeder, whose success depends largely on the number of foals that are produced. On the Taylor Farms this uncertainty has been eliminated to a great degree. If, in spite of the fact that a stallion has been taking Vitamin E, he still shows a lower sperm count than the veterinarians consider necessary for almost certain impregnation, he is immediately put on additional Vitamin E. Remember, the mare is also receiving generous amounts of Vitamin E supplementation in her feed.

Thus, at the Taylor Farms, the level of Vitamin E in the animals' blood is used as a criterion as to whether conception will occur, while increased dosages help to produce more and better offspring. Witness similar effects on humans and on various other animals, as I have reported in the first book and also elsewhere in this book; yet witness, too, alas, how many doctors are ignorant of these relatively simple procedures.

I asked Mr. Darlington if his profits had increased in the last several years since the farm had begun using Vitamin E.

"About one hundred percent increase," he replied.

"Do you think it's due to the Vitamin E?" I asked.
"Or are there other factors?"

"I know that it's due to the Vitamin E," he said. "We are using the same methods as we did before—with the addition of Vitamin E. Also, by controlling the dose of Vitamin E, we can demonstrate its efficacy on both the stallions and the mares. We're also producing the type of foal which is genetically better, as proved by its appearance and its later performance."

He cited the long list of countries that now ship prize mares to be bred at the Taylor Farms. "They can detect a vast difference in both the mare and the foal when they are shipped back," he said. "And, what's more, they are the ones who are winning the races. What more proof do you want?"

E. P. Taylor later sold his interest in the stud farm. It is now called the Winfield Farms. I talked to the present manager, Alan Drizen, who confirmed they are still using Vitamin E in their feeding of both mares and studs, from 3,000 to 5,000 units daily. Prior to the race, the colts and mares are given 10,000 units.

Alan also told me Winfield Farms were exporting horses throughout the world—Europe, South America, and the United States included—and they have increased their earnings by the same figure as we reported in our first book, 100 percent.

It is my particular pleasure to report that my friend, Northern Dancer, whose illustrious record includes winning the Kentucky Derby, the Preakness, and many other notable races, has achieved the all-time record as a stud horse, with earnings of over 20 million dollars. In fact, he has earned more as a father of champions than any other stud known. Wonder why? Both his parents were "Vitamin E horses," as was he.

Good luck, Northern Dancer! I saw you win and I saw you stud, and you did both beautifully, thanks to

Vitamin E—although you were unaware of it. May more humans become aware—such as Muhammad Ali!

Recently, some eminent American researchers, hitherto skeptical, or even hostile, toward Vitamin E, have swung about 180 degrees toward the fact that Vitamin E tends to produce normality in every body cell, including the brain. Most of the critics have died, of course, steadfastly maintaining their skepticism as was imparted to them by the American Medical Association/Food and Drug Administration propaganda.

However, several—including the famous Dr. M. K. Horwitt, at St. Louis University's School of Medicine—have admitted that their prior evaluations of Vitamin E were wrong and are currently pursuing much additional research.

Several years ago, the official publication of the AMA, *Journal of the American Medical Association* (JAMA), considered to be authoritative usually by elderly gentlemen who received their medical education thirty to forty years ago scoffingly editorialized that Vitamin E was a "vitamin in search of a disease." This implied, of course, that since no identifiable disease had been found that Vitamin E would cure, little significance could be placed on it.

Prevention of a noninfectious disease was almost unheard of at that time. To be really useful a vitamin had to cure a specific disease. For example, Vitamin C cures scurvy, but is presumed to be of no help for anything else when taken above the recommended amount. These "facts" were established by doctors who were *disease-oriented* and not aware that vitamins and other substances may be used not only for prevention and cure but also may be pharmacologically useful for many other purposes.

Penicillin, for instance, although not found in the human body, is a natural antibacterial substance and can greatly assist in the body's own natural defenses—

usually the "white" cells—to check infections. So can many other substances quell other diseases: foxglove digitalis (heart), quinine (malaria), buttercups (gout), rauwolfia (high blood pressure), warfarin (anticlotting), to name only a few which are not naturally contained in the body.

In fact, about half of the U.S. Pharmacopoeia is comprised of plant-derived substances, and the search for other as yet undiscovered plants is costing pharmaceutical companies millions of dollars. When found, they would probably have to be marketed in Europe first since laws in the United States against new drugs are so impossibly strict that it usually takes about ten years and multimillions of dollars for them to be accepted by the FDA.

Thus to equate the use of vitamins with a thousand other substances, which were accepted after great reluctance by the medical fraternity is using as narrow a logic as was once employed by our ancestors—the tree, cave, and plains dwellers—but hardly worthy of the present-day spokesmen of science, who control our destiny at the moment. Let me try to prove to you that the tunnel-visioned scientists have deceived not only you but their governments as well—in addition to themselves for the most part.

I digress because I want you to know why you have not received the truth about Vitamin E in the past. Now, however, some of the light is beginning to shine through the dark webs of ignorance, arrogance, greed, and, sometimes, outright conspiracy.

CHAPTER 6

Vitamin E Deficiency Proved Again

In our first book on Vitamin E we stated that “unasailable scientific studies demonstrate that from 5 to 7 percent of adult Americans are deficient in Vitamin E.” (Due to more sophisticated devices for detecting the levels of Vitamin E, it has been determined these figures are below the *actual* deficiency of Vitamin E, as we shall see subsequently.) These experiments were based on the work of our oft-mentioned Vitamin E pioneer researchers, Drs. P. L. Harris and M. K. Horwitt,* among others. The studies were done in the 1940’s and 1950’s, before the advent of the polyunsaturated oil “food faddist” craze, which was foisted upon the gullible American public by certain well-meaning but misled cardiac researchers. They preached, with all the vehemence worthy of our best-known evangelists, that ingesting huge quantities of corn oil or other vegetable oils—while abandoning bacon, eggs, and butter—would lower the blood cholesterol and would therefore prevent heart disease. However, time and further research have shown that all the assumptions contained in the preceding sentence were hardly worthy of the unconscionably prolonged, fantastically expensive, repetitious experiments.

Yet the sophistry of those medical men (who cannot admit a mistake) still lingers on and on. And it will undoubtedly persist until they have gone to their uncertain reward. But think of the millions who blindly fol-

*See Chapter 7, “Researchers Who Find Vitamin E Most Essential.”

lowed their pronouncements. No, let us be charitable. *Don't* think of them.

You see, there have been scores of well-documented studies in the last few years concerning the relationship between polyunsaturated fats and Vitamin E. These investigations corroborated the work of Drs. Harman, Harris, Horwitt, and others. Some are from Japan, the former Soviet Union, England, the United States—in fact, almost every country seriously studying heart disease reports the obvious fact that the more polyunsaturated fats ingested, the more deficient that person (or animal) becomes in Vitamin E. For it is Vitamin E in its role as chief natural antioxidant of the body, combining with polyunsaturated fats, that keeps them from oxidizing and forming hydrogen peroxides. The more polyunsaturated fats, the more Vitamin E is needed to counteract this deadly effect on cellular metabolism. Ironically, although natural Vitamin E is manufactured from these unsaturated oils, the oils do not contain enough alpha tocopherol *in proportion* to the harmful substances to benefit the human body. The exceptions are possibly cottonseed oil or gamma linoleic oil, and you know how popular they are.

Among all the studies which prove this point, perhaps one of the best came from this country under the most authoritative aegis. At a meeting of the American Chemical Society in Los Angeles, Dr. N. R. DiLuzio of Tulane University presented a paper which may well be one of the most significant in the long history of Vitamin E and its relationship to heart disease, aging, and various other ailments with which we are well acquainted.

Dr. DiLuzio's report gives the lie to those who blandly state that the "normal" American diet provides the "normal" American with enough Vitamin E.

Dr. DiLuzio examined the blood of so-called normal persons. He found that 96 percent had abnormally high levels of "peroxidized" fats. He and his associates then

analyzed certain polyunsaturated oils bought from the various supermarkets to determine whether or not the dangerous substances called "dienes" were present. They were. They were in corn oil (the most popular of the various oils), peanut oil, and safflower oils.

Then the diligent researchers determined to find out if the dienes in the blood of the normal subjects could be caused by the relative absence of Vitamin E. The researchers gave supplementary Vitamin E to five of the normal subjects while they ate the same normal diet for seven days. Vitamin E levels and the dienes were measured at the beginning and at the end of the experiment.

In every case, Vitamin E significantly lowered the level of the dangerous dienes and raised the level of antioxidant activity. To put a QED on the experiment, the doctors then took away the Vitamin E. Without exception, the harmful substances returned to the bloodstream because the beneficent effect of the added Vitamin E was no longer there.

Therefore, it should not be difficult to ascertain Dr. DiLuzio's conclusion. The average normal American does have a deficiency of Vitamin E. And a provable one. Not only proved by Dr. DiLuzio's controlled experiment, but, as we have said, by scores of similar studies throughout the world.

One of these similar studies was made in England by Drs. Smith, Kelleher, Losowsky, and Morris. Writing in the British *Journal of Nutrition* they showed that about 90 percent of British diets, although otherwise "adequate," were woefully lacking in Vitamin E. For instance, the average "normal" Briton, eating a "normal" diet, obtains less than 5 milligrams of Vitamin E daily. It has now been acknowledged, even by the majority of U. S. researchers, that 30 milligrams are required by the "average" man—much more if polyunsaturated fats replace the natural fats found in eggs, meat, milk, etc.

Correlate this with the fact that Britain has about the same rate of cardiovascular troubles as the United States and it should make you wonder. It should make the great (and I use the word "great" advisedly) cardiologists do something about an experiment with Vitamin E.

Our point is, did you read or hear about Dr. DiLuzio's studies or any others? The chances are that you did not, but that you read or heard about the "counter meetings" called by the minions of the Medical Establishment to have its "grantees" and their "information dependents"—the media—assure you that extra Vitamin E was not necessary because everybody got plenty of it in the "normal" American diet except in very rare cases; that it could not possibly have an effect in larger doses than those suggested in the National Research Council's Recommended Daily Allowance.

If your doctor is worthy of his title, he will be delighted to examine—or at least skim—the summaries of the reports we cite here and the others we list in the Appendix.*

*See D. Harman (Appendix F-1); A. Tappel (Appendix A); M. K. Horwitt (Appendix A);, and others too numerous to mention here.

Researchers Who Find Vitamin E Most Essential

Dr. Denham Harman: Chief Architect of the Unified Aging Process

Readers will recall that in the Preface I quoted Denham Harman, M.D., Ph.D., Director of Medical Research, University of Nebraska College of Medical Center. Dr. Harman has a host of other credits too numerous to mention here as they would probably necessitate two more volumes about Vitamin E.

Suffice it to say that Dr. Harman is regarded as the conceiver of the free-radical theory of aging (see page 102). In a series of tests beginning in the 1950s, he, along with associates and later, other researchers, established the free-radical theory of aging so convincingly that today few researchers I can find disagree with it.

But Dr. Harman went further than just advancing the theory. He outlined the *why* and *how*, which follows shortly. He showed by many tests that “all components of the body would be subjected to [damaging] chemical change in a more or less random manner.”

These damaging changes are due to the free radicals which are caused by processes occurring within the body, polyunsaturated fats being the chief culprits (corn oil, safflower oil, peanut oil, etc., all of which are promoted as having little or no cholesterol and thereby supposedly reduce the risk of cardiovascular diseases).

More damaging changes come from our *external* en-

vironment: radiation, air, food and water pollution, even cosmic rays, as well as a host of other sources—of which I am sure the reader must be aware by now.

Oversimplified, Dr. Harman's theory* follows: (1) Humans of less than average weight have a relatively long life span. (2) Unsaturated fats decrease the life span of mice and rats and likewise are strongly presumptively so in the human. Certain proteins such as casein decrease the life span in rats, while others (soybeans), which are less easily oxidized, increase the life span. (3) Too much copper is implicated in hypertension since copper, as well as other substances (even calcium), tend to accumulate in us, as well as rats, as we grown older. The brain, heart, and all other organs are affected, thus a plethora of other diseases are involved, including senile dementia and Alzheimer's. (4) Antioxidants such as Vitamin E prevent harmful peroxides from forming and producing the dangerous free radicals which damage every cell in the body, including the arteries, veins, heart, lungs, liver, and brain. (5) Antioxidants—Vitamins E, C, and A (beta carotene) being the most commonly known—enhance the immune system favorably, thus protecting the body against diseases in which the immune system is involved. (6) The autoimmune mechanism causes the immunological protective cells to become abnormal and lose their ability to detect the body's own cells from foreign invaders. Then they attack and destroy the normal cells as well as the invaders, such as in organ transplants. Tests have shown the antioxidants can in some instances change abnormal "protective" cells back to their normal functions.

Here Dr. Harman's report ends, but I'm sure he agrees with what follows in this book.**

*See Dr. Harman's full report, Appendix F-1

**Dr. Harman, of course, advocates supplementation with Vitamin E and other antioxidants.

I have attended several conferences of the American Aging Association (AGE) of which Dr. Harman is Executive Director. I talked with and interviewed him many times. He told me of his discovery of the Free Radical Theory of Aging.

After many long years of study, research, and deep meditation, the answer came to Dr. Harman suddenly on a November afternoon in 1954 in a "flash awareness." It happens to many persons seeking solutions to problems—probably you, too, have experienced this strange, striking phenomenon. It has become known as the "Eureka" effect after Archimedes the famous Greek philosopher-engineer, who while sitting in his bathtub (pondering over the problem of the displacement of water) suddenly shouted "Eureka" (Greek: "I have found it!").

Dr. Harman and other researchers conducted many series of tests on mice and guinea pigs proving that Vitamin E and other antioxidants neutralized the free radicals which cause aging and many other diseases.

Recently (1992), Dr. Harman compiled a huge compendium of hundreds of studies on animals and human patients from all over the developed world. In addition to heart disease and cancer, researchers utilizing still other antioxidants similar to E, C and beta carotene, tested them on Alzheimer's, Parkinson's and Amyloidosis (a fibrous overgrowth of protein tissue in the brain associated with Alzheimer's) and Senile Immune Deficiency.

Actually, these new findings, for the most part, corroborate previously noted evidence in *E: The Essential Vitamin*. (Except, of course, for the new diseases and additional antioxidants.) These antioxidants and other new substances include L-Glutathione, Co-Enzyme Q 10, Phosphatidyl Choline, L-Glutamine, and Superoxide Dismutase. Most of these antioxidants as well as other supplements can easily be obtained without a prescription. Recommended dosages are on the labels of

the bottles. Our readers can start safely on their own, although it would be preferable to be under the supervision of a nutrition-oriented doctor. One particular axiom: "It's always best to begin with the universally accepted E, C, and beta carotene formulas."

Dr. Harman notes that all these diseases recently tested—and their subsequent treatment with the standard as well as the additional antioxidants—further corroborated his Free Radical Theory of Aging and Disease. As if Dr. Harman needed any more factual proof! Let's hope Dr. Harman soon receives his Nobel Prize, which is long overdue.

The bad news is that the Old Guard, namely the Medical Powers, are dragging their well-polished heels.

The good news is that so much evidence has been published all over the world, including the minority of brave, reputable researchers here, that the medical agencies, both governmental and quasi-official, can no longer withstand the avant-garde and probably we will no longer have to flock to other countries where treatments have been used and proved effective for years, yet are not allowed in the United States. But don't hold your breath. Use it for expounding the truth.

Now as I told you, our conceptual *and* practical researcher, Dr. Harman, has evolved a unified plan for combating old age. As we mentioned previously, the chief component is the use of antioxidants which counteract the deadly effects of free radicals generated both inside and outside the body. There are several antioxidants which lock into battle with the intruders, but there is one *natural* leader, Vitamin E, that guides the lieutenants: Vitamins C and A. The troops, too, are most important. What general can win a battle, much less a war, without well-trained troops?

Well, Vitamin E *can* win the war alone, but can win much better with knowledgeable assistance, from C and A (beta carotene) as well as generous amounts of the B vitamins, choline, inositol, minerals (particularly cal-

cium, zinc, magnesium, iron, and the trace minerals such as selenium and manganese). Proper diet and regular exercise are also essential.

Dr. Hayflick and His “Biological Clock”

Several researchers have corroborated the fact that Vitamin E does prolong indefinitely the life of red blood cells—both in the laboratory (*in vitro*) and in the living human body (*in vivo*). This in addition to its multitudinous other tasks at the cell membrane and molecular level. Since the red blood cells carry essential substances: oxygen, vitamins, minerals, food nutrients, as well as many toxins such as carbon monoxide, ozone, carbon dioxide, lead, sulfur dioxide (and many others), it follows that red blood cells are most essential to our life here on this planet, to put it mildly. Every action we undertake from the lifting of a finger to profound thought is controlled by the conveying, nutrient-bearing, toxin-bearing red blood cells. (“The blood is the life,” as the old saying goes.)

I know that multiple physical factors such as vitamins, minerals, and other substances influence the body and mind. (I take all the beneficent substances that I know about, and I hope you do, too.)

Yet I am delighted when I find a study with a *single* factor involved. Put the rats, monkeys, or humans on a diet to which they are accustomed. And *then* bring in *the* added factor. If the added factor produces results, positive or negative, without adding extra substances such as other vitamins or minerals, or whatever, then you have a real, valid, truly scientific test. This does not mean that the proposed material is not efficacious if it cannot prove itself alone. It merely means that it

may need the assistance of other substances in order to fulfill its proposed and/or hypothetical claims.

Such is not the case with Vitamin E. It works alone, yet is aided by other antioxidants—Vitamins A (beta carotene), C, as well as other vitamins and/or minerals. But almost all the studies reported in this book were done with Vitamin E *alone*: no added factors—no extra vitamins—no special diets.

This is the type of study I like to see, so that critics can no longer say, “Well, it may be due to other factors.”

We shall begin at the cellular, or even molecular, level where all health or ill health originates.

We shall find an abundance of corroborating evidence about this hitherto mysterious Vitamin E, both in the laboratory on animals, their cells, and in their lives. We shall find human beings’ cells also analyzed with approximately the same results. We shall see that humans ingesting Vitamin E out of the laboratory reacted just as favorably.

In medicine, many “new” developments are greeted with a chorus of jeers from the establishment or—worse still—a total ignoring of the “perpetrator,” to use police jargon.

Most developments aren’t entirely new. When researched, it is found that there was someone who previously promulgated the original idea or test, but was ignored or unaccepted even when published in “reputable” journals. Science (which includes medicine) builds on itself, it is true; if science had no past, it could not have a present or future.

When Dr. Leonard Hayflick of Stanford University made the discovery that human blood cells can only reproduce themselves (in laboratory dishes) about fifty times and then die, the pall of gloom thickened considerably around proponents of the theory that man should by biological “law” live to be between 120 and 150 years old, or more—as animals do, correlating roughly

five to six times to their maturation and metabolic rates: dogs—two years; cats—two; mice—one; elephants—twenty; birds (depending on species): hummingbird—one; parrot—nine, etc. Using this theory, which seems to be valid, man, whose full maturation is about twenty-five, should live to be between 120 and 150 years instead of the mere three-score-and-ten, as we just noted.

Hayflick's findings *in vitro* show that the human cell can reproduce only about fifty generations before dying or "leveling off" into a state of "undifferentiated" or embryonic cells which resemble cancer cells in that they are no longer capable of functioning as, say, lung cells, and can no longer take orders from central command—the brain. It became axiomatic among most scientists that there was indeed a biological clock which informed each cell when to live and when to die. The body, of course, followed the clock, being dependent on the cells.

There was an abundance of evidence to support this hypothesis ranging from the Bible to modern-day science. Logic, too, was in full agreement, for didn't the young body mature, become middle-aged, grow old, and die within a certain span? Some lived a bit longer than others, but spry as they seemed to be at ninety, nevertheless they, too, soon entered the same "bosky bourne" as had their younger companions.

However, less than three years after Hayflick's classic experiment—and it was classic in that it, more than any other, seemingly demonstrated the inexorable steady march from birth to death—other researchers challenged Hayflick's findings by use of Vitamin E.

For those critics who once thought Vitamin E's effects were due to the placebo effect, we shall shortly present a plutonium-clad experiment which leaves no room for doubt or even debate—except for those whose thoughts and/or livelihood depend on contesting Vitamin E's efficacy on every level no matter what the evidence.

We all know about the placebo effect: that in any controlled experiment, the proposed valid substance will be given to one group of patients, and an equal number of patients are given a harmless, inert substance. At the end of the experiment the patients are examined to find which substance was the more valuable.

Although it would be supposed by those who have read about Hayflick's biological clock for man, the universe, and all "it" contains—animals, plants (above the waters, in them, and below)—that Hayflick would be *the* ultrapessimist, yet he is not but, instead, is an optimist. He thinks that man with his present, and certainly, immediate future, knowledge about cells, neurotransmitters, and other technological findings can alter cells, genes, and 10,000 other factors to make him well nigh immortal.

Incidentally, since science is built on previous suppositions and hypotheses, Hayflick would probably be the first to admit his theory of the biological clock was not entirely original with him, although he "proved" it in the laboratory for the first time. DeBuffon, the French biologist, had advanced the theory for mammals in the eighteenth century and, of course, many thousands of years of folklore had preceded him.

I spoke to Dr. Hayflick at the Vista Hill Foundation's (San Diego) Seminar on Aging in February 1975, shortly after his publication on cell reproduction.

His answer to my positive observations about Vitamin E was: "This work is very interesting. That's what we scientists are here for. To look for the truth in all directions." This is the mark of a real scientist.

Dr. Hayflick contributed much to the favorable evidence of the antioxidative qualities of Vitamin E, corroborating Harman and many others. He it was who supplied Vitamin E (with cell nutrients) to Drs. Packer and Smith for their extensive cell experiments (see page 77).

Dr. Smith-Sonneborn

First Breaks Hayflick's Clock

Hayflick's theory, which had almost become a law, was first upset in 1979 by Joan Smith-Sonneborn, Ph.D., professor of zoology and physiology at the University of Wyoming.

She chose the "lowly" one-cell paramecium for the experiment. Why? One reason was that all life began in the sea before it migrated to the land; and then some species (such as the dolphins and whales) decided the land wasn't for them and returned to Mother Sea. Whether that was a wise move we shall probably learn shortly.

However, the water-living paramecium was elected by Dr. Smith-Sonneborn since it possesses many qualities of higher forms of life, which includes us.

She injured the DNA of the first group with ultraviolet radiation. The second group was untreated.

DNA combined with other substances is responsible for repairing cellular damage, membrane leakage which many scientists ascribe to being partially due to aging of the cell.

The first group died earlier than the second, as expected. Then the doctor tried two more groups, one stimulated with a repair enzyme known to react to a certain light wave. (Both groups had been damaged with ultraviolet light.)

Dr. Smith-Sonneborn reported that not only did the treated animals live much longer than the second damaged group but surpassed it by a whopping 296 percent, although both groups were middle-aged when the enzyme was activated in the first group.

Says Smith-Sonneborn, "What we did was break through the Hayflick limit for paramecia."

A most important breakthrough indeed. The implications for man are vast, as we shall see subsequently.

Drs. Packer and Smith's Research on Human Lung Cells

Very likely our next two researchers were also intrigued by Hayflick's built-in biological clock. In true science, it is the role of researchers to prove or disprove a finding made by a colleague.

These two researchers, Dr. Lester Packer and his colleague, Dr. James R. Smith, at the University of California at Berkeley, reported that they had used the identical method—nutrients, light, heat—used by Hayflick. But they used *two* cultures of cells, equal in every aspect, except in one culture they put Vitamin E in good quantity.

They could not believe the results, except that the microscope proved it. The human lung cells fed Vitamin E lasted 120 generations, while those with normal nutrients died at fifty as predicted. The Vitamin E cells were healthy when the researchers ended the experiment, having exhausted the materials sent them by Dr. Hayflick.

The Packer-Smith report that Vitamin E *alone* could prolong the normal life of the red blood cell apparently *indefinitely*, of course, received wide coverage in the media, including the *New York Times*. However, it seems that no "authority" knew what to make of it, including Packer and Smith. Various authorities were quoted as saying something to the effect: "The work is interesting, but it's too early to tell what it means."

It meant enough to Packer and Smith to begin taking 200 units of Vitamin E a day. We know of several other authorities who also began taking equal or larger amounts, especially when they learned of other conclusive experiments of eminent researchers who had proved that Vitamin E was a most integral part of the long-sought-for search for a preserver of health and, therefore, of life.

Both Packer and Smith are advocates of the free radical theory—as are Harman, Pryor, Tappel, and others—which we explain in the next section.

Later, Packer and Smith reported they could not duplicate their original experiments with Vitamin E. We wonder why. So do the researchers, apparently, since they seem at a loss to explain.*

Dr. Horwitt's Change of Mind

Dr. M. K. Horwitt, noted researcher now at St. Louis University School of Medicine, presented further proof in Japan at the Fourth and Seventh International symposia on Vitamin E, that the need for Vitamin E is directly linked to intake of polyunsaturated fats. He showed that it is *not* the dietary intake of the polyunsaturated fats which reflects the basic need for Vitamin E; it is only by chemical examination of the tissues that scientists can get an accurate picture of the actual need of alpha tocopherol. His conclusions were that 30 I.U.'s of alpha tocopherol were the requirements of most adults, but that individuals vary so much it was almost impossible to set a definite amount for anyone's exact needs.** We have noted this observation repeatedly in regard to *all* the vitamins and minerals, not just Vitamin E.

We agree with most modern research nutritionists: in general, it is far better to take much more of the vitamins than the Recommended Daily Allowance. There are too many variables involved: heredity; illness (acute and chronic); malabsorption; improper diet; lack of exercise; alcohol, drugs (even aspirin); age; allergies;

*They suggest that another unknown substance which was present in the first materials "united" with Vitamin E in the first experiment and was not present in later materials sent to them by Dr. Hayflick. However, both researchers are still ardent Vitamin E advocates.

**See Appendix F-2 for additional information by Dr. Horwitt.

tension, both mental and physical—in short, just about every factor known to the human race. In fact, if the FDA or the AMA could produce very many persons who are perfectly healthy both mentally and physically, who eat a good healthy “normal” diet and therefore don’t need supplemental vitamins—according to their propaganda—why, then, according to this strange logic, we ought to be able to shut down most of the medical schools and close most hospitals tomorrow, except for victims of accidents. You see, healthy citizens who eat normal, healthy recommended diets wouldn’t need doctors or hospitals, would they?

Of course, all the studies made by responsible agencies, including the U.S. Public Health Service, the Department of Agriculture, as well as dozens of others show the picture to be quite different from the propaganda, as we have stated previously. Therefore, whom do you trust? Schizophrenic, erratic, hysterical power (reminiscent of certain dictators), or the calm, dispassionate facts arrived at by thousands of true scientists, some of whom are even employed by the medico-scientific bureaucracy?

Dr. Jeffrey Bland: Vitamin E’s Energizing, Protective Action

Another of our beautifully executed scientific experiments took place around 1976–78 in the laboratory of Jeffrey Bland, Ph.D., associate professor of chemistry at the University of Puget Sound, Tacoma, Washington. The professor is also editor of the *Northwest Academy Review*.

Dr. Bland and his associates penetrated the role of Vitamin E at the molecular and cellular level in a series of the most brilliant studies it has been my privilege to read, demonstrating beyond reasonable doubt the

truth of what hundreds of researchers and clinicians had been saying for twenty years.

His work had been somewhat foreshadowed by the late David Turner, Ph.D. (previously mentioned), who received a NASA award because of his discovery that Vitamin E could counteract the anemia produced in the astronauts since in the early flights they breathed pure oxygen for long periods of time. (Pure oxygen is a slow, deadly cellular poison, although most necessary when combined with nitrogen.)

Briefly, Dr. Bland's results indicated that greater amounts of Vitamin E can help prevent the cells of the body from aging faster than usual.

How did Dr. Bland and his colleagues arrive at this conclusion?

First, the team exposed red blood cells to light and oxygen. They found that the combination was as deadly as Dr. Turner had earlier ascertained—that just by breathing pure oxygen (particularly under pressure) red blood cells would be killed prematurely and thus cause anemia.

What Bland and his colleagues found was corroboration of researchers' earlier work (he gives full credit to all the other researchers, which is much more commendable than certain other rather famous researchers and writers who either cannot, do not, or will not read previous or contemporary medical literature, although it is available to them, perhaps more so than to you).

Yet Dr. Bland et al., went far beyond previous investigators' findings in the laboratory and in human beings.

In one of his first studies, Dr. Bland extracted blood from twenty-four volunteers and exposed their red blood cells to light and oxygen for sixteen days. Result: nearly all died or became abnormal and no longer could perform their functions as red blood cells. (This proved earlier findings that, although oxygen is desperately needed by cells—but, as we mentioned earlier, not

pure oxygen.) You will find that oxygen administered today is only about 20 percent pure oxygen, combined with regular air. As most doctors now know, 100 percent pure oxygen can harm lungs and red blood cells almost as fast as several well-known poisons.

Next, Dr. Bland's team gave the same twenty-four volunteers 600 units of alpha tocopherol a day for ten days. Then they checked their blood again.

To their astonishment, they found that Vitamin E almost totally protected the cells after exposure to light and oxygen for sixteen hours, the same period of time that the unsupplemented cells had been exposed and had died or become abnormal.

They had proved Vitamin E was an extremely potent cell protector.

Further, they discovered that the addition of Vitamin E significantly reduces the rate at which cholesterol, an important factor in the cell membrane, is oxidized to a dangerous compound known as cholesterol hydroperoxide, which weakens the cell membrane. This weakening of the membrane no doubt lets destructive forces both in and out of the cell, thereby opening the door for many diseases or maladies, including most humans' worst fear: old age and death.

But Bland has not yet had his final say about Vitamin E. After all, he is not a graybeard full of pomposities, generalities, and double talk like those, I daresay, you have met in the medical fraternity.

Bland, however, takes his research most seriously, as we have seen. He is a preventive disease enthusiast, a nutritionist, and exercises regularly. He takes 600 units of Vitamin E a day, being a most productive convert to Vitamin E.

Also, we must note that Dr. Bland has also tested long-distance runners. His results show that when runners are given 600 units of Vitamin E, the "peroxida-

tive'' damage (destruction) to their red blood cells was decreased by as much as 50 percent.

Dr. Passwater Surveys 18,000 Persons

Richard A. Passwater, Ph.D., is a renowned research scientist who has contributed many outstanding articles to leading scientific journals. He has also written several books for the general reader. Dr. Passwater's latest book is *The New Supernutrition*. However, the one that interests us most here is *Supernutrition for Healthy Hearts*. In it, he discusses his "favorite research subjects," Vitamin E and the trace mineral selenium (see page 154), both antioxidants, although Vitamin E has a much broader role in human metabolism, as we have seen.

Dr. Passwater is included in our list of quotes of foremost researchers who were drawn to my first book on Vitamin E—and most of whom were encouraged to undertake further experimentation, all of which proved very fruitful.

However, the foregoing is not the reason we are including Richard Passwater in this chapter. We are including him because of his monumental and unique survey among readers of *Prevention* magazine in 1974-75.

As you will see, Dr. Passwater's research buttresses Vitamin E's major role in controlling and preventing cardiovascular diseases, in coronaries, strokes, etc. It adds almost immeasurably to our total assessment of Vitamin E, whether the evidence comes from the laboratory, from rats, or from humans living "normal" lives. We now have most of the evidence supporting the fact that Vitamin E, taken in sufficient dosage for

a period of time, will prevent and/or control cardiovascular accidents.

Dr. Passwater sent out a comprehensive questionnaire to *Prevention* readers between the ages of 50 and 98 who had been taking at least 400 I.U.'s of Vitamin E a day. About 18,000 responded.

To Dr. Passwater's surprise, the results (which were analyzed by a statistician) showed that Vitamin E reduced the incidence of heart disease by as much as 80 percent.

A most impressive figure, indeed, especially since it corroborates scores of other similar reports. Some researchers did not have the large number of participating respondents; others, including the Shutes (with 60,000 patients), had patients with already *proved* heart disease. Now, do you think 18,000 persons would deliberately lie about their condition and also take the trouble of answering a complex questionnaire? You know the answer as well as I.

Many were not supposed *to be alive to answer*, according to statistics accumulated by the various heart agencies, both governmental and semigovernmental.

What did the survey find?

There were 17,894 respondents between the ages of 50 and 98, of whom 3,725 had heart disease. The total number of replies of all ages was 17,256 returned questionnaires which provided 29,336 sets of data.

The major findings are:

1. Heart disease incidence may be reduced by 80 percent.
2. Vitamin E is strongly associated with the reduction of heart disease.
3. The relationship is time- and dose-related.
4. The length of time of taking Vitamin E is more important than the dosage.

More important, the specific findings are:

(a) 400 I.U.'s or more of Vitamin E taken for 10 years or more is strongly associated with reducing the

incidence of heart disease prior to 80 years of age to less than 10 percent of the present rate: 32 per 100 people reduced to 3 per 100 people. Table 10-1 gives the data for this finding. (See Appendix F-3.)

(b) 1,200 I.U.'s or more of Vitamin E taken for 4 years or more is strongly associated with reducing the incidence of heart disease prior to 80 years of age from 32 per 100 people to 10 per 100 people. (Table 10-2 gives the data for this finding.)

Tables of his experiments will be found in Appendix F-3, along with other proofs from many other researchers. (Also see Appendix B for Passwater's comparison of his survey with that of the recent Harvard Surveys.)

Drs. Cheraskin and Ringsdorf's Proofs of Vitamin E Efficacy

Still more evidence of Vitamin E's premier role in cardiovascular symptoms is afforded us by two eminent research professors at the University of Alabama's Department of Oral Medicine. Since the scientific world is enamored of statistics, especially when assembled by computers, we are most pleased to report the study is both statistical and computerized. (Not that we are opposed to statistics or computers, far from it. Both should be utilized in every way possible, but sometimes it may be wise to remember that a computer is not a god or even a man and that its actions, though unbelievably complex, cannot compare with its creator, the human brain. As for statistics, though extremely useful, unless they are logically analyzed and interpreted, they can be misleading. Now with what we consider a necessary explanation, back to our statistics and computer study of Vitamin E.)

This is another undeniably scientific report made by

researchers of unquestionable reputation, which you probably didn't read or hear about. It is another one of those favorable studies on Vitamin E which somehow was ignored, yet studies with far less scientific basis and importance do manage to find their way into the daily newspapers and *Time* or *Newsweek*—and thence to television and radio.

The researchers were Dr. Emanuel Cheraskin, a dentist as well as a medical doctor, and Dr. W. M. Ringsdorf, Jr. Both, together with J. W. Clark, were authors of the book *Diet and Disease* (Rodale Press), a comprehensive work linking nutrition with disease and, conversely, health.

Dr. Cheraskin induced groups of fellow dentists and their families to participate in a two-year Vitamin E cardiovascular experiment. There were 171 dentists and 128 wives, most of whom were between thirty and sixty years of age.

First, they were given a dietary analysis and their average daily nutrient intake was analyzed by a computer system which has been shown to be comparatively simple (for a computer) and yet reliable.

Now, the National Research Council's Recommended Daily Allowance for Vitamin E (alpha form only) is 15 I.U.'s for men and 15 I.U.'s for women. For women who are pregnant or lactating, the amount is 30 I.U.'s, though there are many researchers who claim these amounts are too low. (Among them is Dr. Max Horwitt, who first agreed with these low quantities, but was forced to change his mind, as we report elsewhere.)

However, the computer showed that 82 percent of the dentists and their wives were obtaining even less Vitamin E than the recommended minimum! The amount varied tenfold. Some were receiving only 6 milligrams; a few as high as 50. (Dr. R. H. Bunnell and associates concluded the daily intake of Vitamin E in America varies from 3 to 15 mg, thus coinciding

with Dr. DiLuzio's findings which indicate that 96 percent of Americans are probably Vitamin E deficient. (See page 65.) Now dentists are not poor; most eat well, and most, presumably, have some idea of basic nutrition, and this includes some knowledge of vitamins. They at least eat what the FDA and other nutrition "experts" describe as a normal, healthy diet—which according to the FDA, supplies all the necessary vitamins, minerals, and other nutrients a normal American needs.

Then the subjects were given the Cornell (University) Medical Index Health Questionnaire, which is accepted as "standard." They reported any type of suspected cardiovascular disorders such as chest pains, cold hands or feet, shortness of breath, leg cramps, and doctor-diagnosed high or low blood pressure. Also they reported whether or not a doctor had diagnosed them to be suffering from a heart condition.

As would be expected, cardiovascular symptoms became more prevalent with age. However, in that 22 percent who had been receiving even the minimum Recommended Daily Allowance of Vitamin E, cardiovascular symptoms did *not* increase with age. In other words, Vitamin E was protecting them against cardiovascular disorders.

What follows will be enlightening to those unfamiliar with the action of Vitamin E. Researchers Cheraskin and Ringsdorf informed their experiment-participants of their findings and recommended more Vitamin E intake for those who obviously needed it.

Some heeded the advice, and a year later those who had increased their Vitamin E intake reported their symptoms to be improved greatly or entirely gone. Those cynics who wouldn't change their ways if their lives depended on it—and they do!—still had their same troubles.

The investigators put it this way: "Those who increased Vitamin E intake were paralleled by a decrease

in cardiovascular symptoms and signs. Those who did not increase Vitamin E consumption did not improve with regard to the clinical picture.” They went on to conclude with the hope that their observations “will catalyze interest in this apparently new area and generate additional study.”

It really should, Dr. Cheraskin and Dr. Ringsdorf, but meanwhile, carry on, good doctors, we need more of your kind. Urgently.

Drs. Ayres and Mihan's Innovative Discoveries

Samuel Ayres, M.D., and Richard Mihan, M.D., practicing dermatology in Los Angeles, were intrigued by the possible use of Vitamin E when they saw specific evidence of Vitamin E's efficacy while attending a meeting of the Los Angeles Dermatological Society. Dr. Ayres was (and is) a diplomate of the American Board of Dermatology—a very high honor indeed. He is professor emeritus at the University of California's Medical School at Los Angeles.

Dr. Mihan is assistant professor of medicine (Dermatology and Syphilology); also at UCLA School of Medicine.

At this point our readers might be interested in a letter I received from Dr. Ayres which is self-explanatory.

November 23, 1976

Dear Mr. Bailey,

I am indeed very familiar with your first book, *Vitamin E: Your Key to a Healthy Heart*. My introduction to it was the result of a very interesting dermatological case presented before the Los

Angeles Dermatological Society with a diagnosis of pseudoxanthoma elasticum, a rare disease of the elastic tissues for which no effective treatment had ever been devised. This patient had been presented a year previously; her second presentation followed the administration of Vitamin E during the ensuing interval with the result that her skin lesions were about 75 percent improved, and her eye lesions, which had forced her to give up her work because of poor vision, had been restored to normal to the extent that she could read a newspaper in ordinary daylight. The abnormalities of the elastic tissues occur not only in the skin, but in the retina of the eyes and other parts of the body.

I was so impressed with this patient's response that I went to the Los Angeles County Medical Library to get more information about Vitamin E; up to that time it had been only a name in my consciousness. *The librarian handed me a copy of your book. I might say that this was the beginning of my interest in the possible usefulness of Vitamin E in other dermatological problems* [emphasis mine].

During the past ten years my associate, Dr. Richard Mihan, and I have carried out clinical research on some of these obscure skin problems, and as a result of coincidental observations, also on several muscular abnormalities.

According to your request, I am sending you reprints of most of the papers which we have had published, dealing with our investigations. Some of these cases are so rare that it was impossible to carry out controlled studies, but in many instances the patients serve as their own controls, relapsing when the medication is discontinued, and responding again when adequate doses are administered.

In addition to these reprints, one other paper has just been published in the October issue of the *Southern Medical Journal* in collaboration with Dr. Raymond N. F. Killeen, reporting the dramatic responses of a severe case of polymyositis, after failure to respond to three immunosuppressive drugs.

We have also had another paper accepted for publication by *Cutis* entitled, "Is Vitamin E Involved in the Autoimmune Mechanism?" . . . (See Appendix F-4)

You have our permission to reprint or abstract any of this material that you feel is of value. . . .

As indicated in several of our papers, we do not, by any means, consider Vitamin E to be a panacea. On the other hand, its primary mode of action, which appears to be that of an antioxidant and is intimately concerned with the preservation of the integrity of cell membranes and cellular contents, including enzymes, may account for its wide range of usefulness.

Sincerely,

Samuel Ayres, Jr., M.D.

"Restless legs" is a term for a spasm of the leg muscles usually occurring during sleep. The affliction causes much agony and loss of sleep. Although described several centuries ago, no cure or little relief has been found.

Drs. Ayres and Mihan, then testing skin diseases with Vitamin E and obtaining remarkable success, discovered there were more favorable side effects than they had anticipated.

Many of their patients exhibiting notable improvements and/or cures concerning their skin diseases, remarked that their other ailments—such as restless legs,

leg cramps, intermittent claudication, the racking pain and itching of herpes zoster (shingles, for which there is no orthodox cure), angina pectoris (the first signal of arteriosclerosis), and heart disease—seemed greatly improved or diminished. Wonderful “side effects” in contradistinction to almost all drug therapies.

These observations caused our two intrepid clinicians to widen their activities considerably, although they were still treating skin diseases with Vitamin E, of course.

During the next eight years, they wrote about their favorable findings in several well-known medical journals. Now they had the incontrovertible evidence on hundreds of patients. They were published in such respected medical journals as the *Southern Medical Journal*, *Archives of Dermatology*, and *Cutis*. Further, they found other doctors who corroborated their work in other medical journals. (Due to lack of space, we cannot reprint their results here, but their works are in files and boxes as part of the accumulation of Vitamin E material which threatens to overflow my office/home in Connecticut. As you can imagine, I am the witting victim of a disease, coined, I believe by me, called “paperitis.” The victim becomes overwhelmed by tons of paper and eventually succumbs. There is no known cure. Not even Vitamin E!)

One other pertinent reason Dr. Ayres became most interested in Vitamin E was that both he and his wife had been enduring the pain of leg cramps for some time. Vitamin E ended their muscle spasms quickly.

Then Ayres and Mihaan began their historic trials on all their patients suffering from muscle spasms of the leg and foot, in addition to their skin ailments, naturally.

Some had a history of cramps from five to thirty years or longer with no relief. There were 125 patients in the first study, half of whom received 300 units of

Vitamin E; the others received 400 units. The results were phenomenal. "In 103 cases the cramps were completely, or almost completely, controlled," the doctors wrote in the *Southern Medical Journal*, "and in 13, the response was regarded as good." Out of the 125 sufferers only two failed to obtain satisfactory relief of pain!

Better still, the blessed relief usually came within a week and the clutching pain was permanently abolished.

From this study came several other dramatic, classic findings to which we have alluded elsewhere. The researchers were convinced by the overpowering evidence that it is absolutely necessary to take the vitamin in adequate amounts ranging from 300–400 I.U.'s to 1,600 units, sometimes even as high as 2,400 units—depending on the disease and the patient's absorption. In shingles, for instance, it often takes six months of therapy on high dosage to banish the pain because of the chronic, deeply rooted nature of the disease.

There was no evidence of toxicity in any of the experiments, as has been demonstrated in thousands of previous and present studies conducted throughout the world.

Add another significant *plus* to the already brilliant record of the versatile vitamin.

Yet our seemingly indefatigable researchers were not through with their work.

In recent years it has been suspected, suggested, and (now finally established) by various researchers that Vitamin E is an integral component of the body's defense mechanism which acts against infectious and toxic injuries. It is commonly known as the immunological system. The defense cells organized to fight foreign protein invaders are manufactured chiefly in the spleen, thymus, and liver, and yet often "turn against" the body's own proteins, particularly when they are released by free radicals (page 101). Drug immuno-

suppressors (the chemotherapy drugs currently used in cancer therapy and organ transplantation) destroy the immunological cells thus allowing other diseases to run rampant. Which they do, as proven by the low rate of cancer recovery and other diseases when they are immunosuppressed. (See Appendix F-4.)

Some readers may recall the late Vice President Hubert Humphrey, who, after undergoing chemotherapy, then near death, held up a bottle of chemotherapy drugs and said on a television show: "This is death in a bottle!" It was, and is, in spite of the orthodox groups who continue to use it and will not read the literature or hear the pleas of their benighted patients. Very often, it takes many deaths and many years to compel those medicine men in power at the time to undertake a change in their attitudes.

Shades of George Washington and Robin Hood who haunt us still, having been victims of the medical practice of bloodletting! Although separated by many centuries and thousands of other less famous persons who died by bloodletting from A.D. 500 to 1790, the official medical posture has remained the same—always stubbornly defending the status quo unto absurdity until overwhelmed.

Only the jargon has changed. If only the medical students as well as their professors were forced to take courses in medical history (and nutrition), then perhaps we wouldn't have the same problems we have now.

It did not take clinicians and researchers of the caliber of Ayres and Mihan long to undertake independent research into all phases of Vitamin E's effects.

They were (and are) fearless, truth-seeking researchers. They are of a breed we need most urgently. Meanwhile, good luck, Dr. Ayres and Dr. Mihan. You and other great researchers need it. It doesn't come easy, as you know. Welcome to our ever-growing band of medical pioneers.

Dr. P. L. Harris, Pioneer in E Research

In spite of the medical Old Guard, we are virtually assured of an even more rapid recognition of Vitamin E and its application in the treatment of *all* human beings, including the doctors and others who fought against it.

It should be of interest to readers that Dr. P. L. Harris, who was cited throughout my first book, was for several years, until his retirement, the director of the Department of Nutrition for the Federal Food and Drug Administration (FDA). Dr. Harris is one of the most profound researchers into Vitamin E in all its aspects: chemical, laboratory, and clinical (animal and human). Dr. Harris corroborated the work of various researchers that humans as well as animals were deficient in Vitamin E.

I talked with Dr. Harris many times during our long battle for the acceptance of Vitamin E. On one occasion he had just left Distillation Products (a division of Eastman Kodak), one of the two companies that manufacture Vitamin E in bulk for both human and animal consumption. At that time, the official attitude of the Food and Drug Administration toward Vitamin E was asinine negative. Although Dr. Harris did not hedge on any of his investigations and gave me permission to quote him concerning his favorable work with Vitamin E, I hesitated, having had much experience with bureaucratic medicine and sensing that Dr. Harris had not had the equivalent.

But I think that another significant victory for Vitamin E is evidenced in the fact that when I talked with Dr. Harris again, he gave me carte blanche not only to identify him as being the former director of the Department of Nutrition of the FDA, but to mention his work with Vitamin E. "You can't rewrite history even if you

tried. I'm proud of my work. Conditions are changing, I believe, and Vitamin E will be duly recognized as a most significant factor in human metabolism."

Dr. Harris is now a part-time consultant in biochemistry nutrition and related subjects. He told me that his retirement had no relationship to his work with Vitamin E. He manifests great interest in the hundreds of favorable reports on Vitamin E coming in from all over the world.

"I'll bet you didn't realize your book would be the major spark that lit the Vitamin E bonfire," Dr. Harris commented.

"I hoped it would," I replied. "And I'm glad to see it's coming so quickly. It's not a bad feeling."

No, indeed, not when you know you are at least partly responsible for saving thousands—and soon millions—of human beings in spite of carefully organized attacks from the carefully organized establishment. No, it isn't a bad feeling, Dr. Harris.

There are men and women of great courage such as Dr. Harris, the Shutes, the Cilentos (of Australia), the Kamimuras (of Japan), the Bohlaus (of Germany), the Losowskys and Kellehers (of England), the Butturinis (of Italy), the Anisimovs (of the former Soviet Union), the Loepers and Martins (of France), Drs. Denham Harman, Jeffrey Bland, Lester Packer, and J. R. Smith, Homer Black, Alton Ochsner, and A. L. Tappel, Max Horwitt, Lawrence Machlin, Richard Passwater (of the U.S.A.), as well as scores of other great researchers too numerous to mention here. They will receive their credit elsewhere in this book. However, they and hundreds of other dauntless researchers, are seeking out the final truths about Vitamin E. And I am proud to be an integral part of that work.

Dr. Pfeiffer Corroborates Vitamin E's Effect on the Heart's Electrical Energy

A unique experiment once again proving the efficacy and safety of Vitamin E was carried out by Carl C. Pfeiffer, M.D., Ph.D. (now deceased), with his colleagues, Drs. Singh and Goldstein. These doctors were associated with the New Jersey Neuropsychiatric Institute at Princeton, New Jersey. Their research was sponsored by the State of New Jersey and by individuals interested in the type of original, creative, yet objective scientific research of which Drs. Pfeiffer, Osmond, and other farsighted researchers at this institution have proven themselves capable.*

The report of the present experiment was published in the *Journal of Clinical Pharmacology*.

Dr. Pfeiffer was a brilliant investigator in the physical-electrochemical field as well as the medical-clinical. He was head of the Brain-Bio Center at Princeton, New Jersey.

A previous experiment by Dr. Pfeiffer and his associates had shown that a single dose of lithium produced a rise in the electrical activity of the brain and also the heart. (Lithium is a rare, rather enigmatic metal which is used to "trigger" the hydrogen bomb. It is also prescribed for medical uses as well, including the treatment of depression, particularly the manic-depressive form.)

The researchers then determined to see if other substances would produce similar (or different) results, and if so, which ones would be of value for treatment of human ailments. If they found such a substance, it would be most significant, for any element that alters

*See my book *The Vitamin Pioneers*, Chapter 2, "Niacin: The Cure Doctors Keep Neglecting," for an account of some of their pioneer work.

the total electrical energy of either the brain or the heart—or both—would certainly have vast potential in human application. The electrical energy of the heart, for instance, controls the strength and rhythm of its beat. The brain and the heart are electrically interconnected, of course, with the brain sending messages to the heart and, in turn, receiving messages from the heart, as the great pump's condition varies from second to second, from beat to beat—depending on sextillionth simultaneous functions and activities. To comprehend the complexity of such intricate interactions requires more knowledge than the human brain now possesses, but some humans *are trying*.

In any case, Dr. Pfeiffer and his colleagues selected drugs, vitamins, and trace minerals which they thought might increase or decrease the heart's electrical activity. They selected thirty-four men and three women, all apparently healthy, as volunteers for the experiment. This was a double-blind, controlled experiment, the so-called ideal type that modern scientists are always insisting should be done, but which is rendered somewhat difficult, if not impossible, at times. One plausible reason: human beings aren't rats. In spite of the fact that we know rats to be far more intelligent than ever suspected previously, who ever heard of a rat reacting to a placebo? Humans do. Because of this imagination-becoming-reality, and other infinitely subtle devices the human brain is capable of, the best laid plans of a double-blind controlled experiment frequently end up "astray"—in a quarrel among scientists as to whether the experiment was *really* double-blind controlled or not. And, therefore, valid. Quite naturally the question of ethics and morality also arises. Is it *right* to subject human beings, even *with* their consent, to a highly dangerous—perhaps lethal—experiment? My vote would be yes, provided the volunteers were informed of all possible known reactions, but I am open to other opinions. (For instance, I once volunteered for such an ex-

periment. I do not regret it, although the side effects were somewhat severe. I know that a considerable amount of scientific knowledge was learned which will eventually aid mankind in its quest for triumph over some of its most dreaded diseases.)

Back to the present.

The investigators, through an unorthodox method which they devised, connected an electroencephalograph (EEG) to their subjects' brains and an electrocardiograph (EKG) to their hearts. Then, to obtain the "mean" or "average" heart electrical activity (HEA), they used a "solid state electronic integrator."

Adding "all the values thus obtained"—according to the final report—and dividing by the average pulse rate, provided the researchers with the "mean electrical energy of the PORST complex per heartbeat or the heart electrical activity."

At this point, my readers are probably asking: is this part of the script from *2001*? I must confess, it is. Though not literally a part of the movie or Arthur Clarke's book of that title, the scene *is* one of the future, since our researchers *are* of the future in their thinking and ingenuity.

The complexity of the experiment should overwhelm the mind of that mythical normal healthy American who eats a normal healthy diet and thus obtains all the nutrients he needs to maintain himself as a normal healthy American, whether man, woman, or child—according to the FDA.

The complexity also torments the minds of others, somewhat more devious, who would presume to control the thinking and dietary patterns of the "normal" and "average." It torments *them* because they also cannot realize the complexities, much less the truths derived from whatever source, complex or simple. Yet they can dimly discern the pattern of the future—and they're agin' it, as only bureaucrats following Parkinson's Law can be.

Back to our record of the pioneer experiment, without too much technical data which almost no one understands, but which we had to describe partially so that any reader, science-oriented or not, can know we are dealing with a high magnitude of verifiable facts—*ultra* as they are.

The subjects were given various drugs, vitamins, and trace minerals as well as placebos (for the control group).

What interests us here primarily is that Vitamin E raised the heart's electrical activity (HEA) by 22 percent with a sustained effect which "peaked" at five hours after ingestion of 1,500 units. What is more interesting is that Vitamin E did *not* affect the blood pressure, either systolic or diastolic—in contradistinction to other drugs, vitamins, and minerals which also raised the HEA. Further, Vitamin E actually lowered the pulse rate by nearly ten beats, even while the heart was pulsing stronger.

Now the objective in most heart conditions is to increase the strength of the heartbeat while at the same time not forcing it to beat *faster*, as this creates an added strain upon the already overworked heart. This Vitamin E did, although it had a competitor in the race for better heart remedies. It is called digoxin, a derivative of digitalis, which is the recognized drug of benefit for strengthening the flagging heart, and which, we reported in another book,* was fought by the "cardiac specialists just as bitterly as Vitamin E is being fought by some of the "cardiac specialists" today. The same cardiac specialists (in spirit) in the eighteenth century *knew* with all the knowledge a closed mind can know—which necessarily precludes additional learning—that Dr. Withering was an apostate for declaring that an extract made from the foxglove

*See *Food Facts & Fallacies*, Fredericks/Bailey (New York: Arco Publishing Co.)

plant was beneficial in treating heart disease. Further, he had obtained it from an old "witch-woman." The extract was digitalis, which almost every heart specialist today gives to his trusting patients—at the same time declaring: "There's nothing to this Vitamin E. We'd be the first to know, if it were of any value." Eighteenth century, anyone?

Our indefatigable investigators found that one milligram of digoxin—the same erstwhile proscribed "witchcraft" substance—raised the HEA about 10 points versus alpha tocopherol's 22 points, also did not affect the blood pressure significantly, yet did lower the pulse rate about one-third lower than Vitamin E.

When we consider that all derivatives of digitalis—effective as they were (and are) to strengthen the heart action—are toxic and, therefore, must be used with extreme caution, we have to admit that the nontoxic yet effective Vitamin E is the winner in any cardiovascular contest. Even more so, because according to about twenty excellent reports in recent medical literature from all over the world, Vitamin E will actually protect against "digitalis intoxication." All the doctor has to do is to give more Vitamin E, or less digitalis, in order to alleviate the side effects of digitalis intoxication. Digitalis poisoning is common, as any heart sufferer who has survived will tell you if he knows of what he is the recipient. Among the first to use Vitamin E to detoxify digitalis-intoxicated patients were Drs. Helwing and Hochrein of Germany.

Dr. Pfeiffer told me about his experiment later when I asked him specifically about Vitamin E, since it had been reputed to raise blood pressure initially, then graduate to a normal fall. It does *not*, according to the machines and the competent scientists manning them. Many substances, including certain vitamins, do indeed, raise the heart electrical activity, but those that raise the HEA also raise the blood pressure.

Score another major victory for Vitamin E as being the natural prophylactic and control of cardiovascular disorders.

Dr. Pryor Shows Vitamin E's Unique Role in Anti-Aging

Dr. Hans Selye and his team of researchers at the University of Montreal performed one of the first, and certainly one of the most definitive, studies on aging and the process of reversing it with Vitamin E.

Since that time, researchers all over the globe have duplicated Selye's studies and have amplified and, in some instances, advanced solid workable hypotheses as to why Vitamin E performs its seemingly incredible role in the minutiae of instantaneous cellular reactions. These reactions are, naturally, the reasons why Vitamin E can protect an animal—and a human—from aging.

Perhaps one of the best summations of Vitamin E's part in the body-mind chemistry was presented by Dr. W. A. Pryor, professor of chemistry at Louisiana State University. His article was published in *Chemical and Engineering News*. The following is a partial and necessarily oversimplified synopsis:

A great part of the oxidative process occurs in the mitochondria and organelles (tiny bodies within the cell which aid in the staggering number of metabolic functions that occur every millisecond of a cell's life). The membranes of the mitochondria contain Vitamin E, which prevents harmful substances from combining. *Only* the membrane-bound Vitamin E is effective as an antioxidant in the living body, and in fact, there is no natural replacement for it.

There are many substances which under certain conditions can poison the cell. Some are ozone, carbon dioxide, carbon monoxide, hydrogen peroxide, lead,

and, ironically, oxygen itself. Some of these are derived from polyunsaturated fats, some from the atmosphere (particularly in industrialized areas), radiation, and many other (postulated) origins.

Dr. Pryor points out that the "peroxides" created by these toxic substances probably form what is known as "free radical electrons" which are uncontrolled and, therefore, dangerous to living tissue. However, the natural antioxidant, namely, Vitamin E, captures these "radicals" as they are formed, and converts them to less harmful materials which are then eliminated through natural body processes.

It has been demonstrated that the "old age" pigments seen in the skin, cells, and internal organs (ceroid and lipofuscin) are derived from lipid (oil) and protein fragments which are the result of peroxidation (destruction) of cellular membranes, and the cells themselves.

Vitamin E has been shown to protect animals against the killing effect of ozone, nitrous oxide, lead, sulfur dioxide, carbon monoxide, and other components of smog. Thus, it serves as the best "scavenger" of the dangerous "free radicals" which are implicated in the aging process. Fitting in with its other truly protean tasks, Vitamin E also guards against the damaging effect of oxygen under high pressure, and as we have seen in our astronauts, prevents anemia by protecting the red blood cells against premature destruction. This effect was first demonstrated on rats and mice, then on humans, as we have stated previously.

Pryor thinks that oxidative damage (leading to aging) should not be regarded as occurring only in old age, but as a continuous process (probably beginning with conception) and that either this process occurs faster as the animal becomes older or that aging animals are less able to cope with it because of the accumulations of useless materials (ceroid, lipofuscin) within the cells, thereby rendering them less active.

In any case, Pryor states strongly (after buttressing

his position with proofs from various other researchers) that Vitamin E is one of the most powerful guardians against aging that a living organism can utilize; and that it is probably unique in this respect.

Much credit should also be given to Dr. A. L. Tappel, biochemist, University of California, and his co-researchers; and, of course, the developer of the free radical theory of aging, Denham Harman, M.D., Ph.D., director of medical research at the University of Nebraska Medical Center, whose original concept has been so widely heralded by the scientific community in the last few years.

As we mentioned previously, the free radical theory of aging certainly implicates cardiovascular disorders. The free radical theory seems most logical to advanced scientists, and well may be the key, or at least a paramount key, as to why Vitamin E, in allowing the "loose" (free radical) electrons to unite with the polyunsaturated fat molecules, does indeed delay or halt the atherosclerotic process. This process, if interrupted, precludes the cause of most heart disease and aging.

Dr. Sawyer Advances Electric Theory on Heart Disease

Still another way of stating the electric theory of heart disease and aging was advanced by Dr. Philip N. Sawyer, associate professor of surgery and head of vascular surgical services, State University New York, Kings County Hospital Center, Brooklyn, New York.

Dr. Sawyer spoke before a medical symposium sponsored by the Albert Einstein Medical Center. His theory is that clotting of the blood is caused by electrochemical reactions. It is known that the cells of the blood vessel walls are negatively charged. Some important blood cells are also negatively charged; therefore they

repel each other, thus enabling the blood to flow through its conduits unimpeded. However, Dr. Sawyer says, when a blood vessel is damaged, either through inflammation or "hardening," his studies show that the inner surface of the vessel becomes positively charged, thereby creating an attraction for the negatively charged blood cells (called platelets), which stick to the wall. Then the well-known chemical process of clotting begins. Once the clot is formed (following essentially the same process as when you cut your finger), the power of the blood rushing through the vascular network may cause the clot to break away. In a matter of seconds, it is propelled to the vital organs such as the heart, lungs, or brain, and the victim suffers either a coronary or stroke or lung embolism, depending on the area where the clot settles.

Lady Cilento Confirms Favorable Vitamin E Findings

I received personal confirmation of Vitamin E's efficacy in many diseases from none other than the famed Sir Raphael and Lady Phyllis Cilento, *both doctors of medicine*, among scores of other accomplishments. They now live in Brisbane, Australia.

Sir Raphael Cilento is also a barrister, an authority on tropical medicine, author, historian, and one-time foreign affairs commentator. He was appointed to the United Nations in New York, where his wife attended the New York Health Department's experimental nutrition clinics.

Lady Cilento has an equally interesting background. In addition to being a doctor—a clinician *and* researcher—she is an internationally published columnist and was a university lecturer and welfare worker. The Cilentos met when they were students at Adelaide Uni-

versity, and after receiving medical degrees, worked as a husband-wife team in Malaya, New Guinea, and Queensland.

Lady Cilento wrote: "One of my most humiliating experiences as a young [medical] graduate was being ridiculed in a large London hospital for suggesting that Vitamin D should be given to a child with severe rickets.

" 'You young doctors,' sneered the lecturing professor, 'are too prone to be led away by all this modern nonsense about vitamins. I will only believe in a vitamin when I can buy one over the counter.' " (A most necessary editorial question: How many times is it possible for you to control your rage, contempt, and therefore an urge to violence, when you read the equivalent of that statement today, and know that the same adamant, mind-closed attitude still prevails among "scientific" authoritarians? Answer: Try to sublimate your emotions by supporting those enlightened persons who are working to effect a change.)

Our good Doctor-Lady Cilento continues: "The professor has passed away but vitamins remain. And now you *can* buy them over the counter." (Not, Lady Cilento, in America, if forces behind FDA have their way and transport us back to the age whereof you speak.)

She is aware, as we are, that "at present, vitamins and nutrition generally are not emphasized in the education of medical students." But she knows that "if the medical profession has forgotten vitamins, the public hasn't, but unfortunately the basic good sense of the public in seeking a natural means of maintaining health has been exploited by so-called nutrition experts and extravagant claims have been made by the unqualified."

Our knowledgeable and wise doctor also observes that while caution is necessary in evaluating claims, "there are those medical experts who maintain you can achieve a balanced diet by eating ordinary foods.

Theoretically, this is so, but today it is almost impossible because of long food storage, overprocessing, preservatives, and chemical fertilizers which destroy much of the vitamin content.’’

In spite of their extremely busy, productive lives, the Cilentos have managed to have a happy home life throughout the years. They have six children. One of them, Margaret, is a painter and another, Diane, is a well-known star to motion-picture fans.

Both doctors have used Vitamin E extensively, as the following letter to me will show. Lady Cilento’s letter is self-explanatory, and will add corroborative evidence to the facts already adduced in this book.

Dear Mr. Bailey,

Following on your telephone conversation from New York several days ago, I am sending you copies of several of my articles on Vitamin E, written in a popular vein, for my weekly column in the *Queensland Courier Mail*. As the telephone was very indistinct, I gathered that this is what you wanted for your new book.

I am glad that you are putting out a further edition of this book, as I had purchased it some time ago and consider that it is a most necessary exposé of the active prejudice against Vitamin E. Since writing it, much water has flowed under the bridges and the evidence for its use in many fields, especially coronary heart diseases, is even stronger than when you wrote it in 1964.

Although I have recently retired from general practice, I have had a lot of experience of the tremendous benefits accruing to many people by taking Vitamin E in suitable dosage. My husband has overcome both venous and arterial thromboses in the leg with 800 to 900 milligrams of Vitamin E daily, and his specialist considers the

reestablishment of circulation a "miracle." I myself take 400 to 600 milligrams a day and, at seventy-eight, feel well and full of energy and just as capable of a good day's work as I was twenty years ago.

In this state of Queensland, where skin cancer and keratoses are more prevalent than in any other country in the world, due to the outdoor life of Queenslanders under the ultraviolet rays of a tropical sun, one of the most valuable uses of Vitamin E is in treatment of sunburn and the cure of skin roughness and scaly patches which are often the precursor of skin cancer. As well as internally, I use an ointment or cream with 45 milligrams of d-Alpha-tocopherol per gram, applied night and morning to the affected areas. This smooths out the scales and irregularities on the skin, especially on the face, neck, backs of the hands, and forearms.

Of course, you are conversant with the tremendous value of Vitamin E creams and oils on burns, scalds, and radiation burns. As it is effective against ultraviolet rays of the sun, it must be much the same as on radiation burns; but I cannot find any reference to its use for skin keratoses in any of the literature either in England, the Continent, or the United States. With your vast reading, perhaps you may be able to dig up some reference to this. [I did. Author's note.]

Anyhow, you can use this personal communication of mine if you wish, and I can give you more details if you require them.

Best regards.
Yours sincerely,
Phyllis D. Cilento

Other Researchers Verify Vitamin E's Effects

As we stated elsewhere, many eminent American researcher-doctors have now swung away from arrogant skepticism to soaring optimism regarding Vitamin E's past and present versatile virtues.

There is almost no doubt the immediate future will offer even more wondrous gifts for humankind—judging from experiments already, or nearly, completed.

Still more are being conceived at this moment in countless laboratories and clinics all over the world. The winged horse, after many years of long, constant straining against those who would hold him back, has finally broken his bonds and nothing now can stop his upward flight.

Pegasus still lives, not only in myth, but in the computer-electronic-nuclear age in which we exist.

While we have accumulated evidence for many years on Vitamin E in experiments in the laboratory, in clinics, doctors' offices, and patients' reports, we realize we cannot include all those who need to be recognized.

We have tried to incorporate a few of whom we consider to be paramount in Vitamin E development.

Some of those who cannot be included here due to lack of space, we have listed in the Appendix.

However, we should list a few of those researcher-doctors who during the last several years have entered the so-called Great Vitamin E Controversy. These are the ones for whom we could not allocate a chapter. Yet when you read the list, you will find it will not overstrain your perception to ascertain that what we predicted in 1964 has been, and is, coming true.

1. Ronald R. Watson, Ph.D., Purdue University. Dr. Watson's research corroborated Dr. Harman's findings in the immune response within one week of giving Vitamin E to mice.

2. J. Terrell Hoffeld, D.D.S., Ph.D., researcher, National Institute of Dental Research. Dr. Hoffeld was attempting to corroborate researchers' earlier findings that antioxidants such as Vitamin E block damage to one of the body's protective cells—the lymphocytes. He found not only did the antioxidants stem the damage to the lymphocytes, but the immune response of all the cells increased. Another doctor who corroborated Harman's theory.

3. Dr. Manfred Steiner, professor of medicine, Brown University, Providence, Rhode Island. Following a number of studies with his associates, Dr. Steiner is quoted: "It [Vitamin E] can help prevent heart disease and a large percentage of strokes."

4. Dr. Ronald Mavis, associate professor of biochemistry, radiation biology, and biophysics, University of Rochester School of Medicine. He said he "was definitely excited about the possibilities of Vitamin E in a protective role."

5. Dr. Mary Anne Guggenheim, University of Colorado School of Medicine. She said that Vitamin E therapy has been found to have definite benefit for children with certain neurological problems. "We now have ample evidence that newborns must be assured an adequate intake of Vitamin E for proper physiological development."

6. We rarely mention anyone associated with a commercial company since we could be accused of commercialism (not true, of course) and accused of favoring one company over another. Our readers know we do not promote one company over another—nor have we received one penny from any such companies.

With that disclaimer, I shall insert Lawrence J. Machlin, director of the Department of Biochemical Nutrition, Roche Research Center. Dr. Machlin's chief contribution is that he strongly corroborated earlier findings that prostaglandins (PG) and platelets are greatly influenced by Vitamin E. All for the body's

good, since Vitamin E prevents the platelets from clumping together too much and too fast at the site of an injury, and thereby forming a fatal clot.

Dr. Machlin is distinguished for directing the efforts of the team which helped establish that antioxidant vitamins such as Vitamin E, Vitamin C and beta carotene lower the risk of diseases such as cancer, cardiovascular disease and cataracts. He has had a particular impact in regard to Vitamin E research where he is an internationally recognized authority. His efforts helped make Vitamin E research an important and respectable scientific endeavor. (See Appendix F-5 for abstracts.)

CHAPTER 8

Cholesterol Theory Examined

Cholesterol—that bugaboo that held the stage for more than forty years—is now largely out of fashion with the foremost heart researchers as the primary cause of atherosclerosis. The controversy still lingers on, perpetuated by those die-hard promoters of the hypothesis about two or three decades ago. Many studies throughout the world have shown that the ingestion of cholesterol is not primarily responsible for the rising tide of heart-disease deaths (the deaths still mount throughout the “advanced” countries although millions have switched to the polyunsaturated fats in the last twenty years). As many enlightened investigators have discovered, cholesterol, low level and high level, that is, LDL (low-density lipoproteins) and HDL (high-density lipoproteins and the other fatty substances, chiefly triglycerides, are certainly involved. So is calcium. However, they are probably only one of the steps in the unbelievably intricate electromagnetic-chemophysical-socioeconomic-psychosomatic processes which eventually culminate in “coronary disasters” throughout the developed world.

Of the many studies which show that the cholesterol level in the blood cannot be scientifically equated with heart disease (unless abnormally high), perhaps the most typical, as well as definitive, was made by Dr. Harold A. Kahn of the National Heart and Lung Institute at Bethesda, Maryland. His report was published in the *American Journal of Clinical Nutrition* as early as July 1970.

Kahn studied the dietary pattern of the United States

from 1909—when coronaries were practically unknown—to 1965 when more than half the deaths in the country were (and are still) due to cardiovascular diseases. The consumption of saturated fats (cholesterol) was almost the same in 1909 as it was in 1965. Yet, as we know, for many years during that period, coronary heart disease zoomed to fantastic proportions.

For over three decades we have been told that reducing the saturated fats and replacing them with polyunsaturated fats (vegetable oils, etc.), would lower the level of cholesterol in the blood. Well, sometimes they *do* lower cholesterol levels. But still the deaths continue just as though the heart had not been told that lowering blood cholesterol levels would make it much less susceptible to attacks.

Dr. Kahn concludes that “the increased risk of coronary heart disease . . . is not related to dietary fat changes to a very important degree,” thereby agreeing with the report from the National Academy of Science.

We pointed out this fact in our original book, and also suggested the reason for the continuing rise in cardiovascular problems: the corresponding decline in Vitamin E intake during much of that period. Someday, perhaps soon, the cardiologists may just chance to discover that the brothers Shute were right—after the Medical and Governmental Establishments have spent a few more billions and allowed many more millions to die because of their refusal to face facts about Vitamin E.

The Food and Nutrition Board of the National Academy of Sciences in late 1980 tossed a firestorm bomb into the already sinking ship named *Cholesterol*.

In this chapter, and in my other books, I have repeatedly reported facts from all over our relatively small globe that all was not well with the cholesterol ship launched by those who knew power and public relations quite well, but ignored scientific facts.

The Food and Nutrition Board, chaired by Robert E.

Olson, M.D., of the St. Louis University School of Medicine, concluded after a two-year study conducted via six large-scale tests (in New York City, London, Oslo, Helsinki, New Jersey, and Los Angeles) that “the dietary-cholesterol theory lacks scientific foundation.”

The board stated further: “Intervention trials in which diet modification was employed to alter the incidents of coronary artery *disease and mortality* have been generally negative. . . . In these studies, comprising about 20,000 man-years of observation, in which decreases of serum cholesterol concentrations occurred, there was marginal decrease in coronary incidents—but *no effect on overall mortality.*” (Emphasis mine.)

Dr. Olson added that there had been “a food fad created around the bogeyman of dietary cholesterol. Good food [such as eggs, milk] should not be regarded as a poison, a medicine, or a talisman. It should be eaten and enjoyed.”

Can you imagine the howls of rage and indignation among the vested interests who were dedicated to keeping alive the theory that cholesterol was *the* major factor in heart disease?

The howls and uproar descended in a vast undifferentiated cloud over the prestigious National Academy of Sciences. It was the first time the anticholesterol forces had been hoisted on their own petards; that is, *they* were called “food faddists”—a term they had been using against their opponents—regardless of credentials or lack of them—for the last four decades.*

Millions of dollars have been spent in almost countless experiments to prove the dietary cholesterol theory, which now is largely discounted.

Just about every hypothesis and theory has been tested, with inconclusive results.

The famed Framingham Study, begun in 1950 in Framingham, Massachusetts (a Boston suburb), on

*See Appendix G for additional observations on cholesterol theory.

5,127 men and women aged thirty to sixty-two and free of coronary heart disease, continued—greatly enlarged for a long period. The subjects were checked biyearly for heart disease and atherosclerotic conditions.

The results of the study were—you guessed it—inconclusive. No direct link between cholesterol in the bloodstream and the incidence of cardiovascular disease was discovered. One doctor, supervisor of this first study, W. B. Kannel, writing in *Nutrition Today*, concludes that neither diet *nor any other single factor* can account for our current epidemic of atherosclerotic disease.

In a rather gloomy conclusion, the author says that firm answers to the “puzzles” of atherosclerosis may take decades and cost more than the nation can spend.

Dr. Kannel may not have been acquainted with former President Nixon’s 1972 proposal to “conquer” heart disease and cancer by the expenditure of billions of dollars—expenditures which were passed by Congress in 1972 and are still being spent with the grants going to the “best known” authorities, as usual.

The reasoning goes thusly: All you have to do is spend enough money and the answer to both cancer and heart disease will certainly be found. That’s the way to get to the moon, isn’t it? This type of thinking has not yet worked frequently in medicine, yet there is no question that research money *is* needed—in the right places. But why pour the money down the same old worn-out drains that have not yet produced cures? Or give it to those who epitomize Parkinson’s Law—that 1,000 mediocre minds should equal that of an original thinker, and therefore, it is better to employ 1,000 instead of listening to just one genius? Why use *one* when 1,000 will do just as well? Theoretically, of course.

It’s not quite like propelling men to the moon by spending billions. In that case, we had the men who were proven and the technology which was also

proven. Parkinson's Law was temporarily suspended in this project.

Would that NASA's projects (many—not all!) could be the rule rather than the exception. Then, instead of dealing with an impossibly inept bureaucratic-politico system which controls all of us except the ultrarich, and in many instances, the poor and the elderly, we might escape from the iron collar and conserve what objectives our founding fathers conceived.

The thought apparently never occurred to the ones in charge of shoveling out the money, or those in charge of spending it, that for a mere fraction of the cost involved in massive assembly-line research techniques now being used, a comprehensive review of the vast Vitamin E literature could be undertaken. And that following such a review, some of which we have noted in this present work, carefully organized, objective follow-up studies could be made. Certainly Vitamin E deserves this much attention, even if a researcher were just to glance through the mountains of favorable evidence. But no, we'll follow the "experts," who admit they are almost in a cul-de-sac and keep repeating almost the same experiments even though they continually lead into the land of *Erewhon*—which is Samuel Butler's way of describing Nowhere backward.

Dr. Hermann's Exciting Findings: Cholesterol/Vitamin E

Writing in the *American Journal of Clinical Pathology* (November 1979) William J. Hermann, M.D., a pathologist at Memorial General Hospital, Houston, Texas, describes how he stumbled onto the fact that Vitamin E could elevate the HDL (the "good" form of cholesterol which is considered a low-risk factor in precipitating atherosclerosis).

How did Dr. Hermann make his serendipitous discovery? By experimenting on himself *first*. He was just instituting a new series of tests relative to the amount of cholesterol in the blood of the hospital's patients. Nothing spectacular, because the same tests have been (and are being) done countless times.

However, being aware of the HDL-LDL differential in cholesterol distribution instead of the *total* cholesterol, Dr. Hermann's findings moved him to test his *own* cholesterol levels. He found that his HDL was far below that which is considered normal. He recalled reading that Vitamin E could alter the cholesterol balance between HDL and LDL.

His father, who was taking Vitamin E, persuaded his doctor-son to take 600 units of Vitamin E daily. Dr. Hermann discovered that 600 units of Vitamin E brought his own HDL up to normal—40 percent. It had been 9 percent. These results were phenomenal. Then our good doctor, who had obviously read some Vitamin E literature, decided to conduct a controlled study.

Our dedicated researcher (and, reader of the literature, you may thank your God or gods for such rare researchers) tested his hypothesis on ten volunteer patients. It was a controlled study. Five volunteers received 600 units of Vitamin E; the others received a placebo. Our readers know a placebo consists of a harmless substance, yet inert, and its only power lies in the power of suggestion. Yet this power is very real indeed. The thought of *being helped* engendered in the human mind is so impelling that it *does* help many patients—for a limited period of time.

He started with ten volunteer patients, five of whom had high LDL—the high-risk factor. The others had normal HDL and LDL.

Now, here is the exciting news. It was so exciting to Dr. Hermann that he published it in the *Journal of Clinical Pathology*.

Of the five "high-risk" volunteers, all had raised

their level of HDL to a very significant degree. In fact, four out of five of the "average low-risk" volunteers also raised their levels of HDL most significantly. None of the patients', nor Dr. Hermann's, *total* cholesterol was affected appreciably even when taking 600 to 800 units of Vitamin E daily. This corroborates many modern researchers' findings that it is not the *amount* of cholesterol which lays down the deadly deposit along the arterial walls. It is the type (LDL) of cholesterol. Cholesterol is accompanied by the triglycerides (fatty substances) and calcium which we have reported elsewhere in this book. Many other researchers have corroborated Dr. Hermann's pioneering, original findings. (See Appendix F-2, Horvitt abstracts.)

CHAPTER 9

Vitamin E and Arteriosclerosis

Perhaps the most convincing objective evidence that Vitamin E can actually reduce the deadly arteriosclerosis of the vital heart blood vessels first came from Japan.

In a paper presented before the Fourth International Symposium on Vitamin E, Dr. S. Yoshimura, along with ten other heart specialists and researchers, wrote about their two-year experiment with 535 patients who were suffering from arteriosclerosis. Some had had myocardial infarction (coronaries); some had suffered cerebral apoplexy (strokes); others had hypertension (high blood pressure) which was caused by narrowing or hardening of the arteries; still others had arteriosclerosis produced by diabetes. Altogether quite a large and typical group of persons suffering from what has been called "the disease of living"—especially in a modern society.

The researchers were of that rare species found today, or any day: brilliant, intuitive, ingenious, competent, unprejudiced, thorough, yet fired with a zeal to find the truth. An almost unheard-of combination, yet one which could exist more often here if circumstances were changed in American medicine.

As we shall see, that team of Japanese researchers was to make what I believe to be one of the most explosive breakthroughs in Vitamin E's long, bitter history.

First, they developed a bloodless method of measuring the pulse-wave volume (PWV) of the aorta artery. The pulse-wave volume of blood leaving the heart

through the mighty aorta—the largest single lifeline of the body—indicates the degree of blockage caused by narrowing or hardening of the blood vessels (arteriosclerosis-atherosclerosis). (The narrowing or hardening of the vessels—caused by deposits of fatty substances or calcium on their walls—forces the heart to pump harder according to the degree of the narrowing or hardening, which, in either case, means loss of elasticity.)

Therefore, by measuring the blood volume and the force or pressure that was necessary to send it coursing through the artery wall, investigators can determine the severity of the arteriosclerosis.

Next, our intrepid researchers examined the aorta wall linings of sixty-one persons who had died as a result of arteriosclerosis. These persons were those whose pulse-wave volume had been measured during their lifetimes, so that an accurate correlation could be established between the PWV, the degree of the disease, and the patients' medical history up to the time of their deaths. Almost needless to say of our methodical experimenters, they analyzed the sixty-one aortas for all the factors known to be, or thought to be, useful in the genesis and development of arteriosclerosis.

Assured now of complete scientific verification of their work on patients who had died, they then proceeded with their experiment on the 535 living patients. They gave Vitamin E to some; to others they did not, thereby establishing a "control" group.

By means of their bloodless test, they took PWV's on all patients. After one year, the patients given Vitamin E had *decreased* their PWV values by 0.20 m/sec. and after the second year, still further by 0.26 m/sec., a significant improvement, indicating that the arteriosclerosis was gradually being ameliorated. At the same time, the control group (those receiving no Vitamin E), *increased* their PWV values by 0.35 and 0.45 m/sec.,

thus showing that their arteriosclerosis was getting worse, as would be expected.

Also it was found that blood cholesterol levels had no influence on the Vitamin E patients' decreased PWV readings—still another line written in the epitaph of the cholesterol theory.

While this test by Dr. Yoshimura and his associates is probably the most controlled and scientifically conducted to date, there are many such corroborating experiments throughout the world, particularly in the former U.S.S.R., Germany, and Czechoslovakia.

If we were to allocate to Vitamin E research just a tiny fraction of the billions which are spent in this country on projects that are in essence photocopies of those preceding them, we would witness what every cardiologist says he is looking for: a truly revolutionary method of dramatically controlling heart disease. [Aside: Insiders know the surgeons had rather demonstrate their cutting skills than relinquish their vaunted power to a mere drug—except for those chemotherapeutic drugs which to date have saved few with cancer metastases.

There is one asset to chemotherapy, however. Should you wish to become bald (without shaving your head) within three to six months; have all your metabolic processes change for the worse, including potency, destruction of your red and white blood cells (the immunologic system which protects you against disease), then you have a perfect solution. Take chemotherapy. It will certainly add to your very high cost of dying, plus the agony the drugs themselves produce. But you can tell your friends in the next world: "I did it *their* way!"

Better still, preceding the research methods which are largely nonproductive, medicos could prescribe Vitamin E in adequate dosage—and then we wouldn't need such vast research except to ascertain *fully* why Vitamin E works.

There are many other researchers who have demon-

strated the fact that arteriosclerosis and hypertension can be prevented, and either controlled or cured by vitamins (chiefly Vitamin E) and other nutrients. Among them are Dr. Yousuf I. Misirlioglu, whose many credentials include being a Fellow of the Royal College of Surgeons (Canadian) and a Fellow of the International College of Surgeons. His investigations were conducted while he was senior physician at Connecticut Valley State Hospital (affiliated with Yale University) and as director of the Laboratory of Surgical Research, Unity Hospital, Brooklyn, New York.

In a series of lectures at the Eighth Annual Meeting of the International College of Angiology in Madrid, the St. Mary's Hospital Medical School, London, and at the Royal Graduate Medical School, University of London (who can quarrel with the quality of these forums?), Dr. Misirlioglu presented protocols (medical histories) of 225 patients, all of whom had provable advanced arteriosclerosis accompanied by essential hypertension. (X-rays showed heavy calcification of various arteries: abdominal, aortic, femoral, and others.)

By coordinating information which was already published in the scientific literature, the researcher had advanced a hypothesis that fibrinogenolysis (the first and main step of fibrinolysis, which means dissolution of clots) was the key to arteriosclerosis, and that he could develop a safe, effective method to aid it. (He lists forty-four other well-known researchers in his final paper, all of whom, when put together, support his hypothesis.)

Dr. Misirlioglu used a combination of several natural agents in his therapy, all of which had been reported fibrinolytic. (1) 400 mg of natural E daily before breakfast; (2) 400 mg of Vitamin B-6 daily before lunch; (3) 100 mg of niacin (Vitamin B-3) three times daily after meals, increasing to 300 mg three times daily if tolerated. Exercise and proper diet were encouraged.

The diet stressed avoidance of known drugs and foods which are inhibitors of fibrinogenolysis.

Dr. Misirlioglu noted within two to three weeks an improvement in blood fibrinogen, cholesterol, and a loss of excess water and a drop of blood pressure toward normal. One of his patients had the dangerously high pressure of 280/140, yet all had very high blood pressure. The improvement was subjective as well as objective. Comprehension and memory improved, headaches disappeared, they felt better generally, angina gradually subsided until it was gone after three months; of course, advanced cases took longer to respond, but at the end of six months, all had reacted favorably. When the blood fibrinogen levels dropped satisfactorily (200 mg per 100 ml) and were kept there for two to four weeks, our researcher reports the astounding fact that a complete decalcification of the arteries occurred, and the blood pressure returned to normal—or within satisfactory limits!

Further, the investigator has X-ray pictures of the patients—before and after—to prove his case. And all the protocols which also demonstrate beyond any question the authenticity of his experiment.

Again, we have to repeat the same old dreary refrain. Wouldn't you think that a medical scientist of his impeccable background, coupled with the amount of evidence he presented, would be eagerly corroborated or at least challenged by those who claim to be searching for the answer to heart disease, high blood pressure, arteriosclerosis? His experiments and subsequent lectures occurred over a three-year period, 1963–66. To date, his studies, research as well as clinical, have not attracted the slightest interest in those who are in charge of official medicine. That is, I have not seen any later reports in the scientific literature or the public media, and I read fairly extensively, as the reader may know by now.

Dr. Misirlioglu, with his colleague, Dr. B. G. Shafi-

roff, was published in *Circulation*, Vol. 29, III, 1964, and published again in the *Summary*, December 1967, documented with X-ray pictures of arteriosclerotic lesions being healed, charts on blood pressures, etc.—in short, every proof any scientist would need for verification of the experiment's validity. The *Summary* (now defunct—a victim of lack of funding for Vitamin E research) was one of the few journals in which Dr. Misirlioglu could be published, at least in this country and Canada. Dr. Misirlioglu's effectiveness in reversing arteriosclerosis and essential hypertension by large doses of mere vitamins (E, B-3, B-6) and diet is too hot to be investigated, much less accepted. Need we ask why?

Vitamin B-6 Relates to Arteriosclerosis

While this book is primarily devoted to Vitamin E and its multifarious effects on the body, we, of course, welcome any new hypothesis/theory involving any other vitamin or additional substances which seem to be beneficial in cardiovascular conditions.

Therefore, we present another hypothesis of combating arteriosclerosis as first advanced in 1969 by Dr. Kilmer McCully, professor of pathology at Harvard Medical School. McCully's thesis, necessarily oversimplified, is that an amino acid, methionine, found in protein, is intimately involved. Under ideal conditions—which are very seldom attained—methionine is converted into a useless nontoxic substance termed cystathionine. But it is only converted by adequate amounts of B-6. Much of the B-6 in foods is destroyed by processing, freezing, cooking, and canning. Unless methionine—after rendering its normal role as an

amino acid—is changed into cystathionine, it becomes homocysteine, a deadly substance to blood vessels.

Drs. Edward R. Gruberg and Stephen A. Raymond in their book *Beyond Cholesterol: Vitamin B-6, Arteriosclerosis, and Your Heart* (New York: St. Martin's Press) augment Dr. McCully's premise that the key to arteriosclerosis is Vitamin B-6. They cite studies in England and Australia which show that victims of heart disease have much more homocysteine than normal (a failure of conversion by B-6); that women on the Pill often are depleted in Vitamin B-6 (and Vitamin E, as we have seen) and have a death rate from cardiovascular disease about ten times higher than those not on the Pill. The doctors emphasize that Vitamin B-6 is needed in the changing of methionine into a practically inert substance flushed out by the kidneys. If Vitamin B-6 is deficient, which it frequently is in the amounts needed for the conversion, it cannot participate in the beginning or the end process. No way can women on the Pill obtain enough B-6 from the foods they eat (likewise Vitamin E). The doctors recommend taking a supplementary pill of B-6.

The authors present an excellent case for their hypothesis since they show that homocysteine is at least partially responsible for the laying down of the deadly plaques on the arterial-venous wall. They assert that B-6 is the key factor in the whole complex process leading to cardiovascular diseases. Would they were right; yet as we have seen, it is hard to believe any *one* method holds the *only* answer.

Gruberg and Raymond agree that dietary cholesterol has little to do with atherosclerosis (a fact we pointed out in our first book on the subject, and also here). Yet the doctors ascribe Vitamin B-6—or rather the lack of it—as the key factor in the prevention of cardiovascular troubles.

We know that B-6 is an integral part of the body's whole well-being; yet the researchers have not found

that B-6 *alone* is a curative for atherosclerosis (arteriosclerosis is a more generalized term).

We hope other researchers will follow up on these claims. Someday, somehow, others will “rediscover” these additional benefits of Vitamin B-6—just as they did with Vitamin C, Vitamin E, and more in the alphabet of now-accepted essential vitamins and minerals.

We wonder in almost incredulous amazement why some researchers do not read other researchers’ findings in the *same* scientific field. Nor can we come to any conclusion other than that these researchers are afflicted with tunnel vision and therefore do not possess peripheral vision, which would enable them to perceive others’ efforts in the same scientific endeavor, or even—wonder beyond wonder—understand and incorporate other scientific disciplines within their own hypothetical framework.

Instead, most discoveries are “rediscoveries” and their rediscoverers proudly announce through medical journals and the media, their “breakthroughs.”

CHAPTER 10

Vitamin E Benefits Still More Ailments

The Fourth and Fifth International Symposia on Vitamin E were held in Japan. Both symposia corroborated the findings on Vitamin E which we reported in our first book, but each progressed much further, as we will show. The researchers have now expanded about twentyfold and include the most famous in every advanced country. The distinguished scientists meeting at beautiful Lake Yamanaka exchanged research findings on Vitamin E. They came from the U.S., the Netherlands, Switzerland, Thailand, Germany, Canada, and, of course, Japan.

A mountain of material was presented, all corroborating the basic value of Vitamin E, in both deficient and normal cells.

The research and clinical evidence presented dramatic proof of Vitamin E's efficacy in such conditions as anemias of many types, aging, malfunctioning of the liver, kidneys, and other organs, arteriosclerosis (and associated heart disease), diabetes, and several other major afflictions affecting man as well as those we choose to designate as the lower animals.

In addition, it was again confirmed that Vitamin E greatly increases the "shelf life" of stored blood as well as the life span of red blood cells in the body; maintains the integrity of the capillaries, both in the skin and everywhere else in the body. And in a definitive study by two German doctors, V. Behlau and E. Behlau, 60,000 efficiency tests (conducted on healthy, sick, and convalescent persons) showed that heavy

doses of Vitamin E (900 milligrams a day) significantly improves the general efficiency of the body. The patients were checked before and after the experiment by almost every recognized test known to medical science.

Vitamin E and Skin

There has been much corroborating evidence that Vitamin E used as an ointment (and preferably orally, too) will hasten the healing time of wounds and burns. It acts primarily as a *stimulant* of normal circulation.

Probably the first, and certainly the most thorough, experiment as to the way Vitamin E is absorbed through the skin was performed by Dr. Mitsuo Kamimura and Dr. Tohru Matsuzawa of the Department of Dermatology, Sapporo Medical College, Sapporo, Japan.

Dr. Kamimura, along with his associates, has conducted more than forty experiments with Vitamin E as an ointment.

The classic one we are describing here was published in the *Journal of Vitaminology* in 1968. There were others published later, all corroborative. First, the experimenters obtained seven hospitalized volunteer patients. All had their hair cut short so that the researchers could conduct their experiments efficiently.

The Vitamin E was radioactivated, then sprayed on the subjects' heads. To avoid possible contamination by the radioactive substances, the Vitamin E solution was inserted through a tiny hole in the skin five-tenths of a centimeter in diameter. The solution was then dried (antiseptically) and covered by Saran Wrap—which apparently has more uses than just as a food protector.

The wound was then sterilized at four- six- and twenty-four, hour intervals. The reason for this is that the dermatologists were operating during these periods, excising “necessary quantities” of all layers of skin. This proce-

ture, though it may seem somewhat drastic, was absolutely essential to prove on the human being that Vitamin E penetrated the skin, at what depth, and how long it remained. X-rays showed that the skin is highly permeable to Vitamin E; it enters quite easily, makes contact with various circulatory conduits, and concentrates around their hair roots and follicles. Vitamin E remains evident for at least twenty-four hours.

We congratulate Drs. Kamimura, Matsuzawa, and their brave volunteers for establishing such a basic, definitive study which naturally indicates why Vitamin E ointment is not only effective against wounds, burns, and indolent ulcers (bedsores) but also against many other skin conditions.

Wounds and Burns

Although, incredibly, Vitamin E is still not recognized by most authoritarian medical groups in America, including the burn centers, we refer you to an impeccably controlled study, again by the team of Japanese doctors headed by Dr. Kamimura.

It is an impressive scientific experiment. It was later confirmed to me by Dr. Kamimura in private correspondence (as if his letter were needed, Dr. Kamimura being a foremost scientist of world renown and having been published in scores of reputable medical journals).

Dr. Kamimura and associates, after their classic Vitamin E skin experiment, previously mentioned, went on to demonstrate Vitamin E's efficacy in treating burns and wounds of many types. Hundreds of burn victims were treated for several years with amazingly favorable results.

These results were fully corroborated by Dr. Shute, who, of course, had pioneered wound/burn treatment. Dr. Shute took "before-and-after" color films proving rapid healing of burns and wounds treated with Vitamin E. It

should be noted that Dr. Shute tried to show the undeniable evidence at AMA convention exhibits. His efforts were rejected as usual by the top AMA brass who chose not to see the truth—and certainly not to let it be known.

It merely confirms that the Medical Establishment of this country, being so immersed in specialized techniques such as reattaching a severed arm or leg, cannot or does not read the literature of other scientific disciplines.

I have seen the aforementioned pictures, together with the protocols, and they are all still extant—having been published in the *Summary*, December 1974–75. I know they would be available from the Shute Institute by any qualified group wishing to learn the truth about wounds and burns which was—and still is—suppressed!

There are many other similar reports in the medical literature, some of which are controlled and reinforce the basic premise that Vitamin E will and does affect wounds and burns to a remarkable degree.

The Japanese studies are unassailable from any scientific viewpoint and Dr. Kamimura's experiments prove the point conclusively.

Personal Observations on Wounds and Burns

I can also speak personally about how Vitamin E acts against burns.

Some years ago I was placing a log on our fire when a large glowing ember flew out of the fireplace and lodged between my fourth and fifth finger.

During the few seconds it took to dislodge the ember, it had burned through the skin down to the bones of both fingers—I could see the bones of both fingers clearly. After my first shock of disbelief, followed by intense agony, I cleansed the burn/wound.

Ordinarily, of course, it would have been a hospital-

doctor case, being a third-degree burn, but I decided upon another method, having read about and witnessed its efficacy in almost similar situations. After cleansing the burn, I broke open an ampule of Vitamin E (400 units) and let the contents run directly into the wound. Next I applied a loose gauze bandage around the two affected fingers, knowing that the loose gauze would continue to let Vitamin E drip into the injured area. It did.

Some would call the process miraculous, others would call it the placebo effect; nevertheless, the wound/burn was healed within three days. Today I defy any surgeon to tell where the scar is, because there is none. Placebo? Ask any surgeon who has suffered a similar experience where he would go. The hospital, of course.

A summertime neighbor of mine, Fred Swierlein, retired as Assistant Fire Chief of Bridgeport, Connecticut, also has proof of Vitamin E's efficacy in burn and wound healing.

Fred and his wife had read my book on Vitamin E and were taking several hundred units of Vitamin E in addition to other suggested supplements. They admitted "feeling better," but the real proof was to come a few months later.

Fred, who was also a sergeant, is, moreover, a jack of all trades and good at most; he also likes to cook.

But this evening the grease splashed from the pan. Several large globs hit him in the face. Fortunately, the grease did not strike his eyes, which would have blinded him. Instead, the flow of steam and food struck our ex-sergeant in the forehead. Anyone who has been scalded severely knows the almost instantaneous reaction—as fast as the nerves can transmit to the brain and back again. Fred felt that shock, but instead of running for the butter or another home remedy, he remembered Vitamin E. He broke an ampule and rubbed its contents on his burning, aching forehead.

As he told me several days later "I couldn't believe

what happened. The pain disappeared. Two or three days later all symptoms of burns had gone.

“If I never believed Vitamin E worked before, I no longer have any doubts.”

He has no trace of a scar. His neighbors corroborate his story with some degree of awe. Anecdotal? Of course, but so many similar anecdotes exist, they comprise overwhelming evidence. In fact, most of the treatments now used in modern medicine are derived from folklore “anecdotal” medicine. Source: *Journal of the AMA*.

Breast Cysts

Millions of women are afflicted with breast cysts. Although most are benign, almost all are painful and often lead to cancer.

A study of twenty-six woman conducted by Dr. Robert S. London, director of reproductive endocrinology at Mount Sinai Hospital in Baltimore, was reported in the September 5, 1981, issue of the *Journal of the American Medical Association*.^{*} The report stated the complete relief of pain as a result of Vitamin E therapy in 38 percent of the women, and the researchers noted that in ten women, “the lumps went away and the patients noticed tremendous clinical improvement.”

Vitamin E had “profound effects” in normalizing the high level of female hormones which are usually found in most women who have the lumps (cysts).

Other studies are being conducted and, judging from the preliminary results, are reaching the same conclusion as did Dr. London and his researcher team.

At any rate, we already have a merry dish to set before the kings or those who told us that Vitamin E was “in search of a disease.” First admitted in the

^{*}See Appendix H.

orthodoxy was premature babies; second was "intermittent claudication"; third was "Buerger's disease" (related to peripheral circulatory malfunction); fourth was mammary "lumps"; and now a strong inference is emerging that Vitamin E is invaluable in "open heart" surgery and in the follow-up of bypass surgery instead of the currently popular aspirin therapy. All of the preceding information has been published in reputable medical journals—the *JAMA*, the *New England Journal of Medicine*, the *Southern Medical Journal*, as well as many others.

Would that Evan Shute were alive to read these reports. Perhaps he is watching from another dimension and smiling in joy. We hope and think so.

Vitamin E and Diabetes

As we pointed out in our previous book, diabetes is not a simple disease, and it was not "conquered" by the discoverers of insulin, Banting and Best—important and exciting as their discovery was. The great E. P. Joslin, a profound student of diabetes, said that insulin merely prolonged the life of the diabetic so that he could die of other diseases, primarily cardiovascular. However, insulin is of utmost importance in treating most diabetics, and has prolonged the lives of millions.

Now, as was first ascertained by our medically ubiquitous brothers Shute, the secondary effects of diabetes such as gangrene, blindness, ulcers, intermittent claudication, and heart disease were found to be greatly helped by Vitamin E. Since then, other researchers (approximately fifty now throughout the developed world) have corroborated the Shutes' findings.

Since the condition of the eyes known as retinopathy (breakage of the small blood vessels in the eye) frequently indicates the first signs of disaster often associated with diabetes, we cite a study that shows Vitamin

E can improve this condition—which heretofore was unimprovable, and as a result, more than half of the afflicted go blind within five years.

Also, there are many other studies which relate to the same basic reason why Vitamin E controls or even reverses other conditions which afflict the diabetic: gangrene of the feet and legs; cardiovascular diseases, kidney and liver diseases—in short, just about every ailment known to man which is caused by failure of the body to get enough nutrient/Vitamin E-laden blood to the capillaries and therefore the cells. The cells, of course, are the final arbiters of our life and death.

In one of these first studies, Drs. Kojima, Okajima, and Suzuki (need we say, of Japan?) gave 90 to 180 mg of Vitamin E daily to thirty-eight patients with diabetic retinopathy. With continued dosages, up to nineteen to twenty-four months of treatment, a significant improvement was noted in 67 percent of *all* patients—including those in the middle stages. More plainly: a reversal of the steadily deteriorating process. I know certain researchers who would have said the dosage was too small, and that is why it took such a long time for these retinopathies to respond. Yet respond they did, 67 percent more improvement than any other agent known today can do. As stated previously, there are many more such studies concerning diabetes and Vitamin E, not just retinopathy, but we cannot recount them all here. It should also be remembered that our original researchers findings are not only factual but are applicable today. They explain to the reader some facts about diabetes unknown even to many diabetes “experts.”

Vitamin E and Premature Infants

The AMA, overwhelmed by a tidal wave of evidence from virtually every country in the world that prema-

ture babies (and full-term, too) suffer anemia due to lack of Vitamin E, finally agreed that babies need extra Vitamin E.

This fact has been documented hundreds of times throughout the world during the last twenty years. Also, you have just read that the Japanese researchers proved beyond a doubt that adults who were going blind because of their diabetic condition were able to see again due to Vitamin E therapy.

Now comes the wonderful news *twenty years later*, that premature babies are also prevented from blindness by high dosages of Vitamin E. No less authorities than from the University of Texas and Baylor College of Medicine found that Vitamin E does indeed prevent blindness in "preemies" when given in high dosages in the first twenty-four hours after birth and daily for eight weeks thereafter.

Score again for Vitamin E. It only took twenty years for American medicine to catch up with Japanese research. Just about the average time. You will note no credit was given to the Japanese, of course.

Vitamin E and Sterility

There have been many studies on male and female sterility using Vitamin E. These studies, when conducted properly and comprehensively, certainly provide us with a definite conclusion, to wit: Vitamin E in adequate dosage for both male and female for the three or four months prior to conception has been shown to be extremely potent in the creation of that fluid-absorbing, nutrient-taking parasite which metamorphoses into what we accept as a human being.

However, of all the experiments which I have examined, there is none so thorough, so scientific, and, therefore, so convincing, as that of Dr. Bayer of Germany,

which we reported on in our first book. Dr. Bayer's experiment will remain the model for all time for any such experiments on sterility in the human. He observed all the elements of the scientific method, and finally achieved 100 percent favorable results. The other experiments, as reported, do corroborate Dr. Bayer's findings. Dr. Bayer's studies were conducted on 100 married couples.

Although I personally have seen several examples of fertility being stimulated by Vitamin E (in addition to the literature proof), the following one interests me most.

Robert and Pat Ruland were friends and neighbors of ours for many years. They had one son, Brad. Shortly after Brad was born, Pat developed cancers in two places, one on the eyelid and the other in her fallopian tubes. Both were perceived and removed before the usual systemic spread. (The doctor's name is on file, so he can verify this account.)

The doctor cut the tubes as high as possible so as to eliminate any chance of possible cancer metastases. He told Pat: "I'm sure we've got the cancer, but I'm also sure you'll never have another baby. There is only one case on record where the tubes were cut this high and birth occurred. However, we'll wait for a year and then if nothing happens, your chances are zero, I'm sorry to say."

A year passed and nothing happened. In fact, seven years passed. No hope, although they both desperately wanted another child.

Enter Bailey mounted on his White Horse Charger, named Vitamin E, bearing a copy of his book, also appropriately titled *Vitamin E*. Advice: husband takes 400 units, wife takes 800.

The advice was followed. A year later, a healthy daughter was born, much to the bewilderment of their doctor, who said (I am paraphrasing): "I can't believe it! Who ever heard of such a thing?" How many times, dear readers, have we heard this refrain?

Intermittent Claudication

“Restless legs,” Buerger’s disease, phlebitis, intermittent claudication, all related to poor circulation in the legs, have responded, in most cases, to the properly administered dosage of Vitamin E. (See *Drs. Ayres’s and Mihan’s Innovative Discoveries*, page 87)

We should note that the work of Dr. H. T. C. Williams, together with Drs. Fenna and Macbeth in the treatment of intermittent claudication, has now been fully corroborated.

The experiment lasted ten years (1957–67) and was published in the highly respected *Surgery, Gynecology, Obstetrics* in 1971. The study was double blind with placebos, naturally. Conclusion: for those who had poor circulation in the legs because of inadequate arterial and capillary function, “there is good evidence that alpha tocopherol is valuable in the treatment.” So valuable, in fact, that those taking Vitamin E had a 243 percent improvement, compared with an improvement of 112 percent by those taking placebos. (See other similar reports in the *Summary*, Appendix A.)

There have been thirty-five confirmatory reports since Drs. Shute and Vogelsang presented their favorable results on intermittent claudication in the same medical journal in 1948. Fortunately, due to this overwhelming evidence the AMA listed intermittent claudication among the diseases to be treated with Vitamin E, as we have noted previously (See page 130.)

Vitamin E and the Pill

Hope for women taking the Pill and manifesting symptoms of thrombophlebitis is afforded by Dr. Evan Shute. The Pill is chiefly composed of synthetic estrogenic hormones which are sometimes known to produce adverse symptoms. Dr. Shute used Vitamin E for

this condition for years. He advises that women taking the Pill should certainly take generous amounts of Vitamin E, as Vitamin E counteracts the sometimes harmful effects of the Pill without lessening its effectiveness for birth control. Being an antiestrogen, anticoagulant, in addition to its other qualities, Vitamin E controls the tendency toward clotting often observed in women on the Pill. Also, other vitamins such as B-6 and C should be taken concomitantly, as they work synergistically with E.

Vitamin E and Smog

We must mention one more common complaint which affects all living beings on this planet. Since Vitamin E has been found to be effective against a wide variety of diseases, it is only natural that it be utilized against one of our worst threats to health: air pollution. Smog. We all live with it more or less, since we and most other industrialized countries are for the most part urbanized. Even the suburban and rural areas are not free from the enveloping blanket of chemicals released daily by factories, automobiles, ships, and diverse other machines—in fact, just about every artifact that modern humans employ in their pursuit of “civilization.”

This is no discredit to us; we did not comprehend the consequences of pollution (we weren't even aware of the words “ecology” or “pollution” until recently). Of course, no one really wants to foul his own nest, which happens to be Mother Earth at this present stage of human evolution.

So there is smog. And so there is Vitamin E. While we are cleaning up the polluted mess we have unwittingly created, we will be able to alleviate some of our own personal physical problems relating to air pollution.

Several groups of scientists at universities and elsewhere have been experimenting with Vitamin E to ascertain if the vitamin could protect lung tissue from the ravaging effects of air pollution.

The first study, to our knowledge, was made by Drs. J. B. Roehm, J. G. Hadley, and D. B. Menzel, then at the Battelle Memorial Institute, Richland, Washington. It lasted three years and was conducted for the U.S. National Air Pollution Control Administration.

Their findings: Vitamin E-fortified rats subjected to smog conditions similar to those existing in Los Angeles, Tokyo, New York, London, Stockholm, and other major industrialized cities, lived *twice* as long as their normal, yet unprotected, litter mates. Vitamin E guards against obstructive lung diseases caused by smog, such as emphysema and lung edema. (Emphysema is known to be a frequent precursor of lung cancer. Several of my friends and relatives have followed the tragic course from emphysema to lung cancer.)

Perhaps the best *testimonial*—we dislike that word, not only because the medical profession scorns it, often correctly, but because we like *objective* proof rather than *subjective*—is the fact that the originators of the test, Drs. Roehm, Menzel et al., are themselves taking large doses of Vitamin E to prevent lung diseases caused by our polluted atmosphere. (Smoking also should be included on the list of pollutants, since tobacco smoke contains just about all the pollutants we mentioned plus many more.)

... And Still More Diseases Aided

There has been monumental confirmatory evidence corroborating Vitamin E's value in treating various other diseases and conditions, but since the major thesis of

this present work is primarily concerned with aging, cardiovascular problems, and diseases which are of paramount concern to the largest number of people, we cannot devote more space to the other areas in a work of this necessarily limited size. In addition to the massive data accumulated on the heart and other related problems—which alone would fill (and do) two large rooms—there are also data on other “minor” diseases that occupy the same amount of space.

For example, I have scores of favorable reports on the various diseases reported on previously, such as indolent ulcers, intermittent claudication, varicose veins, muscular dystrophy, sickle-cell anemia, arthritis, thalassemia (a genetic blood disease peculiar to Mediterranean peoples), and Down’s syndrome. There are also very promising reports on kidney diseases, liver diseases, peptic ulcers, and arthritis. The published results vary in size and scope, and for the most part are strongly corroborative of our reports of several years ago. Some of these studies will be found in the Appendix.

Most of these experiments I consider very important, as, say, the fact that Vitamin E has been found to be probably the most efficient agent the liver uses as a detoxifier. The liver is responsible for keeping the body in order even when it is subjected to heavy assaults by alcohol, carbon tetrachloride, and the ingredients of smog, which we have just mentioned, *and* every other poisonous substance to which an animal (or a human) can be subjected. Vitamin C and B-6 also should be included as detoxifiers and should be taken along with Vitamin E. There are other vitamins and minerals which should be taken as well. We shall refer to them elsewhere.

These results show that almost every disease should be carefully studied in relation to Vitamin E, since we now know its omnipresent, nearly omnipotent, part in keeping the cellular structure functioning.

Heart Deaths Declining!

The United States Public Health Service (USPHS) stated in its series of reports from 1968 through 1990 that cardiovascular diseases declined approximately *twenty* percent. The decline continued through 1992, according to the USPHS.

Yet cardiovascular diseases *also* continued their deadly march upward during that same period—topping that of cancer and other diseases by far. In 1990 cardiovascular diseases claimed more than 930,000 American lives. Cancer deaths exceeded 500,000, while the third leading cause of death, strokes, accounted for over 145,000 victims. (From 1980 to 1990 the death rate from heart attack declined over 32 percent.)

Can heart disease soar and still decline? Yes. The answer to this seemingly contradictory question is that while heart diseases are still the leading cause of death, they have been (and are) declining from their highest *ratio* three decades ago to the present time of the last USPHS report (1990). In other words, more people are getting and dying from cardiovascular problems than cancer because more heart disease victims are living longer. See page 141 for our partial explanation: that more and more people now are getting increasing amounts of Vitamin E, chiefly by supplementation.

Since my first book was published, there have been few major developments in the prevention or cure of man's cardiovascular problems—certainly few advanced by the orthodoxy that have altered the dismaying epidemic of deaths due to heart disease.*

*See Appendix I for confirmation.

The rate of decline is still an astounding 20 percent. But don't let statistics fool you—and lull you into a sense of false security. Statistics won't mean a thing to you, if you're among those 70 million victims.

Just keep on taking your Vitamin E and other antioxidants as we have advised you many places throughout this book. And stay away from those dreadful statistical tables, both figuratively and literally! I'll guarantee it will be a more pleasant way to avoid them—if you take our *advanced* researchers' advice.

The following figures are in approximately the same declining *ratio* for 1990 as for 1980, as shown in the latest USPHS report. This is why I did not have to change or rewrite the *real essence* of this book. Just the *elaboration* of the numbers—which turned out to be somewhat of a problem because if you are like me, it's difficult to become excited over numbers and statistics. Important as they are. Amen—yet bear with us, friends.

For instance, the death rate from hypertension (high blood pressure) fell 21.6 percent from 1980 to 1990. Ischemic heart disease (destruction of various parts of the heart) death rate for the same period dropped 20 percent.

And strokes (cerebrovascular disease) declined an amazing 32.4 percent. Atherosclerosis, which is long known to be related to strokes and heart attacks, also declined.

The dreaded atherosclerosis (“A man is as old as his arteries”) has fallen along with the other cardiovascular diseases and deaths. It declined more than 40 percent.

Of course, no one would quarrel with moderate exercise and medical checkups as benefiting the entire body. Exercise should be done gradually, systematically, and—certainly, at the beginning—with a medical examination. Yet, as we have seen repeatedly, exercise itself is not a guarantee against coronaries. To my knowledge, there are no papers in the literature categor-

ically demonstrating that exercise, medical checkups, or lowering dietary cholesterol will prevent heart ailments or prolong life.

Now back to the cause of the 20 percent decline in cardiovascular diseases in the last decade. Since no studies have definitely proved that any *one* of multiple factors are the direct cause of cardiovascular afflictions, we hereby proffer our own solution to the multi-phased problem.

Based on thousands of experiments, on animals, reptiles, and man, we offer the theory—not hypothesis—that Vitamin E is responsible for the decline in cardiovascular deaths and/or episodes. In defense of our Vitamin E theory, we present the fact that our book, first published in hardcover in 1964, sold about 80,000 copies. The paperback version sold over 1,500,000 copies during the next fourteen years, and as the media have stated, was the key factor in attracting millions of persons, as well as many researchers, to begin taking Vitamin E. At least ten other books were written on Vitamin E, all of which were based on our original research. Altogether, the books and articles reached multimillions of people. We know that millions began taking Vitamin E by the dramatic increase in sales of Vitamin E.

The researchers started taking Vitamin E, as we stated, and we know the book writers began as well. How do we know? Statements in their books and in their appearances on various TV and radio shows.

“The Cure Is Just Around the Corner”

How many times have you heard those beautifully advertised programs (usually with a child and a tearful mother) pleading for money for research, saying that

the cure for cancer, heart disease, or multiple sclerosis is “just around the corner”?

What they neglect to tell you (purposely) is that they will not grant one penny to geniuses such as Linus Pauling,* twice a Nobel Prize winner, and Dr. Szent-Györgi (now deceased), who was awarded the Nobel Prize for his discovery of Vitamin C. He was working on an immunological cure for cancer, but could not get one Indian copperhead from any of the national or private agencies proposing to “fight” diseases.

These world-famous researchers are not alone. There are hundreds of other worthy scientists who have advanced notable research projects to these “disease-fighting” agencies—only to receive not one former *sou* (worth less than the American penny), but also to be attacked subtly or savagely for “unorthodox” procedures—meaning that what money the agencies had left from their administrative and fund-raising expenditures, they would give to their “pets,” who certainly knew upon which side of the bread their cholesterol-free imitation butter was spread.

Dr. Pauling, eminent as he was, and is, could not believe *he* would *not* be believed by the establishment. Yet it happened to him and many other well-known—as well as little-known—researchers who dared to confront the medical bureaucracy’s “consensus of opinion”—as they usually refer to their researcher-petserfs who wear their iron collars, either consciously or ignorantly.

Yet of the billions of dollars being allocated for medical research, where do you think most of the money is being spent? Just as one instance, vast research was done here and abroad on two anticlotting drugs called

*Dr. Pauling, after being refused repeatedly, was finally granted a “pittance” of \$50,000—which means nothing in modern research. To shut him up and thereby justify their existence? If you don’t know the answer by now, after reading this book and other books whose authors were dedicated to promulgating the truth, you deserve to get what you deserve.

streptokinase and urokinase. Both drugs are extremely expensive, so you, as taxpayers, paid for the experiment.

Without detracting from the merits of streptokinase and urokinase (of which I have a bundle of research material from Germany and elsewhere), may I suggest a far cheaper, safer, easily procurable, and already proven substance which is not only anticlotting, but possesses an almost inexhaustible store of benefits for the body? Vitamin E!*

Another instance of overlooking Vitamin E: the NIH awarded millions to study the effects of aspirin as a "fighter" against heart disease. According to Drs. Horwitt, Passwater, Machlin, among many other researchers, aspirin should be replaced by Vitamin E as a preventive measure for heart disease. (See page 186 for Vitamin E displacing aspirin.)

Proof piles on proof. The amount and the quality staggers an open mind, or even an objective one. But a closed or prejudiced mind—there is no amount or quality of proof that will open a closed mind or change a prejudiced one. Schiller said: "Against stupidity the very gods themselves contend in vain." I disagree. It is possible to guide or direct what we call stupidity, but it is rarely possible, short of Thor's thunderbolt, to penetrate a sealed mind. You may say scientists *have* changed their minds, but they are somewhat rare. Most follow the "scientific consensus" of the times.

*Since this was written both substances have been mainly abandoned. Streptokinase is now used temporarily in dissolving clots immediately after a heart attack, before the patient is placed in intensive care. Even so, its use is limited to surgeons and hospitals.

Healthiest Nation?

This section will probably prove to be one of the most important in this book, particularly since so much corroborative evidence has been amassed concerning the various conditions for which Vitamin E is effective, apart from the cardiovascular.

If the reader will recall at the time of our original publication (1964), there were 183 million inhabitants of the "healthiest nation in the world," yet withal, 74 millions of those healthiest persons suffered from one or more chronic diseases. Those figures were issued by the United States Public Health Service.

Now, sad to relate, the last report (1990) of the USPHS is worse. In an estimated total population of 250 millions, there are now over 90 million victims of chronic diseases—exclusive of those unfortunates in nursing homes and, probably more important, alcoholics (now estimated at 10 to 15 millions) and drug addicts (no reliable statistics, but probably 2 or 3 million hard-core addicts). Hard-core means cocaine and "mainline" amphetamine [speed] users and/or barbiturate-tranquilizer addicts, of which there are many more millions. We do not include marijuana users as hard-core, since repeated studies have shown the weed not to be addictive per se; it is not physiologically harmful (no really provable damage to the liver or other vital organs); therefore, as a drug it is probably far less dangerous than alcohol, heroin, and "speed." (For instance, coffee [caffeine] will break chromosomes just as marijuana will. Yet any substance known to man can be abused, and is.) Marijuana is certainly less con-

ductive to violence than alcohol, cocaine, and "speed," as even the "hard-core" authorities will now admit.

However, back to our premise: the "healthiest" nation in the world is rapidly becoming among the unhealthiest. Deaths from heart disease (with the exception we mentioned), cancer, and all the other major diseases are keeping pace with the growth (with minor exceptions) in the population, just as if there were no attempts to check them whatsoever. The same ratio applies to the "minor" diseases such as arthritis, muscular dystrophy, kidney and liver ailments, diabetes—just to mention a few. (We don't mean "minor" except in terms of numbers dying from the disease. Certainly these diseases are not minor to those persons afflicted by them.)

We wonder, we wonder, why do we pay the most monstrous bills for medical care in history? Help, help, Hillary, what can we do? Write to our multimillionaire senators, some of whom are not so far from earthly death themselves and who are not *that* interested in much except their reelection? Certainly not their own increasing medical costs, since *their* medical bills are paid for by us—the taxpayers.

Could it be, for instance, that certain persons in the health agencies have run out of excuses, and are indeed weary of them? Everyone knows beyond any question, should one be of the middle class, either white or blue collar, that he or his relatives had better not contract a major illness which will entail a lengthy hospital stay. No matter how much medical insurance he has, it won't be enough, as so many hundreds of thousands have found out, to their sorrow and bewilderment. It is only the ultrarich and the truly poor patients who can afford to have a major illness or accident. I do not think it is necessary to prove this statement or to corroborate it by listings in the Appendix. Only those who are profoundly foolish or profoundly greedy could fail to see the necessity of some form of a national health insur-

ance plan in order to provide *everyone* with health care in our free enterprise American [democratic] Republic. The health insurance plan probably could be managed by private companies under governmental supervision, like the postal service, public utilities, railroads, airlines, etc. (Some form of national health insurance will probably be passed by the time you read this.)

CHAPTER 13

How Doctors Are Programmed

You have probably heard that doctors are willing to try anything that is effective, and that if anything, *anything*—regardless of origin—works, the doctors will be the first to try it on themselves or their own loved ones. Nonsense. Most persons in such a rigidly controlled, totalitarian-oriented group would react exactly as most doctors react. They would rather die than see the hated foe defeat them—defeat meaning changing their conditioned ideas.

This is a characteristic of all well-disciplined organizations, or “in-groups.” To serious students of history and science, examination of doctors and scientists of any discipline reveals that there is just as much innate jealousy, bigotry, love, hate, generosity, compassion, and eccentricity as there is in every human being. For example, any General or Private, any steelworker, corporation president, sanitary engineer, chicken sales mogul, TV comedian, salesman, or housewife would have the same reaction patterns.

In short, there is very little difference, generally, between any class of human beings, all being subject to the same old emotional linkage and bonds. One of the differences between classes of people is the degree of sincerity, pretense, and pomposity which separates one in-group from another.

This observation does not apply to those very few individuals who have managed to escape man’s peculiar entrapment, viz., the failure to use more than one-tenth of his available 15 billion brain cells.

Nor does this observation apply to those even fewer

cosmically aware persons who have—by insight or prodigious mental work—caught a fleeting glimpse of the multitrillion, multidimensional factors which comprise what we call our universe.

But back to our analogy.

Until recently, doctors were able to conceal their own prejudices and jealousies to a considerable degree, because they were in a commanding position. They healed the rest of the population. Ever since man could talk, the act of healing was considered a sacrosanct, mysterious gift or art, rivaling those emissaries of gods who sought to teach men how the gods themselves wanted them to behave. Therefore, anyone who could *heal . . . !*

It was, of course, necessary to trust this august, mysterious person, because he or she kept you and your loved ones alive, and many times the affliction would vanish completely, due to his secret powers. Naturally, the more you trusted him and his methods of healing, the more he could heal.

Later, as medicine became “scientific,” still no one seemed to want to learn what made the doctor function. He was superhuman, scientific, certainly above all human failings.

There were, however, “outsiders,” “mavericks,” “apostates” who, somehow having been trained in the unorthodox way of doing things, attempted to demonstrate that they had better ways of healing.

But the orthodox doctor-healers established a tight inner circle and therefore control over the people, because of the power of numerical superiority. They took oaths to protect their patients and the public, supposedly even at the risk of their own lives. Ergo, how could they possibly discount the work of some colleague if it was calculated to save human lives? Theoretically, being incapable of jealousy, bigotry, greed, and all the other vices afflicting the rest of the human race, they certainly would be sympathetic to anything

original. Being so dedicated to science and humanity, they should be expected to test on themselves or their patients any new development if rationally presented—even if it seemed to threaten their own ideas and teachings about the subject. This is the “ideal image” which the doctors have projected about themselves.

Alas, readers, if you didn’t know by now—which I would find it almost impossible to believe judging from the intelligence level of the letters I receive—the picture just depicted in the preceding paragraphs is about as far from reality as I can conceive.

Many doctors *had* rather die—along with their families, relatives, friends, and patients—than admit they could possibly be wrong. Of course, to the few hundreds (now thousands) of doctors—truth-seekers who have either researched Vitamin E or have prescribed it to their patients—we say, “Welcome to the truth!” And as for those hypocrites who take Vitamin E themselves without prescribing it for their patients, well, what else can we say?

How My Books Are Researched

Queen Victoria's classic phrase, "We are not amused," spoken when someone attempted to entertain her and she felt the attempt was in poor taste, or not apropos, bears a faint resemblance to the comment this investigative writer-researcher uttered when yet another—the eighth—book on Vitamin E arrived upon his desk. I am quite sure, however, that Victoria Regina would not have approved, or been amused by, my American vulgarity.

The fact that the other Vitamin E books were hurriedly concocted because of the "Vitamin E boom," which was started by my first book, did not disturb me.

The *New York Times*, the *Washington Post* syndicate, and several other leading newspapers, magazines, and TV-radio shows gave the book credit for attracting public and medical attention to the long-maligned vitamin. (See Appendix B: *New York Times*, *Washington Post*, *Chicago Sun-Times* syndicate [Irv Kupcinet], *Glamour* magazine, and *New York* magazine.)

My initial reaction to the first three or four such books was, "Wonderful, finally after all these years, we're getting more and more people to write the truth about Vitamin E." (At that time, my book, both hard cover and paperback, had sold nearly a million copies and I had received thousands of letters from people attesting to Vitamin E's effectiveness.)

What did disturb me after *reading* some of those books, one in particular, was that I found that my research and words were coming back to me—without acknowledgment. It took me three years to research

and write my book about Vitamin E. I think it is fitting that I tell my readers something of the procedures I undertake before I release a book for publication.

My personal research-in-depth consisted of interviews with researchers, talks with persons who had benefited by Vitamin E, scores of trips to the medical libraries, examinations of thousands of reports. I have personal contacts in other countries who furnished me with information, including studies from Germany, Austria, Canada, and the former Soviet Union. (Recently, Japan and Australia have been included.)

I sent one of my best researchers, Neal Thorpe, vice-president of Man's Frontiers, to various medical libraries in the United States to examine *all* the available literature on Vitamin E. There were hundreds of foreign reports, though almost none were translated into English. Neal, of course, made extensive notes on the studies from the U.S., England, Canada, and other English-speaking countries. Having a smattering of a few other languages, I selected those articles which I thought should be translated. The translations were done by a young researcher then working at *Life* magazine. Also, I found excellent abstracts in the *Summary*, the medical journal published by the Shute Foundation; and in the comprehensive *Annotated Bibliography of Vitamin E*, published by Distillation Products Industries.

After collating and examining all the thousands of reports, I selected over 300 to be used and then winnowed out about fifty or so which I would use as examples in the book itself.

The selection was purely a personal choice, as I had thousands of other reports equally as sound scientifically. However, following intuition, I chose the ones that I felt *I* could best write about.

Therefore, after reading some of the later Vitamin E books, I was struck by the not-so-remarkable fact that several of the authors had used exactly the same examples I had used. Out of thousands of reports available—

available only with years of careful personal investigation, buttressed with the help of good researchers, assistance from astute doctors—some of the other writers had “given me the honor” of making the same choices I made. One writer didn’t even bother to change the style or the phraseology to an appreciable degree. Now, thousands of other reports on Vitamin E have been published since I wrote the original book in 1964 (which is one reason for this new version), yet only a few of these were utilized by some of our industrious, if not original, workers.

But here is perhaps the most humorous example of fast work at the typewriter. There were very few mistakes in the original book; however, there was one significant mistake which somehow managed to slip by me and others. It is now corrected, but in the original book, Chapter 2, it read that synthetic Vitamin E possessed only *one-fifth* the potency of the vitamin derived from natural sources. It should have read only *one-half* the potency of the natural Vitamin E (d-alpha tocopherol)—quite a significant difference. Now these figures are easily obtainable from the National Research Council, among many other sources.

Yes, you guessed it, it is *one-fifth* in at least one other book. Here is a tip, authors, correct it in your next edition, please.

As a friend remarked to me the other day: “If imitation is the sincerest form of flattery, consider yourself the most flattered author of this decade. They even copy your mistakes!”*

Yet I must say that in eight major books—most of them controversial—which I have written, I have not been challenged by anyone, except in one instance. That challenge occurred prior to publication, when a lawyer who had not read the Appendix wherein lay the *absolute* proof, forced me by a series of seemingly

*See page 49 for later information on synthetic versus natural Vitamin E.

clever maneuvers to prove, almost line by line, everything I had written in the book. I suppose he had to justify his being on the payroll. At any rate, I won the case, because I proved everything in the book was true. Result: not one paragraph in the book was changed.

This episode caused the book to be delayed for a year and no doubt cost the publishers thousands of dollars, as well as loss of publication time.

While I certainly believe in lawyers as being necessary to argue the law, some are overzealous and contribute to the "law's delay," which was one of Hamlet's chief grievances in his most famous soliloquy.

Would that he could see the law's delay in America as it is practiced today; he would probably take poison himself instead of waiting and taking a hit from Laertes' poisoned sword.

How to Help Yourself and Others

This chapter will contain somewhat the same information as in my previous book, but with some very valuable additions, I am glad to say.

A trace mineral, selenium, acts as a potent antioxidant and it abets the action of Vitamin E. Selenium is deficient in various parts of the country, particularly the eastern states, although states such as Indiana and Oregon are also selenium deficient. (See Appendix J.) Since selenium, as with almost any substance, becomes toxic with overdosage, nutritionists recommend only slight supplementation, viz., 50 to 200 micrograms a day.

Vitamins A (beta carotene) and C, as we have mentioned previously, also aid in Vitamin E's role in preserving the body's cells and tissues from harm by free-radical molecules. They also afford protection from such potentially harmful substances as polyunsaturated fatty acids—in large amounts (the vegetable oils so widely promoted for the last two decades by most medicos and the food manufacturers as being helpful in preventing heart disease), smog, radiation, toxic substances, and many other environmentally produced factors.

Vitamin E, of course, has many usages other than that of an antioxidant, and almost certainly has wondrous potentials as yet undiscovered.

For instance, twenty years ago, no "orthodox" scientist would admit the possibility of life on other planets. Today it is hard to find a scientist who does not believe that life exists elsewhere. We cite the aforemen-

tioned as a typical, common example of how science, including medicine, changes so rapidly that even science's foremost practitioners cannot always be aware of new developments in their own specialties.

So how can doctors, even cardiologists, know how to treat cardiovascular diseases effectively when they cannot or often will not read the literature of their own specialties?

If you have the luck or the insight (you'll need both) to acquire a doctor who is truly nutrition conscious and practices holistic medicine, that is, treats the patient as a whole entity, composed of staggering numbers of interactions—and doesn't charge the equivalent of the jewels of Opar (many do), then he will be amenable to vitamin-mineral therapy, both for prevention and treatment. I must warn you, these doctors are, as yet, in the minority.

I have just received information from a friend who wrote to several of these doctors and/or clinics. The replies she received confirmed my suspicions: the fees were, I think, unconscionably high, much higher than even the "average" doctors' fees, which are high enough.

My advice: Don't get sick, don't go to a hospital unless you are ultrarich. The poor can go and receive fairly reasonable treatment, depending on the hospital.

But not the middle class, either upper, medium, or lower. That is, unless they pauperize themselves and usually their loved ones. Being sick with a major illness has wiped out more families than any war, depression, or recession the United States has ever encountered. (I also include the War Between the States, since that war was more devastating to both sides than any other.)

Persons who have suffered a catastrophic illness in the family will get my message very loud and with the clarity unequaled by even the best of stereos. For instance, the cost of a hospital bed in New York City is about \$800 a day, and, of course, that doesn't cover

intensive care, the doctor's fees, or special operations such as bypass heart surgery, which costs more than middle-class families can save in their lifetimes.

However, the high cost of dying from chronic (or acute) heart disease is equally matched by dying from cancer and several other chronic "lingering" diseases. Twenty to twenty-five years ago, it cost only \$15,000–\$20,000 to die from cancer or heart disease. Today it often costs more than triple that, as almost everyone knows. Blue Cross and Blue Shield, which were at first so bitterly fought by the AMA as being "socialistic," are now, in effect, approved by the AMA. Both BC and BS have now become a grim joke since neither usually pays off more than a modicum of the full cost when a long-term illness involving a long stay in the hospital occurs. (The other "insurers" are as bad, if not worse, than Blue Cross and Blue Shield.)

With these warnings over (which I am sure you already knew), I can now make a few recommendations.

First of all, go to your doctor—if you have one—show him this book, ask him to read it with all its undeniable monumental evidence (especially the Appendix). Then if he shrugs you off, tell him to go to the inferno which Dante described so vividly and seek another doctor who might be more intelligent.

You should try to find an intelligent, perceptive doctor who knows about and will prescribe Vitamin E along with the other supplements which we have mentioned.

There is an ever-growing group of clinics as well as doctors in private practice who are aware of Vitamin E's value. Of course, those living near the Shute Institute may wish to go there. (Shute Institute for Medical Research, 367 Princess Avenue, London, Ontario, Canada N6B 2A7.) This clinic is the most experienced in the world in treating cardiovascular problems. Further, it is not expensive and the diagnosticians are excellent,

qualified physicians who carry on in the same tradition as that great humanitarian, Dr. Evan V. Shute, who organized and was director of the Shute Institute and Foundation for many years.

Dr. G. E. Desaulniers is medical director of the Shute Institute, heading a team of two other doctors, as well as several nurses and other medical personnel. Janet Shute, daughter of the late great Evan Shute, is the knowledgeable, considerate administrator. Janet told me the Institute is also using various other antioxidant-nutrients such as Vitamin C, beta carotene, selenium, zinc, calcium, magnesium—in short a well-balanced nutritional program.

Your last and least expensive alternative lies within you. Do as I did: treat yourself. Millions of Americans have followed that advice for the last thirty years. To my knowledge, all have benefited. I have received hundreds of letters to prove the point. *Prevention* magazine and the Shute Institute have an equally favorable response.

Some persons complain of the oily aftertaste of Vitamin E. I would advise these complainants to take the dry form of Vitamin E. There are, of course, many excellent products on the market now.

If you are experimenting on your own without medical advice, as are most people who take Vitamin E, then doctor-researchers suggest that you begin with 100 units of Vitamin E daily and gradually build up to the optimum dosage for you. You will know when the optimum occurs, and only you can tell. It may take 600–800 units or more. (I have been taking about 2,000 units for the last twenty-eight years.)

Please remember to take only the natural form, since, as we have mentioned previously, the synthetic Vitamin E is only one-half as effective. Remember, also, *d* is natural and *dl* is synthetic.*

*See Appendix J for more on natural vs. synthetic.

Then, when you return to your doctor who scoffed at the suggestion of Vitamin E for cardiovascular conditions, and after he examines you, finds definite improvement, and throws up his hands when you tell him it's due to Vitamin E, don't be rude to the poor arrogant man. Just leave quietly and don't go back. Also, don't forget the premonitory warnings given previously. These warnings apply to diabetics taking insulin; to heart disease victims taking digitalis; those with dangerously high blood pressure; those who have a damaged heart due to rheumatic fever which was not treated with antibiotics (now very rare). All of these aforementioned conditions should be treated with Vitamin E, but under medical supervision.

I suggest that, ideally, as advised by our *advanced* researchers, you should take a combination of Vitamins E, C, A (beta carotene), as well as multivitamin/mineral capsules, which contain all of the important vitamins and minerals (including zinc, selenium, calcium, and magnesium), plus a generous amount of choline and inositol.

Herewith is the suggested dosage of Vitamin E as recommended by the Shute Institute.

Schedule of Dosage

Natural Alpha Tocopherol (Vitamin E)—in the treatment of Cardiovascular-Renal Diseases by physicians as suggested by the Shute Institute for Clinical and Laboratory Medicine, London, Canada.

1. Acute coronary thrombosis—800–2,400 I.U. a day started as soon as possible and maintained.
2. Older cases of coronary thrombosis—800–2,400 I.U. if systolic pressure is under 160—otherwise 800 I.U. for the first four weeks, particularly if a hypotensive agent is used concurrently.

3. Anginal Syndrome—800–2,400 I.U. if systolic pressure is under 160. Otherwise start on 400 I.U. for four weeks then 800 I.U. for four weeks, particularly if hypotensive agent is used.
4. Hypertensive heart disease—400 I.U. for four weeks, 800 I.U. for four weeks, then cautiously increase. Should be used with hypotensive agents.
5. Thrombophlebitis and Phlebothrombosis—800–2,400 I.U. daily.
6. Diabetes Mellitus—Same schedule as for cardiacs.
7. Acute and Chronic Nephritis—As for cardiac patients.
8. Burns, Plastic Surgery, Mazoplasia—800–2,400 I.U. daily, using Vitamin E Ointment or Vitamin E Spray as adjunct.

More on Antiaging

In addition, I should mention another important factor in antiaging. It was the subject of one of my later books: *GH3—Will It Keep You Young Longer?* (New York: Bantam Books, 1977).

In case you have not read the book, I'll describe briefly what GH3 is and how it can alleviate depression and, also, how it can prevent or even reverse the signs and symptoms of aging.

GH3 is Gerovital H3, whose antiaging, antidepressive, antiarthritic properties—as well as a host of other beneficent qualities—were first advanced by Professor Dr. Ana Aslan (now deceased), Director of the National Institute of Gerontology and Geriatrics, Bucharest, Romania.

Thousands of ailing folk flock to Romania every year to receive the rejuvenating GH3 treatment in Romania's 144 clinics where GH3 is available cheaply. Scores of

thousands of others also take GH3 in the various countries where the substance is not only legal, but usually may be purchased over the counter without a prescription. In fact, the United States is virtually the only developed country where Gerovital H3 is not obtainable legally. This is, of course, due to the FDA's conflicting policies of both overprotecting and underprotecting the consumer as well as the producers (usually the pharmaceutical houses, large and small; particularly the small company if it does not have the prestige, the time, the power, or, naturally, the money necessary to comply with FDA rules and regulations). As we mentioned previously, the time for approval of a new drug is about ten years, the cost in the multimillions; the effort in animal and human experimentation almost immeasurable.

Fundamentally, GH3 consists of procaine (known in America as novocaine), which has been administered by dentists and physicians for the past eighty-five years without harmful side effects. (One person out of 5,000 has a mild allergic response.) However, Dr. Aslan found that procaine had far more propensities than just as a local anesthetic. She discovered that when properly buffered by various ingredients, such as potassium, sodium bisulfite, and also by altering the PH (acid/alkaline) value to the point where procaine no longer was an anesthetic, it metamorphosed into an antiaging, anti-depressive, antiarthritic substance.

Procaine itself is compounded from two natural elements: para-aminobenzoic acid (PABA, one of the many factors in the Vitamin B complex), and diethyl-aminoethanol (DEAE, a material also found in the body which is a safe, natural "energizer" according to Dr. Carl Pfeiffer, brilliant researcher previously mentioned for his Vitamin E electrical experiment; others have confirmed his DEAE findings).

Now while we could cite Dr. Ana Aslan's experiments during several decades proving in controlled

studies on many thousands of persons of all ages that GH3 did, indeed, curb or reverse signs and symptoms of various afflictions, including senility, we will not because GH3 belongs in a separate stratum, which we have already reported in our GH3 book.

Therefore, I will not report in any detail on the many favorable tests conducted in the United States under the supervision of that wonderfully organized bureaucracy known as the FDA. We will only say that certain officials in the FDA sabotaged the GH3 project in the U.S. How? By changing the rules near the successful conclusion of the experiment. One such so-called official guardian of our health said in effect to American sponsors of GH3: "Okay, suppose you've proved GH3 is a good, safe antidepressant. Now you'll have to prove GH3 is an *antiaging* agent before we'll approve it."

Such is the mentality and morality of some of our FDA bureaucrats, who freely spend our money to work against us, as well as to foster their own economic and political ambitions.

If you can afford it, I would strongly urge you to go to Romania for the GH3 treatment. The cost is minimal and the results are phenomenal.*

*The FDA with its usual police-Gestapo tactics has raided all the known outlets for GH3 in this country. All clinics and laboratories were closed, all materials confiscated, owners arrested and charged with illegal possession of drugs. (See Chapter 16 on FDA.)

The FDA: Protection, Yes—Oppression, No!

The Food and Drug Administration (FDA) has introduced yet another bill (1993) designed to severely curtail, or ban altogether, your access to dietary supplements. The intent of this bill is to lower potencies of vitamins and minerals to those found in foods and to classify the higher dosages as unsafe food additives. Further, to classify herbs as drugs and to require a prescription for all amino acids, as well as the higher—or more effective—doses of vitamins.

The “protective” actions by the FDA seem to be cyclical—this bill is virtually a carbon copy of the several previous proposals to control almost every vitamin-mineral nutrient supplement known—and yet to be discovered.

Remember it was only thirty years ago, in June 1962, that the FDA proposed to prohibit any vitamin manufacturer from listing Vitamin E on the label in *any* amount because it was not a nutrient “. . . [not] recognized by competent authorities as essential and of significant dietary-supplement value in human nutrition. . . .”

Even after the FDA was forced to accept Vitamin E as necessary to human nutrition, the agency tried to ban the sale of *all* vitamins and minerals, except in minimum quantities. With your money, the FDA mounted a huge campaign to close out vitamins, including Vitamin E, of course, except by prescription from doctors.

This tactic was easily discerned by various watchful groups and individuals. Chief among those who opposed the FDA ban was the National Health Federation, an organization devoted to the right of the individual to choose his own form of therapy.

Among the eminent persons who testified against the FDA's proposed vitamin-mineral ban were Miles Robinson, M.D.,* and the notable nutritionist, Carlton Fredericks, Ph.D. After long months of haggling, postponements (at which the government lawyers are particularly adept since they don't have to worry about money—they have all the money, time, and authority necessary to “wear down” even the richest of opponents), the FDA lost! It's an old story; using our own money to try to defeat us.

Nevertheless, inspired by a defense approaching genius, in addition to hundreds of thousands of letters from enlightened citizens, Congress voted overwhelmingly to reject the FDA proposal.

Now all persons could relax (they thought) in the knowledge that they could buy the harmless vitamin-mineral supplements without a doctor's prescription—which would be almost meaningless since the “average” doctor knows little about vitamins and nutrition other than what is taught at the college level. Reason: only about 5 percent of the medical schools offer a course in nutrition, which is an elective in any case and which few future doctors elect to study. Fortunately, a few doctors have finally discovered some facts about nutrition, vitamins, and minerals and are using the information to excellent advantage for their patients.

Now wouldn't you think the FDA would listen to the voice of Congress and the people?

No, the FDA was merely seething in the wings, concocting (with our money, of course) still another devi-

*Readers of my first Vitamin E book will recall that Dr. Robinson wrote the brilliant Foreword.

ous plan to deprive us of one more of our inalienable rights: freedom to choose between various forms of medical treatments, whether it be Christian Science, faith healing, allopathic, homeopathic, chiropractic, holistic, acupuncture, chelation, or other forms of alternative medicine—or a combination of all. (This proposal was advanced in the early 1980's—and again defeated.)

The 1993 proposal the FDA has advanced does not essentially differ from the previous ones, except it is more cunningly worded.

In essence, the proposal would deprive us of the right to buy vitamins, minerals, and other supplements without a doctor's prescription. It's *still* the same sad old story.

Again, the only solution is to inundate your senators and congressional representatives with protests. Ask these *elected, sworn* protectors of our inalienable rights to take away the FDA's police powers with its pistol-totin', shotgun-brandishing agents (and sheriff's deputies), who raid legitimate clinics and laboratories and confiscate legal, widely prescribed medicines and drugs. Also, ask Congress to demand that the FDA cease its Gestapo tactics against reputable medical practitioners—which no other developed country permits in the guise of “protecting the public.”* (Although by the time this book is published, it may be too late for action on this particular bill, yet your ongoing vigilance is most necessary.)

The Table on the following page is from the *Townsend Letter for Doctors* (May 1993) and shows number of poisonings in 1991 from commonly used vitamins, herbs, and pharmaceuticals. The total reported poisonings from vitamins/minerals/herbs are zero; the total

*Dr. Jonathan Wright of Tacoma, Washington, was arrested in such a raid. He is a prime example of many such FDA actions. See his “History Repeating Itself. . . ,” Appendix B.

reported for pharmaceuticals is 9,805! Now, FDA, *what* are you protecting us from by your proposed bans and restrictions?

An effective organization, deserving of your support, which is fighting to protect consumers' access to dietary supplements is the Nutritional Health Alliance, Box 267, Farmingdale, New York 11735.

Action by an informed, educated, and organized public can, and will, win against stupidity, arrogance, and greed. Turn your resentment which these traits naturally engender in you, into constructive, integrated, positive action.

Table 1

Herbs .. No. of Poisonings 1991	Pharmaceuticals*
Ginseng (Ren Shen)0	No. of Poisonings 1991
Echinacea0	Analgesics2669
Dandelion.....0	Antidepressants517
Angelica (Dong Gul).....0	Antihistamines412
Astragali.....0	Antimicrobials953
Rehmanniae0	Asthma Therapies257
Bupleuri.....0	Cardiovascular Drugs.....370
Polygoni Multiflori0	Cough & Cold
Goldenseal.....0	Preparations1526
Hawthorn0	Gastrointestinal
Total.....0	Preparations619
	Hormones & Hormone
Vitamins	Antagonists488
No. of Poisonings 1991	Anti-Anxiety/Anti-Psychotics 888
Vitamin A0	Topicals1106
Vitamin C0	Total.....9805
Vitamin D0	
Vitamin E0	
Iron.....0	
Zinc0	
Vitamin B or B-Complex0	
Calcium.....0	
Magnesium0	
Potassium.....0	
Total.....0	
Townsend Letter for Doctors, May 1993	

*Some of the poisonings may be considered suicide attempts. The poison control centers do not differentiate such factors when gathering yearly totals.

Recommendations For a Healthy, Longer Life by Advanced Researchers

Suggested procedures to prevent, treat and control/cure: the diseases, afflictions, and ailments common to all humankind. Try to:

1. Eat well, but in moderation.
2. Exercise regularly.
3. Sleep as much as you need—if you can.
4. Take supplements recommended throughout this book. All the essential antioxidants are easily obtainable in *one* or two capsules. In general, follow instructions on the label. Remember warnings and cautions for those who are, or should be, under competent medical supervision.

Note to smokers: It's true that Vitamin E will protect you somewhat against the ravages of smoking, just as it will against smog. However, against *both* smoking *and* smog? Probably, in very heavy doses (2,000–2,400 I.U.), but no researcher I know would advise such a drastic procedure except if there is no other way out for you. The data against smoking are overwhelming!

Well, our story ends here, friends. We have tried to bring you all the latest available information on Vitamin E and several other substances which will probably help to prolong your existence in this dimension—if you wish it.

I, unfortunately, have few facilities for answering letters (I have to spend most of my time writing and researching, else there would be no more books such as this one). Since I put in everything I know that is pertinent to the subject in the books, a careful reader can usually discern the answers to his or her questions. I welcome letters, however, but please do not expect a reply.

Let us hope you do wish to extend your life, and that you do what you can to aid others on this earthly sojourn before you travel elsewhere.

Original research is included in the Appendix along with the most recent to prove that most of the later research corroborates the first discoveries of Vitamin E's versatility.

H.B.

Appendix A

Excerpts from: *The Summary*, Vol. 28, January 1977.
Published by the Shute Foundation for Medical Research, London, Ontario, Canada.

Therapeutic Note on Tocopherol Therapy

by John K. Mackenzie, M.D., C.M.
Islington, Ontario
Canada

I am happy to contribute to this issue of the Summary. Being interested in and having observed the effects of reasonable dosages of vitamin E over the last twenty years, I think it might be of interest to describe some (few) cases that came under my care and showed a dramatic response.

The first of these examples concerned a man, Mr. H., who was referred to me in February 1970 because of severe impairment of vision. When first seen this man was sixty-one. He had a history of having experienced what he called "sudden whiteouts" about the year before I saw him. He was subsequently seen by an ophthalmologist who found his blood sugar elevated and evidence of haemorrhages upon fundusoscopic examination. He was instructed to have the blood sugar brought under control by use of an oral hypoglycaemic agent. Mr. H.

had known of his diabetic condition for twenty years. The haemorrhagic areas cleared subsequently. No adequate explanation could be given for the gross loss of vision.

Later the patient was referred to a neurologist and admitted to hospital where an intensive investigation was carried out. The consensus of medical opinion following that investigation in hospital was that the impaired vision was due to diabetic retinopathy. A very gloomy and extremely poor prognosis was given. Impairment involved primary central vision.

This man was a commercial artist and when seen by me was in the throes of giving up his job because he could not cope due to the marked impairment of vision, and he had his car up for sale. There was a family history of diabetes which included his mother and his maternal aunt. He had been a smoker all his life (3 packs a day since 1945). He changed to a pipe, but never did give up the habit. He had bilateral deafness with occasional tendonitis. He had symptoms suggestive of intermittent claudication which gradually increased over two years.

On examination, the funduscope revealed evidence of diabetic retinopathy. The blood pressure was slightly elevated at 155/70. The cardiac rhythm was regular and heart sounds were normal. Posterior tibial pulses could not be felt and the dorsalis pedis pulses were weak bilaterally. Bilateral femoral bruits were present. Vibration, position sense, cold sensation, light touch, and pin prick were all reduced below the knee in both legs.

At two hours p.c. blood sugar was 213 mg per cent. He had been left with peripheral vision and used a cane for guidance as well as support. In my office, while standing less than three feet from my medical diploma hanging on the wall, he was quite unable to read any of the script. He said he had the impression he had been medically abandoned.

After a thorough medical examination and extensive blood studies, he was placed on 1600 units vitamin E initially and was later increased to 2000 units. He was followed closely and frequently. In a matter of a few weeks he reported that he could read the upper-case print in *Reader's Digest*. His exercise tolerance increased. He was able to walk greater distances with less pain in his legs. His diabetic condition came under better control. With the recovery of his vision, his state of well-being increased considerably. He purchased another car and returned to his former capacity as a commercial artist. He was instructed to arrange an appointment with the ophthalmologist who had seen him several months before. The ophthalmologist, in reviewing his file for comparison, expressed astonishment at his recent findings using the words "dramatic" and "miraculous."

The other case concerned an elderly diabetic suffering from extensive ulceration of the left foot with gangrenous changes including the toe. Mr. L. was seventy years of age. He had been a known diabetic for

many years and the condition was quite severe, necessitating his taking very large doses of injectable insulin. He was a fisherman and lived in Halifax, Nova Scotia. He was seen by me at the request of his relatives.

When I was asked my advice, I suggested the man be taken to the hospital in Halifax where he could be brought under some degree of management with a strong possibility amputation might be recommended. This advice was not taken nor kindly received, primarily because there was a disinclination to leave the home to go into Halifax. As a compromise, we agreed to have Mr. L. follow through on a liberal dosage of vitamin E and the man agreed to this.

It was requested that photographs be taken periodically to indicate any progress. The story had a happy ending. A man whose daily habit was to sit in the kitchen with his left leg supported on a kitchen chair became fully ambulatory and the ulcerative areas completely healed. The areas of blackish discoloration disappeared and a grossly swollen foot became the match of its counterpart.

My experience over the years seems to indicate that the large body of vitamin E agnostics is shrinking. As they should. For those cases which I considered had a reasonable chance to respond to vitamin E intake, one has never really had a failure yet. I also believe that in most cases, vitamin C should be taken concomitantly with vitamin E and that also in adequate doses.

Vitamin E has travelled far from London, Ontario, and the world literature is now heavily endowed with multiple references on vitamin E therapy. I admire the pioneers who have shown the way through much unnecessary and unwarranted adversity, because it was only through their perseverance that vitamin E is becoming so well recognized as an established therapeutic agent.

Requirement of Vitamin E in Human Diet—

a letter to the *Toronto Star*
(Dec. 16, 1976)

As a physician interested in nutrition, I am compelled to comment on your paper's syndicated column of Dec. 1, 1976, "Eating Right" by Ruth Fremes and Dr. Zachary Sabry. Their claim that vitamin E in the diet is adequate is no longer valid. Responsible nutritionists are rightly concerned about protecting the public from unwise and extreme dietary habits. However, fads develop because exact information about nutrition is often lacking. Nutritional science is still in its infancy, although most nutritionists have great difficulty acknowledging the fact.

Dr. Sabry's comments echo the conclusions about vitamin E human

requirements which have been adopted by expert consensus in the U.S. during recent years. It is useful to recall that until 1959, this vitamin was not officially declared essential for man; that only in 1968, we were told how much we need (a maximum of 30 units daily in adults). This figure was revised downward in 1974 to 15 units (ie., 10mgm) daily, partly because analysis showed that American diets contained far less than the 1968 figure, and partly because dietitians complained bitterly that they were simply unable to plan diets containing as much as 30 units.

The above figures were influenced by two erroneous concepts and one large hang-up. The first dubious concept is that one can obtain reliable information about the amount of stored vitamin reserves in our body by measuring the level of vitamin in the blood plasma, an idea which has been challenged for a quarter century, but which still dominates our thinking. The second faulty and very dangerous assumption is that in the absence of obvious signs of vitamin deficiency in the population, the diet must therefore be adequate. That is, one just measures the amount of vitamin E in the diet, and that must be the amount we require. However, optimal health is clearly more than the mere absence of disease; optimal nutritional requirements are, in many circumstances, significantly greater. The large hang-up? Nutritionists as a group seem to be committed to holding the line and not giving an inch to the health food crowd. The quality of our diet has deteriorated progressively for decades, and every new survey shows it. It takes a big person to be the first to admit the situation is far from satisfactory. Vitamin E has always been a particularly hard nut to swallow. We "fortify" with thiamin, riboflavin, niacin and iron. Why not vitamin E?

Dr. Sabry might have mentioned that some top experts in this field are very concerned about the problem of dietary adequacy. Daniel B. Menzel, of Duke University, found that human red blood cells had greater resistance to ozonides (in air pollutants) when 200 units of vitamin E were given daily than when 100 units were given. He declared that "the present dietary intake of vitamin E is inadequate to provide maximal protection against ozonides!" Menzel's work has been confirmed by Jeffrey Bland, Ph.D., of Tacoma, Washington. Bland now recommends supplements of vitamin E be taken daily.

Perhaps the most important opinion comes from Dr. Max K. Horwitt, Professor of Biochemistry, Washington University, St. Louis, who was Chairman of the 1974 Minneapolis Conference on vitamin E, and who conducted the only long term (8 years) human vitamin E deficiency study ever carried out. In the past year, Horwitt has warned of the potential danger to those individuals exposed to smog, and to those whose diets are high in unsaturated oils. He emphasized that in view of the losses of dietary vitamin E from processing, and the lack of toxicity of the vitamin, supplementation of daily amounts of 200-800 units are recommended. The *American Journal of Clinical Nutrition* of May, 1976 contains Horwitt's 8-page "Vitamin E: a reexam-

ination" in which he concludes that "the recommended dietary allowance of 15 International Units (10mgm) is too low."

David Turner, the brilliant Canadian biochemist, received a NASA citation for his part in solving the space anemia problem in astronauts by correcting a vitamin E deficiency, caused by prolonged breathing of pure oxygen. Dr. Turner indicated to me, shortly before his untimely death in 1971, that he was concerned that ozone from automobile exhausts and industrial pollution (smog) would greatly increase our need for vitamin E, and that our dietary levels were inadequate to cover this need.

Dr. A. J. Tappel, Professor of Nutrition, University of California at Davis, Cal., has found support for Turner's concern experimentally, showing that protection was increased in proportion to the logarithm of the dose of vitamin E added to the diet. This, combined with the fact that absorption of high doses of the vitamin is low (an added safety factor) explains why it appears necessary to increase our intake many times higher than the dietary level.

Evidence of widespread subtle human vitamin E deficiency was provided by Dr. N. R. DiLuzio, who found diene fats in 78 out of 81 persons tested. These substances, formed in the absence of sufficient vitamin E to prevent peroxidation, have been correlated with heart disease. DiLuzio noted decreased amounts of dienes after supplementation with vitamin E.

Horwitt had emphasized that it is impossible to perform experiments to evaluate the full effects of prolonged (30-40 years) mild vitamin deficiency. Most persons over this age may in fact be unwitting candidates, without having any satisfactory way to identify it. Some suggest that accelerated aging may represent a sign of prolonged mild deficiency. Our increasing incidence of degenerative diseases occurring in ever younger populations may represent another sign of it. It is of interest that Dr. Lester Packer was able to extend the life of human lung fibroblast cells growing in tissue culture from a span of 50 cell divisions (generations) to 116 generations by merely adding extra vitamin E to the culture medium. It is no myth that E is important in retarding aging of our cells.

Vitamin E supplements are particularly important for an estimated fifty million world-wide users of the contraceptive pill, to minimize risk of venous thrombosis. Those exposed to X-ray treatment for cancer therapy as well as those requiring prolonged oxygen therapy are also candidates for benefit from supplements of vitamin E.

We should think of nutrition as our prime means of "rust-proofing" our bodies. However, the problem is compounded when nutrient losses are high in our modern life-style. Foods depleted by processing, transportation, storing, freezing, thawing, cooking, and draining, or by poor food selection cannot hope to protect us in these times of increased nutrient demand. Our over-use of aspirin, antibiotics, alcohol, diuretics, sugar, tobacco, and tranquilizers, plus the stress created by illness, injury, surgery, domestic and work pressures have been shown to increase specific nutrient require-

ments. It is not surprising that more and more people are discovering that they feel better when they supplement their diets regularly.

A word about potential hazard from vitamin E. Personal experience has absolutely confirmed (following initial skepticism) the warning of Drs. Evan and Wilfrid Shute that most chronic rheumatic heart disease patients, some other patients in severe congestive heart failure, and patients with uncontrolled high blood pressure are seriously aggravated by daily doses in excess of 200 units.

Thus Dr. Sabry's advice that 10 mg of vitamin E in our diet is sufficient is not in our best interest in the light of current research findings.

The choice lies between gambling on the marginal protection contained in our diet, hoping for the best in a situation that is impossible to fully assess, or to recommend taking a safe 200 unit, daily capsule. Until more complete answers are available, the only reasonable course becomes clear. It is to be hoped that nutritionists will not get stalled over their large hang-up, but will update their recommendations regarding the need for vitamin E supplementation.

Dr. J. H. Barker, M.D.

Abstracts and Abstract Comments

Heart

Vitamin E

Rosenvold L. (U.S.A.)—Drop Your Blood Pressure, (Pyramid Books), p. 86

When used with caution, 400 to 800 milligrams per day (of vitamin E) appears to be satisfactory daily dosage. There is little to lose by using vitamin E properly, but much to gain.

That vitamin E is beneficial in cardiac states is shown by the fact that digitalis dosage can often be reduced by one-half in some heart patients who take vitamin E regularly.

Vitamin E Decreases Erythrocyte Fragility After Whole-body Irradiation

Hoffer A. and Roy R. M. (Canada)—Radiation Research 61:439, 1975

Erythrocytes from mice in an irradiated deficiency group usually exhibited greater fragility. The least fragile red blood cells were gener-

ally found to be those from subjects in an irradiated and vitamin E supplemented group. A statistically significant radiation effect and vitamin E effect could be detected.

After exposure to high radiation dose levels, vitamin E-deficiency enhanced the rate of radiation-induced hemolysis of red blood cells. Vitamin E has also been implicated in heme synthesis, hematopoiesis, and in maintaining the integrity of the erythrocyte membrane as manifested by increased red blood cell fragility.

The dramatic decrease in fragility displayed by the erythrocytes from the irradiated and supplemented group may have indicated two modes of interaction between radiation and tocopherol. Orally administered vitamin E decreased the fragility of red cells from mice exposed to whole-body irradiation as compared with cells from unirradiated animals. Absence of the vitamin potentiated radiation damage. In the past *in vitro* irradiation at considerably higher dosages has indicated membrane effects on erythrocytes; the present study demonstrates a significant radiation effect on erythrocyte fragility *in vivo* at a dose level approaching the LD50/30.

Vitamin E Shown to Prevent Premature Aging of Human Cells Medical Tribune Report—J. Bland (U.S.A)—August 18, 1976

Vitamin E has been shown to prevent "blow-outs" in human red blood cells. Lipid peroxidation is caused by direct exposure to light or oxygen. Vitamin E apparently slows down the process called "budding of cells," caused by peroxidation of cholesterol into cholesterol hydroperoxide. Bland comments: "We were amazed to find that exposure of the cells to light and oxygen for 16 hours, conditions which in the absence of vitamin E regime would have led to totally budded cells, gave only a small number of budded cells." He recommends "enhanced intake" of vitamin E in the presence of environmental oxidants such as smog and cigarette smoking.

Vascular-Renal

Prevention of Cardiovascular Disease: An Urgent Canadian Problem

Milliken J. A. and Rautaharju P. M. (Canada)—C.M.A.J. 114:185, 1976

Cardiovascular disease causes over half the deaths in Canada each year. How much of this is the result of misplaced priorities?

We are too oriented to hospital care and established disease in our current health care. We do not stress prevention. Indeed, we hardly pay it even lip service.

Over three million Canadians must be hypertensive; less than 50 percent are aware of their disease; less than 20 percent are receiving adequate treatment.

We need individualized care to prevent strokes, kidney failure, heart failure and heart attacks.

One of every four myocardial infarctions is silent or is uncovered only by the E.C.G. However, cardiologists often find it difficult to assess borderline ST segment abnormalities in electrocardiograms.

In view of the epidemic of cardiovascular disease in this country and the United States, should we wait longer? Detection and prevention programmes could be established now.

We should establish a "demonstration" study inside an organized health care delivery system to evaluate the new technology for the prevention of cardiovascular disease.

We are being left far behind in the prevention of cardiovascular disease.

Comment: *Yes—until we all use alpha tocopherol. Like other food factors it prevents what it relieves.*

Influence of Vitamin E on Platelet Aggregation and Thrombocythemia in the Rat

Machlin L., Filipski R., Wallis A. et al. (U.S.A.)—Proc. Soc. for Exp. Biol. and Med. 149:275, 1975

Platelet aggregation was less in blood of animals receiving vitamin E than in blood from vitamin E-deficient rats.

Animals fed vitamin E had a lower platelet count. "The increase in the count of vitamin E-deficient animals could contribute to increased aggregation."

Controls:

Editorial—Lancet, 1:347, 1976

Comment: *Very important article. It suggests that much postoperative deep vein thrombosis is really the tail end of preoperative deep vein thrombosis. This opens up a hope that widespread use of alpha tocopherol taken as a prophylactic before operation might get rid of much trouble in good time. [See Oschner corroboration, page 43.]*

*Muscle-Nerve***Therapeutic Value of Vitamin E on Postherpetic Neuralgia**

Kamimura M., Ohtsuki K., Imura S. (Japan)—*Rinsho Derma* (Tokyo) 17:545, 1975

The authors have treated 9 patients suffering from postherpetic neuralgia with vitamin E (alpha-tocopherol nicotinate and acetate), administered orally, with gratifying results. Of the 9 patients, 7 had excellent results (in 7 and 14 days no neuralgic symptoms), and 2 had had good results (in 14 and 21 days needing only slight anesthesia). Consideration is given to the mechanism of the therapeutic effect of vitamin E, especially alpha tocopheryl nicotinate.

Prevention and Treatment against Postherpetic Neuralgia by Vitamin E Ester

Mitsuo Kamimura, Kuniji Ohtsuki, Ryoza Takahash and Harumitsu Imura (Sapporo Medical College, Sapporo, Japan)—personal communication—Dec. 1976

Twenty-eight postherpetic neuralgia and 111 herpes zoster patients have been treated by vitamin E esters. The very excellent effect by giving dl-alpha tocopheryl nicotinate was noted: no side effects. The authors speculate that the accelerating effect of vitamin E on the microcirculation and less local anoxia are helpful for this disease and neuralgia of other types.

The Effect of Vitamin E on the Toxicity of Alkyl Mercurials on Nerve Tissue in Culture

Kasuya M. (Japan)—*Toxicol. and Applied Pharm* 32:347, 1975

The results are in accord with an interaction of alkylmercurials with cell membranes of nervous tissue. This produces a degenerative change in the cells.

Many workers have noticed that vitamin E stabilizes hepatocellular plasma membranes, erythrocyte and leucocyte membranes, and lysosomal membranes. Furthermore, vitamin E may form a stable complex with membrane phospholipids containing arachidony resi-

dues, and provide a mechanism for anchoring vitamin E in biological membranes.

In Minamota disease the results now suggest that vitamin E may be one of the important agents modifying the toxicity of mercury compounds. The vitamin seems to shift the dose-response relationships of living tissue and these toxic substances achieve a more tolerant state. [*Note: First evidence that Vitamin E protects against mercury poisoning.*]

Biophysical Studies Help Explain Muscle Disease **Medical News—J.A.M.A. 235:239, 1976**

Myotonia and progressive muscular dystrophy occur spontaneously in the quokka, a rat-like Australian marsupial, and both conditions can be reversed by giving the animals vitamin E. [*Note: First evidence of Vitamin E's positive effect on muscular dystrophy, [MS].*]

Skin

Epidermolysis Bullosa: Treatment with Vitamin E, Preliminary Results

Degreef J. and Flour M. (Belgium)—Arch. Belges de Dermatologie, 30: 2: 83, 1974

Treatment with Vitamin E (alpha tocopherol) is related.

The first dose was (chosen empirically) 300 mgm daily. This was doubled each two weeks until an optimal effective dose was reached. In addition to the usual seasonal fluctuations in the course of the disease the results seemed to be dose and time dependent.

The effective daily dose ranged from 1,200 mgm to 2,000 mgm, with a higher dose required in summer.

Following an interruption of treatment there seemed to be a latent period before a repeated high dose became effective.

The following preliminary results of tocopherol therapy were noticed:

The blisters were less numerous, smaller, and healed quicker; the patients resisted mechanical trauma better (activity, occupation, games, manual labour); relatives who suffer from the disease use the same treatment, which points to, at least, a subjective conviction of improvement.

*Basic Studies***Extension of Lifespan of Cultured Normal Human Diploid Cells by Vitamin E**

Packer L. and Smith J. R. (U.S.A.)—Proc. Nat. Acad. Sci. 71:4763, 1974

Inclusion of synthetic vitamin E (d1 alpha tocopherol) in a culture medium for human diploid cells greatly prolongs their lifespans in vitro. The addition of 100 mgm of d1 alpha tocopherol per ml of medium has allowed the authors to date to culture cells for more than 100 population doublings. (These cells normally have an in vitro lifespan of only 50 ± 10 population doublings.) Cells at the 100 population doubling still have a normal diploid karyotype, and appear to behave in all respects like young cells, indeed, are still actively dividing. This supports the free radical theory of aging.

Hayflick has reported that normal human fibroblasts are capable of 50 ± 10 population doublings when cultured in an adequate environment. The authors have been able to grow cell cultures to 100 population doublings in the presence of 100 mcg of d1 alpha tocopherol per ml of medium.

At the 100th population doubling level these cultures still have the normal karyotype and are growing as rapidly as cultures at the 20th population doubling level. The percentage of cells capable of synthesizing the DNA is equivalent to cultures with more than 70 percent of their in vitro lifespan remaining.

Human fibroblasts may be capable of a very large number of divisions, perhaps an indefinite number.

The present investigation suggests that senescence in tissue cells is associated with oxidative damage occurring during their proliferation in culture.

The cultures treated with 10 and 100 mcg of tocopherol per ml of medium stopped growing at the 112 and 117 population doubling levels respectively.

Cultures from which the tocopherol was removed at the 73rd population doubling level were able to undergo essentially the same number of population doublings as mentioned above.

Effect of Tocopherol on the Growth and Development of Rotifers

Gilbert J. J. (U.S.A.)—International Symp. on Vitamin E, held in Minneapolis, U.S.A., September 27 and 28, 1973, p. 1005

The author's theory is that alpha tocopherol may be essential for male fertility in this rotifer as it is in some mammals and in some

insects. It insures that when males are produced their mothers will have a sufficient supply of this vitamin for their sperm.

Protection Effect of Oral Alpha Tocopherol in Rats Exposed to Toxic Levels of Ozone and Nitrogen Dioxide

Tappel A. L. and Fletcher B. L. (U.S.A.)—Environmental Res. 6:165, 1973

All antioxidants protective here, as well as alpha tocopherol. Alpha tocopherol is protective against lipid peroxidation. The amount of this peroxidation in the lung was proportional to the log of the dietary alpha tocopherol.

Rates of Reaction and Quenching of Alpha Tocopherol and Singlet Oxygen

Foots C. S., Ching Ta-Yen and Geller G. G. (U.S.A.)—Photochem. and Photobiol. 20:511, 1974

The total rate of scavenging ($6.7 \times 10^3 \text{ M}^{-1}\text{s}^{-1}$) by alpha tocopherol makes it an effective protective agent against photooxidation mediated by single oxygen.

Age, Parity and Vitamin A Supplements, and Vitamin E Requirement of Female Rats

Ames S. R. (U.S.A.)—Am. J. Clin. Nut. 27:1017, 1974

A large increase of vitamin E intake is needed as rats age—this is not due to decreased absorption but to differences in appetite and activity.

Blood Levels of Alpha Tocopherol in a Disorder of Lipid Peroxidation: Batten's Disease

Siakotos A. N., Koppang D. V. M., Youmans S., and Bucana C. (U.S.A.)—Internat. Symp. on Vitamin E held in Minneapolis, U.S.A., Sept. 27 and 28, 1973, p. 1152

Batten's disease is characterized by the onset of progressive mental regression and seizures. The brain cells of patients with this disease show the accumulation of large masses of lipopigment, ceroid.

Studies have established that the accumulated lipopigment, ceroid, is clearly different from "age" pigment, or lipofuscin in a number of parameters.

The standard treatment for this group of disorders is the administration of a mixture of vitamin E ascorbic acid, methionine, and butylated hydroxylated toluene. Patients on this therapeutic regimen have shown some improvement.

Obviously this "peroxidative disorder" progresses in spite of an ample supply of the biological antioxidant, alpha tocopherol. If alpha tocopherol is a significant factor in the inhibition of peroxidative damage, then alpha tocopherol may not act exclusively as an antioxidant. This had been proposed by a number of other investigators.

Vitamin E Holds Hope for Victims of "Mediterranean Anemia"

Editorial—Prevention, 28:86, 1976

For some as yet unexplained reason the red blood cells of thalassemia sufferers are abnormally thin and brittle.

Destruction of red blood cells in these persons occurs faster than the body can produce them. Severe anemia develops. Jaundice, leg ulcers and gallstones also occur. The spleen and the liver become abnormally enlarged, as they must remove useless blood cells from the system. And because the bone marrow must work more to replace the destroyed erythrocytes, the bones become enlarged and malformed. "Mongoloid" facial deformities are common. Osteoporosis weakens the bones even further. Frequent transfusions cause large amounts of iron to be deposited in the heart, enlarging it abnormally. Together with the anemia, this can result in heart failure and death. Sufferers of thalassemia rarely live beyond puberty.

Vitamin E can almost normalize the texture of the red blood cells weakened by thalassemia.

Treatment in such cases is begun with 750 I.U. of vitamin E continuously each day for three to six months. This treatment results in a doubling and tripling of vitamin E levels in the blood of the supplemented patients.

Compared to the thalassemia patients who had not received vitamin E the supplemented group's score was near normal.

Vitamin E

Bieri, J. G. (U.S.A.)—Nutrition, Rev. 33:161, 1975

Clinical evidence of vitamin E deficiency is almost entirely restricted to premature infants.

Current information indicates this anemia in premature infants is hemolytic.

Adult men force-fed for three years displayed no symptoms even though plasma tocopherol fell to very low levels. A slight decrease in red cell life span was found, but there was no anemia. Vitamin E deficiency occurs in individuals who, for various reasons, have a defect in their ability to absorb dietary fat.

Blood levels of a-tocopherol may not reflect accurately for level of intake tissue storage.

But plasma tocopherols correlate well with plasma cholesterol or total lipids.

Plasma tocopherol values may have little meaning if not accompanied by values for plasma lipids. The ratio may indicate plasma total tocopherols below 0.5 mgm per 100 ml.

In the presence of adequate selenium, vitamin E status did not affect glutathione peroxidase in liver, kidney, and lung. But lower activity was found in adipose and muscle tissue from alpha tocopherol deficient animals.

In man, absorption of 20 to 30 percent of small doses of alpha tocopherol was found by lymphatic cannulations. Although gamma tocopherol is initially taken up by the tissues as efficiently as is alpha tocopherol, it turns over more rapidly. An exception lies in fat which progressively accumulates the vitamin. When the vitamin was removed from the diet almost one-half the alpha tocopherol in liver and heart was lost inside of two weeks.

The relatively nonlabile fraction may represent alpha tocopherol in cellular membranes.

Reduced activity in deficient rats is restored to normal 12 hours after an oral dose of vitamin E.

Vitamin E is not directly involved in the electron transport mechanism.

Supplementary alpha tocopherol should decrease toxicity from a high oxygen atmosphere.

Similar protection from ozone and nitrous oxide has been demonstrated also.

Claims for dietary supplements of vitamin E remain unsubstantiated and the Food and Nutrition Board has stated that such claims are inadequately documented. The general medical impression is that for most persons daily doses below 300 I.U. are innocuous.

A recommended dietary allowance (R.D.A.) for vitamin E was first set in 1965. But a fixed allowance for any age and sex group is not realistic. A range of 10 to 20 I.U. of total vitamin E activity should be present in adult diets supplying 1,800 to 300 kcal. Generally, however, as the PUFA in diets increases the vitamin E follows suit.

Typical balanced diets of about 2500 kcal. average from 6.4 to 9.0 mgm (9.5 to 13.4 I.U.) of alpha tocopherol daily. U.S. diets now contain considerably more gamma tocopherol than alpha tocopherol, due to the trend to substitute vegetable fats and oils for animal fats.

The predominant vegetable oil, soybean oil, contains about six times more gamma than alpha tocopherol. Even though gamma tocopherol has only about 10 percent of the biological activity of alpha tocopherol, the large amount of gamma tocopherol in diets was estimated to contribute as much as 20 percent of the total vitamin E activity. Detailed analyses of whole cereal grains found that barley, oats and rye contain from 0.9 to 1.7 mgm per 100 g of other vitamers, alpha tocotrienol and beta tocopherol, which are about 40 percent as active as alpha tocopherol.

In calculating the vitamin E content of mixed diets, values for alpha tocopherol should be multiplied by a factor of 1.2 to give more accurate estimation of their total vitamin E activity, expressed as alpha tocopherol equivalent.

When the hydroxyl group is replaced by a methyl-amino group these analogues of beta and gamma-tocopherols become fully as active as alpha tocopherol and are stored in tissues as well as or better than alpha tocopherol.

Vitamin E

Tappel A. L. (U.S.A.)—Nutrition Today 8:4, 1973

More than four hundred papers have been published about vitamin E in the past year alone.

Since it is found in nearly every food we eat it is not likely that a vitamin E-deficiency could arise, says Tappel. In this respect vitamin E stands alone. Even boiling it at 100°C, does not affect the potency of vitamin E.

Many months of deprivation would be needed to deplete the body stores. Indeed, ten times the R.D.A. is required to double the tissue concentration of vitamin E.

The clues to the activity of vitamin E within cells suggest that we are dealing with a substance of great potential. Cellular membrane, endoplasmic reticulum, and mitochondria often comprise over one-half of the cell mass. In these and other membranous parts, vitamin E interdigitates with phospholipids, cholesterol, and triglycerides, the three major structural elements of membranes. In the mitochondrial membrane, the ratio of vitamin E to polyunsaturated fatty acids (PUFA), which are mainly in the form of phospholipids, is about 1:1000.

Vitamin E spares polyunsaturated fats from oxidative deterioration and the other parts of the cell and tissues from damage by free radicals.

The pigments in aging cells are variously called age pigments, lipofuscin, or ceroid.

Vitamin E protects vitamin A, which tends to oxidize freely. Vita-

mins A and E also act together in affecting membranous parts of the cells.

The similarity between the nutritional effects of vitamin E and very small amounts of selenium in our food has been puzzling.

The role of the sulfur amino acid, cystine, as a constituent of glutathione probably underlies the nutritional relationships among these sulfur amino acids, selenium, and vitamin E.

The oxidative production of prostaglandins apparently can be inhibited by vitamin E and the related glutathione peroxidase system. Inhibition of prostaglandin production could account for anti-inflammatory effects of high levels of vitamin E and of similar anti-inflammatory properties of selenium.

Vitamin E may protect against lung damage by oxidant components of air pollution. Thus vitamin E and the related protective systems now appear to be very important defensive systems against oxidant insult to lungs via air pollution.

It has been found that nitrogen dioxide caused damage to lung polyunsaturated fats, and that large doses of vitamin E provided protection against such damage.

Probably, for adults, a level in the range of 20 to 30 I.U. per day could be recommended for those exposed to photochemical smog.

Horwitt concluded that the polyunsaturated fat intake was the most important determinant of vitamin E need and that may range from 5 mgm per day to as high as 30 mgm per day.

Some oils do not contain sufficient vitamin E to balance their polyunsaturated fat content. The assessment of dietary intake of vitamin E from food tables can lead to important error. Processing and storage losses in vitamin E can be significant. If prepared foods are held in frozen storage large losses of vitamin E arise. In general, losses of vitamin E from vegetable oils or in foods deep fat fried in vegetable oils are caused by oxidation of vitamin E. Other important losses of vitamin E occur in the processing of wheat and other grains, especially in milling. This removes the wheat germ, and bleaches to produce white flour.

Blood levels in adults in the U.S.A., Canada, and Europe have remained essentially the same for the last three decades.

Both premature and full-term newborn infants may have relatively low tissue and blood levels of vitamin E but cow's milk is lower in vitamin E. Milk formulas may be based on cow's milk or simulated milk that contains polyunsaturated fats in the form of vegetable oils and formulas often contain added iron that can have a pro-oxidant effect.

Iron supplementation alone is not efficient in prevention of infant anaemia during the first ten weeks of life. Iron apparently accelerated polyunsaturated fat oxidative deterioration and the red blood cell membrane was weakened with a resultant greater trend toward hemolytic anaemia. However, in the infants with vitamin E-supplemented formu-

las, the red blood cell membrane was protected and the infants showed higher hemoglobin levels. Fifteen infant formulas were examined and it was found that nine had vitamin E to polyunsaturated ratios of less than 0.4. The need for vitamin E at nutritional levels is related to malnourishment, poor diets, or malabsorption problems, found in humans with cystic fibrosis, liver cirrhosis, post-gastrectomy, obstructive jaundice, pancreatic insufficiency, and sprue.

Vitamin E probably has no effect on coronary heart disease. Apparently, however, there will be reassessments of any potential role of vitamin E in the prevention and treatment of the cardiovascular diseases.

Effects of Alpha Tocopherol on the Rate of Photohemolysis of Human Erythrocytes

Bland J. et al. (U.S.A.)—Physiological, Chem. and Physics, 7:69, 1975

Vitamin E in vivo prohibits production of cholesterol hydroperoxide. It also reduces the degree of cell membrane deformation when the blood is exposed to light and water. It prohibits the production of cholesterol hydroperoxide which has an application in electron microscopy (the "budded" cell).

Does Vitamin E Slow Cellular Aging?

Bland J. (U.S.A.)—Prevention, 28:80, 1976

Human erythrocytes can be "aged" rapidly by exposure to light and oxygen, producing a strange "budded cell" type. Exposing red blood cells to ozone causes them to be very fragile as a result of cell membrane destruction. There is not merely one membrane surrounding the cell, but a number of them. These membranes actively transport all the vitamins, minerals and trace metals, nutrients, hormones, and other chemical messenger substances on which the cell depends. These vital substances don't merely "ooze" through the cell membrane to reach the place where they are needed. There is a mechanism in which the membrane actively selects what it is going to take in and what it is going to expel. There may be particular "ports" on the cell membrane. Indeed, these may be programmed to "recognize" various substances—like molecular doormen.

In diabetes it appears that the cell membranes cease to take up glucose. In cancer, it seems that something happens to the cell membrane which prevents abnormal cells from being attacked and destroyed by the immune system.

Such antioxidants as vitamin E reduce the rate of "peroxidation" or oxidative damage due to chemicals such as the ozone in smog.

Vitamin E has a great affinity for cell membranes, because these membranes contain large amounts of unsaturated fatty acids and other fats and vitamin E is fat soluble.

The cell membrane is a kind of sandwich of fat-like molecules with cholesterol lying between them and selected proteins trapped there. Vitamin E sits in this fatty bilayer and protects its fats.

Exposure of cells to light and oxygen for 16 hours, conditions which in the absence of vitamin E regimen could have led to totally budded cells, gave only a small number of budded cells. Human cell cultures grown in a medium containing large amounts of vitamin E were able to divide and live much longer than cells grown in ordinary cell cultures. There may be an optimal level of vitamin E in the red blood cell to lend it maximum protection against membrane destruction (between 400 and 600 units per day).

Other factors lead to increased cellular oxidation, too, and should be countered with even greater intakes of vitamin E! It is incorporated predominantly in the fatty acid-rich cell membrane and inner membranes.

Vitamin E reduces the rate at which cholesterol, a very important component in the red blood cell membrane structure, is peroxidized by oxidants to a product known as cholesterol hydroperoxide. On 600 units of vitamin E a day, this peroxidation of the blood cells of distance runners is decreased by as much as 50 percent.

Influence of Vitamin E on Platelet Aggregation and Thrombocythemia in the Rat

Machlin, L. J., Filipinski R., Willis A. L., Kuhn D. C. and Brin M. (U.S.A.)—Proc. Soc. Exp. Biol. and Med. 149:275, 1975

Platelet aggregation was less in the blood of animals receiving vitamin E than in blood from control animals. In older animals the platelet count was higher in deficient animals than in those fed vitamin E. The increase in count could be due to increase in aggregation. The reduction of aggregation by vitamin E could be related to an antioxidant role in prevention of peroxidation of arachidonate.

Are the effects of vitamin E on platelet aggregation and platelet count involved in the etiology of vascular pathology found in vitamin E deficient conditions or in the development of human thrombosis?

After prolonged deficiency (15 weeks) both platelet aggregation and platelet counts were elevated in vitamin E deficient animals. [*Note: Again, this proves precedence by Machlin.*]

Readers' Report on Vitamin E in Patients 70 to 79 Years Old

Passwater R. A. (U.S.A.)—Prevention, 28:61, 1976

A questionnaire was sent out to nearly 20,000 people of all ages who had taken vitamin E for years. The trend which emerged from analyzing the 4,060 replies in this age group is that people who have taken 300 I.U. or more of vitamin E daily for ten years or more are "not likely to develop heart disease." Neither are people taking at least 1,200 I.U. daily for more than three years.

"The statistics indicate that there is only one chance in a thousand that the observed association could be in error, and that there is a 90 percent positive association (not causation) between increased vitamin E dosage and time-usage with freedom from heart disease."

(For those interested in statistics, the Spearman-Rho test shows rho is 0.92 where the significance required is only 0.01, and the Kendall-Tau test shows tau to be 0.89).

Vitamin E—Nature Feeds Us

Shute E. V. (Canada): Modern Press, Saskatoon, Canada, p. 169

Alpha tocopherol really is "the most versatile of all the vitamins." As I said at the Venice Symposium years ago, add up all your needs in a heart and vascular drug and you will come up with a substance identical with vitamin E.

We do not think of it as replacing vitamin E deficiency, although it does. But it can be used in "normal" persons with burns or phlebitis, for example, without demonstrating a deficiency before using it. People take digitalis for heart lesions without being deficient in digitalis and quinine for malaria without first being shown to be quinine deficient. So it is with vitamin E. It can be and is most often used as a specific drug, and, thus used, its dosage level is usually high.

Vitamin E: A Reexamination

Horwitt, M.K. (USA)—Am J Clin Nutr 29:569, 1976

There is now evidence to support investigators who claimed that supplementation with vitamin E effects coagulation.

Although one must be concerned with the levels at which vitamin E is safe, so much has been used by so many for so long that one may be certain that supplementation with 750 mg/day rarely has any toxic effect. There will

always be some physiological effects. A study conducted at the National Institutes of Health of individuals taking 800 units over an average of three years found no evidence of toxicity. Considered judgment of any prescribed level of warfarin is recommended in the patient supplemented with vitamin E.

There is good reason to believe that cardiologists could use vitamin E to replace aspirin. Vitamin E could prove useful in decreasing the risk of thrombosis in thousands of women now using oral contraceptives.

[Note: This establishes Dr. Horwitt's "change of mind" regarding Vitamin E's efficacy in many diseases. H.B.]

Appendix B

(Chapter 1)

Health Freedom News, March 1993

Vitamin E Protects Against Heart Disease

By Dr. Richard Passwater

A survey of 18,000 readers of *Prevention Magazine*, performed by our research group in 1974, and published in 1976, showed that Vitamin E may reduce the risk of heart disease by as much as 80%(1). At the most recent American Heart Association meeting in New Orleans, two Harvard studies were presented that showed very similar results. I smile a lot these days thanks to the recent confirmations of my research on heart disease, cancer prevention, cholesterol irrelevance, and the trans-fats in margarine making it inferior to butter.

In 1991, Dr. Fred Gey published his study showing that a deficiency of vitamin E is the major and most reliable indicator of heart disease risk(2).

It wasn't until later that Vitamin E was established as the protective mechanism that protects the particles that carry cholesterol in the blood from being damaged by an undesirable reaction with oxygen. Studies presented to the American Heart Association showed that vitamin E prevents oxidized-LDL from forming. Oxidized-LDL is the only agent that contributes to developing atherosclerotic lesions commonly called

“cholesterol deposits.” Neither LDL nor cholesterol form these atherosclerotic lesions, and, thus, do not play a major role in heart disease(3).

Two Harvard studies presented to the American Heart Association’s annual scientific session in New Orleans in November, 1992 confirmed Passwater’s study. Here are the major points of all three vitamin E studies.

1. That vitamin E supplements (the amount of vitamin E in the diet is not sufficient to produce a significant protective effect) dramatically and significantly reduce the incidence of heart disease.

2. That the protective effects of vitamin E supplements is greater than those obtained by reducing dietary cholesterol, dietary fat, blood cholesterol, and other (currently popular) approaches.

3. That the length of time of supplementation is critical, and that at least two years of supplementation is required before substantial benefit can be measured.

RESULTS:

Dr. Stampfer’s group at Harvard found that women who took more than 100 IU of vitamin E daily (time period not specified) had a 27% lower risk of heart disease (4).

Dr. Rimm’s group at Harvard found that men who used more than 100 IU of vitamin E daily (time period not specified) had a 37% lower risk of heart disease (5).

Dr. Passwater’s study showed that those who took even more vitamin E over the long term produced even greater results. The amount of heart disease in any age group decreased proportionately with the length of time that vitamin E had been taken. In fact, the length of time was more important than dosage after a minimum of 400 IU daily was taken.

In all instances where persons consumed 400 IU or more daily of vitamin E for more than 10 years, their rate of heart disease was significantly lower than normal (3 per hundred, compared to 32 per hundred). Those who took 1,200 IU or more of vitamin E daily for four or more years reduced the incidence of heart disease from 32 per hundred to 10 per hundred.

STUDY DETAILS

	Passwater(1)	Stampfer(4)	Rimm(5)
Number of persons	17,894	87,245	51,529
Sex	M & F	F	M
Ages	50-90		
Subject group	Prev. Readers	Nurses	Health Pros
Study period	retrospective	8 years	5 years
Yr. study begun	1974	1984	1987
Yr. study presented	1976	1992	1992
Corrected for	Age & Sex	Many	Many
Possible bias	self selection	participation	participation

The participants in this study were also more likely to consume significant amounts of other antioxidant nutrients which would enhance the synergistic effect.

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Dr. Richard Passwater is highly respected in the preventive health field. His new book, "The New Supernutrition" (Pocket Books, NY) is recommended reading for everybody; the book received much praise in Health Freedom News, and received a Four-Star rating in the Preventive Medicine Update.

Medical Tribune, November 26, 1992

Vitamin E May Cut Heart Risk

Use for at least two years reduced risk in women by 46%, men 26%

By L.A. McKeown

NEW ORLEANS—Taking supplements of vitamin E for two or more years may significantly reduce the incidence of cardiovascular disease in healthy individuals, according to results presented at the American Heart Association's 65th Scientific Sessions here.

In two studies of over 130,000 people with no history of cardiovascular problems, vitamin E lowered heart disease risk in women by 46%

and in men by 26% when the vitamin was taken in the form of supplements of 100 IU or more per day for at least two or more years.

People who took megadoses of the vitamin had no greater reduction in risk than people who took 100 IU. The protective effect of vitamin E appeared to be independent of beta carotene or vitamin C.

The data on women were taken from the Nurses' Health Study and were reported by Meir Stampfer, M.D. of Harvard School of Public Health. The study has followed 87,245 women ages 34 to 59 since 1980. The men's data were reported by Eric B. Rimm, also of Harvard. That study has followed 45,720 men ages 40 to 75 since 1986.

Dr. Stampfer expressed surprise at the findings and said while the antioxidant properties of vitamin E are promising, he is not prepared to recommend universal vitamin E supplements.

Co-investigator Charles Hennekens, M.D., agreed. "The only way to find out if vitamin E is truly beneficial in reducing risk is by conducting large-scale randomized trials."

The Women's Health Study, which is just beginning, will compare the preventive benefits of vitamin E, beta carotene and aspirin use in 44,000 women. Dr. Hennekens said that study should result in the most reliable data to date.

Victor Herbert, M.D., scientific editor of the *Mt. Sinai School of Medicine Complete Book of Nutrition*, said he thought it was reasonable to suggest that vitamin E supplements are protective against cardiovascular disease, but said more data is required to show an overall trend toward decreased morbidity and mortality.

Let's Live, April 1993

History—And Humans— Repeating Themselves

By Jonathan Wright, M.D.

In 1633, Galileo Galilei was found guilty by an official Church court of having taught heresy when he said that the earth orbits the sun. Ignoring the evidence provided, his inquisitors demanded that he recant.

In 1992, the Church formally announced that Galileo Galilei had been correct, thus bringing the Church into official agreement with every schoolchild's textbook.

In 1952, Doctors Evan and Wilfrid Shute were summoned before a committee of the College of Physicians and Surgeons of the province of Ontario. Ignoring the evidence provided, the committee declared: "The committee is convinced that vitamin E has no place in the treatment of cardiovascular disease."

In the late 1930's, the Shute brothers had begun working with vitamin E. Their work was interrupted by and resumed after the war, but, by the early 1950s, they had successfully treated thousands of individuals with heart and blood vessel disease. By 1952, there were 46 published scientific reports supporting their findings about heart disease, atherosclerosis, and vitamin E.

A group of grateful patients of the Shutes organized the Cardiac Society to publicize the doctors' work. From an office in Detroit, they sold the Shutes' book, as well as vitamin E capsules. The American Medical Association persuaded the U.S. Postal Service not to deliver letters sent to the Cardiac Society, getting them to return them, marked "FRAUDULENT."

In 1991, the World Health Organization released a report which found that low blood levels of vitamin E are a more important risk factor for death from heart disease than high cholesterol or high blood pressure. The report stated that low vitamin E levels were found in 62 percent of men who died of ischemic (lack of blood flow) heart disease.

The headline in *Family Practice News* (January 15, 1993, p. 25) read, "Vitamin E Supplements Tied to Lower CHD Risk in Large Studies." Quoting word for word, from the top:

Vitamin E supplements appear to greatly reduce the risk of coronary heart disease, according to the results of two huge prospective epidemiologic studies presented at the annual scientific sessions of the American Heart Association. Dr. Meir J. Stampfer reported on more than 87,200 women, ages 34-59, who were followed for eight years in the Nurses Health Study; all women were free of heart disease at the outset of the study. Those who took vitamin E supplements had a 38 percent lower risk of developing coronary heart disease during the study period after statistical adjustment for age and heart disease risk factors.

Among women who took supplemental vitamin E for more than two years, the risk of coronary heart disease was 46 percent less than in those who did not take supplements, added Dr. Stampfer, of Brigham and Women's Hospital and Harvard Medical School, Boston.

In a separate presentation, Eric B. Rimm, Sc.D., of the Harvard School of Public Health, reported on 45,720 men followed for an average of close to four years in the Health Professionals Follow Up Study. Those who took vitamin E supplements had a 24 percent lower risk of coronary heart disease, dropping to 37 percent in those who took 100 IU/day for more than two years.

Results of both studies held up after adjustments for intake of vitamin C and carotene, which, like vitamin E, are antioxidants. Vitamin E is thought to protect against coronary disease by preventing oxidation of LDL cholesterol into its more atherogenic form.

Dr. Stampfer said the cardioprotective effect was seen in sub-

jects taking more than 100 IU of vitamin E daily—more than can be obtained from dietary sources alone.

It should be quite obvious, even to the university scientists, that Drs. Evan and Wilfrid Shute were correct in their pioneering reports on vitamin E. It is sad that not one university researcher anywhere has publicly acknowledged that the Shute brothers were the true vitamin E pioneers. Although it is “correct” (and even obligatory) for researchers to note the work of predecessors in research reports, Evan and Wilfrid Shute continue to be snubbed and ignored by the “scientific community.”

We can only hope that it will not take 359 years for the Ontario College of Physicians and Surgeons (or the American Medical Association) to officially admit their error, and apologize to the descendants of the Shute brothers for all of the public criticism and mistreatment they provoked. Long before then, every schoolchild's textbook will state as simple fact that vitamin E supplements are key elements in the prevention and treatment of heart disease.

Dr. Wright is in private practice in Washington state. His Tahoma Clinic Nutrition and Allergy Hotline answers inquiries. Please call 1-800-825-8924 between 3 and 5 p.m., Pacific time, Monday through Thursday.

Journal of Orthomolecular Medicine, Vol. 7, No. 4, 1992

The True Cost Of Cynicism

by A. Hoffer, M.D., Ph.D.

Thursday, November 19, 1992, *New York Newsday* carried a report that *vitamin E had decreased the risk of heart disease between one-third and one-half*. The studies reported were conducted at the *Harvard School of Public Health*. In one study, Dr. M. Stampfer found that *during an eight year follow-up, women who had taken at least 100 IU of vitamin E daily for two years had a 46 percent lower risk of having a heart attack*. This was based on a *population study involving 87,245 women*. The second study, on men, by Dr. E. Rimm, based upon *51,529 subjects*, showed a *37 percent lower risk*. They found that there was *not enough vitamin E in food*; Dr. Stampfer was so convinced by the data he is taking the vitamin himself.

These findings, of course, are not surprising to anyone familiar with the history of vitamin E and heart disease. I suggest that you read Dr. Shute's book. *Drs. Wilfrid and Evan Shute* began to treat large numbers of patients with megadoses of vitamin E, usually above 800 IU daily. Their clinic

had experience with perhaps 30,000 patients who came from all over North America to receive their treatment. Their work was a model of clinical research, but the idea was so novel that their work was discounted entirely and they were considered quacks for recommending these doses for a disease "known" not to be a vitamin deficiency disease. Fifty years ago, about the time they began their studies, hardly anyone knew what vitamin E was, and it was not considered important or relevant.

However, the medical profession, instead of doing its duty, which is to investigate claims made by reputable physicians and scientists, persisted in downgrading their work. In fact, one of the publications with the *Harvard* name on it, *The Medical Letter*, did a hatchet job on the Shutes' work many years ago. By doing so it effectively killed interest for many decades. In this *Medical Letter* they reviewed four studies published between 1940 and 1950 which they claimed were final definitive studies which proved that vitamin E had no therapeutic value for treating heart disease. I examined the four original papers, something which few doctors seem to have done. I found them to be inept, and so badly done that even journals willing to publish vitamin papers, would have rejected them. They did not follow the directions described by the Drs. Shute, using too little vitamin E for too short periods of time.

Now scientists from the same university report a 35 to 50 percent reduction in heart disease. It is important to think about this and to convert this into real costs.

Today, 40% of all deaths are caused by heart disease. Each day 2,000 people, or about 750,000 persons per year, die from heart disease. Let us assume that the reduction in risk is exaggerated, and that in reality there is only a 10 percent reduction. This means that each year about 50,000 fewer people would have died, a saving of about 200 patients daily. It is difficult to calculate overall how many would have been saved if the *Harvard* group had taken their responsibility seriously and examined the vitamin E claims in 1950 instead of waiting until 1992.

This is real cost of medical cynicism. Had they been merely skeptical they could have done the studies to satisfy their own curiosity, but they were so convinced the Shutes' findings were meaningless they went out of their way to destroy them. They succeeded. I wonder if anyone at *Harvard Medical School* today has given any thought to the cost of this type of inexcusable delay. This happened several hundred years ago when Sir James Lind proved that citrus fruit could prevent scurvy in British sailors. The Navy began to issue limes 40 years later. In the meantime 100,000 sailors died. This again illustrates the true costs of delay in examining seriously claims made by physicians.

The medical establishment consoles itself by claiming that the onus for proving new findings is on the original investigator. This is merely an excuse for doing nothing. The price is enormous. How much longer will society per-

mit doctors the luxury of doing nothing, especially when the suggested treatments are *safe, economical* and, in the opinion of doctors who follow these treatments, *so effective*? Harvard Medical School should be ashamed of itself, and *owes the American people an apology*.

The recent Harvard findings illustrate again the rapid advance of the orthomolecular paradigm in overthrowing the anti-vitamin paradigm—which has done so much harm in the past twenty years.

Medical Tribune, December 24, 1992

100 IU may help heart, but mortality data hasn't firmed up

Wanted: Stands On Vitamin E

By Bill Ingram

We want your stand on vitamin E because the supplements are so controversial.

Recent large, scientifically rigorous studies have suggested that at 100 IU, vitamin E can prevent or delay heart disease in both men and women.

Concededly, the eight-year studies have not as yet reached end-points. That is, not enough deaths have occurred to validate mortality data.

Conservative experts argue that the current recommended daily allowance of 30 IU should not be changed until these end-points are reached and until large controlled interventional trials confirm the findings.

Some authorities even support cutting back the RDA to 14 IU, closer to the amount necessary to develop the blood levels found in average, healthy Americans. The FDA is expected to propose such a cutback next spring.

Other authorities, many of them nutritional researchers, believe the RDA should be raised instead of lowered.

Siding with the liberals, about eight out of 10 doctor-respondents tell this newspaper they take E themselves and recommend it at 100 IU to their patients.

They cite effective use of the vitamin to treat fibrocystic breast disease and prevent decubitus ulcers, as well as to avert heart disease.

If you haven't declared yourself yet, please do so now by phoning our hotline at 1-800-937-8838.

Some doctors ask for solid data on E deficiency states and toxicity

risks. The National Research Council notes that E deficiency only occurs in very low birth-weight infants and in those with hepatobiliary and lipid transport abnormalities.

As for excess E, the FDA once calculated that a dose of 56,000 IU over six months would still be safe. The vitamin is excreted in LDL via the liver; some is absorbed by the intestine.

The Q & A is over. Hit the hotline and take a stance.

Medical Tribune, December 24, 1992

MDs Overwhelming For Vitamin E At 100 IU

By Bill Ingram

Scores of readers have responded to our query soliciting their views on vitamin E. Nearly eight out of 10 look favorably on the vitamin (and often on vitamin C and several other supplements).

Please join the issue on vitamin E now and phone your views to the hotline at 1-800-937-8838.

So far, virtually all those giving thumbs up are for the 100-IU dose, more than three times the RDA.

The unconvinced—the skeptics and cynics—have cited the sketchiness of the published data, a failure to detail the design of the two studies and the need for large clinical studies to corroborate the findings and completely eliminate confounding variables.

The sharp polarization is par for the course. Read what Linus Pauling has to say below about the longtime reluctance of pro-vitamin doctors to come out of the closet.

They want to avoid being labeled quacks, pseudo-scientists, rejuvenation wackos (Remember all the barns that used to trumpet “Dr. Pierce’s Golden Medical Discovery—Makes Red Blood?”)

One reader told us it’s as stupid to ask doctors whether they believe in vitamins as it is to ask whether they believe in religion.

Another said that to ask for a response based on such superficial data is “like something from USA Today.” But a third physician tells us she is going “to cut out the article, photocopy it and send it to all my patients.”

Frankly, since the pros so far outnumber the cons, we’d like to hear from more cons. If you traditionally wait until five years after THE DEFINITIVE ARTICLE appears in the New England Journal of Medicine to prescribe or recommend ANYTHING, tell us.

We had one internist confess, "Frankly, I don't want my colleagues seeing me spout off in a more or less unscientific area."

Hey, even people who bite their nails come out of the closet today. Come out, come out wherever you are. You can sign off as "Anonymous," you know.

Pauling not really surprised that so many doctors use E

It's interesting that *Medical Tribune* is getting a heavy, positive response from a broad group of doctors after asking whether they believe in vitamins and favor using vitamin E at three times the RDA.

In the current research you are citing, vitamin E was given at 100 IU dosage. For many years, I have been taking 800 IU daily (along with 18 grams of vitamin C).

I'm reminded of something that happened at a big party I attended at a Stanford University faculty member's home at least 25 years ago.

Something like 25 doctors were at the party. They were quite cordial, talking with me at some length about ongoing vitamin C research, but they really weren't expressing any personal opinions about ascorbic acid one way or the other.

And then I struck up a conversation with an assistant dean from the medical school.

He turned to me at one point and said, "I suppose you're wondering about all these doctors and what they really think about vitamin C. Let me tell you something. You'd be surprised at how many of them are taking it but won't admit it."

Now that was 25 years ago, but it strikes me that the situation with vitamin E now is similar to the situation with vitamin C then (except that doctors today appear more forthcoming, at least when asked).

And of course there's been a lot of research in between, preceding the recent Harvard studies.

Dr. James Henson and I published a paper 10 years ago on a followup of 700 elderly Californians who had answered a questionnaire on vitamin C and E. There was a statistically significant difference between the mortality rate of those who took vitamin E and those who did not. The difference with vitamin C was not as big and was not statistically significant.

I have heard there is a legislative move under consideration to make vitamins a prescription drug wherever they are available at doses of more than twice the RDA. Makers would have to get FDA approvals of the vitamins at the higher doses.

That would kill the whole thing. Nobody makes enough money to spend the amount needed to get FDA approval for this.

*Linus Pauling
Palo Alto, Calif.*

Each person should try to find optimal level in benefit

In reply to your query about vitamin E, I do believe in vitamins and for my own benefit, as well as my patients', I have been pulling lists of supplements out of the literature for the last 20 years.

I think the ideal dosage of vitamin E may lie between 400 IU and 2,400 IU. (The RDA is 30 IU). I take 2,400.

Most of the excellent investigational results seem to be achieved at 400 IU or less.

I'm familiar with vitamin E's mechanism of action as an antioxidant, i.e. surrounding the LDL molecule to prevent its attachment to the endothelium in the coronary artery. The vitamin inhibits or perhaps even reverses atherosclerosis.

One of the major problems in the conventional approach to supplements is always trying to utilize them with the idea of preventing a deficit. Instead, they should be utilized to maximize health—without going overboard.

The amount of (any particular) vitamin required to do that is probably 10 times the RDA.

I recommend supplements to patients at three levels. At the first or multivitamin level, I believe you can raise vitamin E up to more than 2000 IU, vitamin C up to the 2000-4000 mg range and beta carotene to 25,000 IU.

At the second level, I favor adding extra magnesium and an herb and at the third level, certain prescription drugs or "smart drugs" that may help mental functioning.

I believe there is a range of benefit that varies individually and each person finds their own optimal dosage. If it's 2,000 IU and you feel better, then you're not going to hurt yourself, though you can't tell somebody else they ought to take that amount.

*Donald C. Thompson, M.D.
Morristown, Tenn.*

With vitamins, it's more evaluation than belief

In the context of the November 26 article, "Vitamin E may cut heart risk," I find your question, "Do you believe in vitamins?" kind of funny.

I would be for recommending 100 IU of vitamin E, the amount taken in the Nurses' Health Study. But when the risk of heart disease appears to be reduced by 46% in women and 26% in men, it's merely a case of believing in things we're not sure about.

So it isn't so much a belief as a statistical measure. In traditional medicine, it shouldn't be a matter of believing or not believing something that was developed after \$300 million worth of study.

*Michael Burruano, M.D.
Brewster, N.Y.*

Using E and C, supposed to have heart surgery (haven't)

I've been using and recommending vitamin E to my patients for 16 or 17 years. Counting me, there are two doctors here in Fulton County (population, 40,000) who have been recommending vitamins all along.

I've also been taking vitamin C (as you know, Linus Pauling has been recommending it for a long time) and numerous others.

I was supposed to have heart surgery about six years ago (but haven't).

I believe the big mistake in the dosage of vitamins is the RDA. I don't think it's adequate at all.

I could talk to you for hours about this.

*Franz Strauch, M.D.
Canton, Ill.*

Believer says 'we probably need more than 100 IU'

Yes, I certainly believe in vitamin E and I think we should certainly be recommending that our patients use much more than 100 IUs a day.

Vitamins are involved in the (integrity and health) of cell membranes. Without them, no life could exist on earth.

*David Darbro, M.D.
Indianapolis, Ind.*

Can't judge the study results without knowing the design

My problem with your article about vitamin E is that it describes nothing about the study design.

All it tells us is how many patients were enrolled and how long they were followed.

We have no idea how the study was performed. We don't know anything about the results except that the risk was decreased by 46% (for women) or 26% (for men).

We don't know what the absolute value of the risk was, so we can't judge how important that decrease was.

I find it ludicrous that you've asked us to respond to this. It seems more like something from *USA Today* than a physician's newspaper.

*Chris Lahart, M.D.
Houston, Tex.*

Your criticism is unduly harsh, considering the size of the study, the clear statement of relative risk, the stature of the Walter Willett-directed group at Harvard and the ground that had to be covered in our 350-word report.

Both groups were free of heart disease at baseline. In the 87,245 women, 552 cases of CHD were confirmed during 8 years of follow-up. Relative risk for vitamin E users was 0.64 (95% CI 0.47-0.87) after adjusting for age, smoking, obesity, exercise and other CHD risk factors and multivitamin use.

With the men, during 159,923 person-years of follow-up, 880 cases

of CHD were documented (258 non-fatal MIs, 143 fatal MIs and 479 bypass surgeries or angioplasties).

After adjusting for intake of vitamin C and carotene as well as age, smoking, hypertension, diabetes and other risk factors, further adjusting for multiple vitamin use and exclusion of vitamin E users of less than two years duration, relative risk was 0.74 (95% CI 0.61-0.90).

Editor.

Definitely for 100 IU

In regard to the query, 'To E or Not to E?' I'm definitely in favor of vitamins and would definitely be for 100 IU a day of vitamin E.

Benjamin L. Gil, D.O.

Akron, Ohio

Take vitamins and haven't had a sick day in ten years

Yes, I do take vitamins, including vitamin E, a multiple vitamin and a copious amount of vitamin C.

I am in the medical field and I am presented with 100 different viruses throughout the winter and I have been working at the same office for 10 years and I have never had a sick day.

I also believe it has more to do with my attitude and my attitude is I do believe in vitamins. And if I believe in them, they will work.

My age is 50.

Jackie Sullivan

Medical Technician

Jupiter, Fla.

I've seen the 100-IU dose reduce cholesterol

I have used vitamin E for some 40 years and I have seen some remarkable things at the 100-IU dosage.

I have seen it reduce cholesterol from 608 to 110.

I have seen it reduce blood pressure over a period of 12 or 14 years to a point where a patient didn't have to take antihypertensive pills anymore.

One hundred units is a proper dose because the 100-unit form is said to be 18 times more potent than any other on the market.

Thomas O. Quinn, M.D.

Los Angeles, Calif.

Westerner thumbs up

I'm in favor of vitamin E.

Richard L. Gleason, M.D.

Oakland, Calif.

Another strong supporter

Just calling to say that I think vitamin E at 100 IU a day should be recommended.

Anonymous M.D.

Use it for heart—and PAD

I'm in family practice and I'm very much in favor of vitamin E for heart risk and peripheral vascular insufficiency as well.

*John Cunniff, D.O.
Burlington, Tex.*

Question data, but it's worth trying

I question the percentage of risk reduction in women and men and how investigators arrived at the figures. But I think it's worth trying.

*Jack Right, M.D.
Mesquite, Tex.*

Long Islander for 100 IU

I absolutely agree that taking an extra dose of vitamin E will make a difference and I recommend increasing the units to at least 100 IU a day.

*Robert Schoenfeld, M.D.
Plainview, N.Y.*

Medical Tribune, January 7, 1993

Doctor's Groundswell For Vitamin E At Megadoses

By Bill Ingram

In yet another round of communications, doctors have belted out a vitamin E affirmation with the decibels of a Broadway chorus singing Oklahoma. The vitamin definitely appears to prevent heart disease, they opine. More than half favor megadoses. Almost all take E themselves.

A couple of callers take strong exception to the wind roarin' down the plain, however.

Keep those hotline calls coming to 1-800-937-8838.

The Achilles' heel of all the favorable response, of course, is that the observations are anecdotal. Throw in the Hawthorn effect, the Kantian caution that "too often, in science, we see what we want to see," the usual shamanistic zealotry and you hit the wall of doubt.

Illinois' Dr. Acosta (see below) comments cogently that when added

to multifaceted lifestyle modification, vitamin E appears to make heart disease completely preventable.

“Aren’t you forgetting something?” asks James Lieberman III, M.D., chairman of family and community medicine at the University of Delaware. “It’s genes—my biggest protection. My 90-year-old father is currently irate because he can’t go out every day and play nine holes of golf.”

Okay. Add genes. Now we’re preventing 135% of heart disease!

Vitamin E doesn’t cut risk of cardiovascular disease

I don’t believe vitamin E reduces the risk of cardiovascular disease.

If we go by statistics, we reduce the CV disease risk by about 36% with Vitamin E. But according to your own newspaper, smoking cessation would cut the risk by 50%, (proper) diet and (controlled) alcohol intake by another 20-30% and the level of exercising (an additional amount).

And I am sure all Americans are doing one thing or another. So we have effectively eliminated cardiovascular disease, according to the statistics.

But according to the patient population in CCUs and hospital emergency rooms and doctors’ offices, we have not made a dent in cardiovascular disease.

*Sergio M. Acosta, M.D.
Elmhurst, Ill.*

We have made quite a dent in CV disease, actually, but that takes nothing away from the excellent point you make about extravagant claims for all the lifestyle components.

CV deaths dropped 24% from 1980 to 1988; the rate fell from 588 to 448 per 100,000.

The contribution of smoking to CV risk that you cite sounds inflated, relative to diet and exercise. But when it comes to overlapping risk factors, please be assured that the vitamin E investigators calculated risk reduction after adjusting for age, smoking, obesity, exercise and other CV risk factors.

It’s hard to believe that vitamin E’s apparent reduction of CV risk is an artifact.

Editor.

Considering a 400 IU dose for females, 800 for males

Yes, I do believe vitamin E prevents coronary problems. We’ve seen this in our own practice and in our own families.

A higher RDA than 30 should be recommended.

We’re thinking about 400 IU for females and 800 IU for males.

Anonymous M.D.

Already putting patients on it

I've been placing patients on vitamin E for the last two years and would applaud the studies on vitamin E as an antioxidant.

*William R. Craig, M.D.
Greenville, S.C.*

Med student backs 100 IU, glad for nutrition enrichment

I feel strongly that giving supplementation of 100 units of vitamin E a day is quite reasonable.

I also want to congratulate you guys on the excellent coverage you've been providing on nutritional issues recently. I'm in medical school, getting a total of 10 hours of formal education on nutrition; how pathetic that allotment has always been.

I think it's important because of all the recent research on nutrition affecting things that we have overlooked in the past.

I've been trying to keep up with the field and you've been helpful.

*Michael Carletons, M.D.
Santa Rosa, Calif.*

99% of those up there in age can benefit from 400-800

I'm a family physician who is into nutrition and I certainly vote for taking vitamin E.

The article on vitamin E and heart disease is revealing evidence (based on years of follow-up). So, certainly, extra E, 400-800 IU a day, can safeguard people with uncontrolled high blood pressure or people with severe heart valvular disease.

Other than that, 99% of people getting up there in age can benefit from it and take it safely and have no concern about toxicity.

*Paul Bobeleas, M.D.
Laurel, Md.*

Use it in hypercholesteremia

I use vitamin E in my practice with hypercholesterolemic patients (as well as generally) and I will continue using it, so clearly I think it is a good idea.

*Darryl L. Fortson, M.D.
Gary, Ind.*

Info needed on forms of E

I think vitamin E is a great idea. 100 IU or more daily is probably beneficial.

I think the form of E is probably important, too. Studies should be done on the various forms, specifically a nonsteroidified form.

*Richard Menashe, M.D.
Edison, N.J.*

Agree on dose in studies

I would agree with recommending at least 100 IU a day.

*Dwight McKee, M.D.
Aptos, Calif.*

Concur on positive effect

I do agree with the vitamin E article. I do believe that based on the data presented in these studies as well as others, the vitamin may have a positive effect in reducing cardiovascular risk of both men and women.

*William Boone, M.D.
Dallas, Tex.*

Helpful for arteriosclerosis

I do believe vitamin E is helpful for arteriosclerosis.

*Alan Magaziner, M.D.
Cherry Hill, N.J.*

Photocopying the article and sending it to all my patients

I want to compliment you. I think this is an outstanding article.

You're talking about preventive medicine, something that patients can do to help cut their heart risk.

I'm going to photocopy the article and send it to my patients.

I find this kind of reporting is so beneficial. We spend so much time taking care of problems instead of preventing them.

The other front-page article in the November 26 issue, "Fiber in fruit protects against hypertension," was also outstanding, as well as the story on vitamin A for prostate cancer.

There are ways that we can prevent disease and live healthier lifestyles that the medical profession must focus on.

The front-page stories not only caught my eye but made me read the whole newspaper. Normally, I wouldn't sit down and do that.

*Serena Friedman, M.D.
Canoga Park, Calif.*

More than 50% of illness remediable by nutrition

I think vitamins should be more available and that more information on them should be provided to physicians. That would help foster the expectation of physicians to in turn educate their patients about vitamin supplements.

I'm a family physician who works with nutrition. I have for 20 years and my belief is that 50% of illness is strictly caused by nutritional problems that could be remedied by diet and supplementation.

The ground in this country, the earth, is devitalized. U.S. Department of Agriculture research is showing that the produce has half or less the nutrition it did 40 years ago. I suppose you can say the more pollution, the less oxygen.

We need vitamins until we can clean up and renew the earth.

*Jess Handley, M.D.
Malibu, Calif.*

For doses higher than RDA

Regarding 'To E or Not to E,' I definitely believe that vitamins in doses greater than the recommended daily allowances have a positive effect.

*Pat Hennessy, M.D.
Columbia, Mo.*

100 IU not at all a large dose

Vitamin E 100 IU is certainly a small dose, even though it appears to be dramatically protective against heart disease.

That goes for the 44 mg (66 IU) dosage of vitamin E in a multivitamin, multimineral regimen given to the elderly, which reduced infection. The article was on page 4 of the same issue (Nov. 26) containing the vitamin E story.

*Phyllis Moeller, M.D.
Glendale, Calif.*

Favor use up to 400 IU a day

I'm heartily in favor of vitamin E supplements up to 400 IU a day. That's something we're recommending to patients in general in our practice.

While the clinical evidence is fairly weak, there's a lot of animal work suggesting that vitamin E's effect as a tissue antioxidant is quite prominent.

Since we live in an oxidant atmosphere in L.A., I think it's very reasonable to suggest small doses.

But I think doses greater than 800 IU are probably not indicated and may lead to myalgia and weakness syndrome.

*Charles Sharp, M.D.
Pasadena, Calif.*

Beta carotene; vitamin C, too

I have been recommending vitamin E 400 IU daily along with beta carotene 25,000 units and vitamin C, 1 g daily.

I think the antioxidant effect and the effect on the immune system will turn out to be beneficial.

*Steve Myerson, M.D.
Miami, Fla.*

A 'no' that leaves little doubt

I am against vitamin E period.

*K. P. Channabassappa, M.D.
Madison, Tenn.*

Recommend it to all adults

For the last two months, based on previously reported studies, I've been recommending vitamin E to all my adult patients in an effort to cut down on heart disease.

Of course, I will continue to do so. I'm happy to see that in your November 26 report, 100 IU or more seems to be helpful.

*Duane Lawrence, M.D.
Virginia Beach, Va.*

No downside to antioxidant

I've believed in antioxidants even before reading the vitamin E article. This just adds further evidence.

I believe there's essentially no downside to the use of these vitamins. I feel comfortable in advocating their use for the prevention of heart disease.

*Greg Brian, M.D.
Portland, Ore.*

Favorably inclined toward vitamin for some years now

I'm an internal medicine physician at a Kaiser HMO responding to the question: To E or not to E?

I believe that vitamin E has a beneficial effect as an antioxidant and I have been favorably inclined toward it for some years.

Moreover, I'm interested in more in-depth investigations regarding its benefits and how they are achieved.

*Thomas Spethmann, M.D.
San Diego, Calif.*

Long Island believer

Yes, I believe in vitamins.

*Penny Budoff, M.D.
Bethpage, N.Y.*

Use high-quality vitamin E, about 720 IU and emulsified

Yes, we agree that vitamin E is quite important in our practice and we use it every day.

We use a very good quality vitamin E.

It's emulsified and figures out to about 720 IU if taken every day.

We have worked with cardiovascular patients and we have had very, very good results.

The bottom line is that we highly recommend it.

We certainly agree with the report. Continue to write about vitamins and their effects in preventing disease. We've seen what supplementation can do for us.

Keep up the good work.

Anonymous M.D.s

Routinely recommend 400

I routinely recommend 400 IU daily of vitamin E.

*John Mundall, M.D.
Connell, Wash.*

Medical Tribune, February 11, 1993

FDA Criticized For Vitamin E, Food-Labeling Policies

By Bill Ingram

The vitamin E beat goes on. Those bongos just won't stop and 99% are positive.

In the latest series of anecdotes, E's use is associated with erythrocyte robustness that can be visualized in live-cell studies—but in an unusual twist, with fatigue by a disenchanted Californian.

Most respondents are convinced of and quick to cite the vitamin's cardioprotective effects. It gets credit for doctors and/or patients never missing a day's work, rarely even feeling punk and enduring long hours of daily practice.

A Texas old-timer swears by it at high dosage even though he had to have a CABG six years after getting on the supplement.

The FDA takes it on the chin for its RDA conservatism in several communications.

Doctors commonly report using vitamin E at 400-800 IU and claim it can benefit epileptics, women with fibrocystic breast disease and older patients with peripheral arterial disease.

Science is concededly lacking, but there's nothing like first-hand experience.

Just keep hitting that hotline at 1-800-937-8838.

FDA has proven a poor nutritional evaluator

There's a lot of data to suggest that vitamin E improves health and longevity. I think that cutting back the RDA would be bizarre.

Basically, I suspect the government is exhibiting the same attitude it has in remaining unconvinced that smoking causes medical problems (sufficient to curtail the makers). They (the government) continue to subsidize the tobacco industry and allow cigarette advertising that is available to children.

Considering that policy, I'm not at all surprised that they're thinking of cutting down the RDA. The FDA is really not equipped by knowl-

edge, temperament, or ability—and certainly not by its history—to evaluate nutritional treatments.

I think this function should be taken away from the FDA and given to an agency with the background and open-mindedness to perform the function.

*Jacob Teitelbaum, M.D.
Annapolis, Md.*

FDA has shelved the Reference Daily Intake idea, which would cut the RDA for vitamin E in half—from 30 IU to 14 IU. But the agency is dead set against increasing RDAs.

Editor.

And a poor label enforcer with cukes and apples

I'm sorry to learn that the FDA is trying to lower the maximum level of vitamin E when there's no definitive research of dangers at current allowance levels.

Yet the FDA doesn't enforce its own labeling laws, allowing wax on our cucumbers and apples. And the agency permits exposure of food to ionizing radiation, in many cases without labels. This, despite numerous studies linking irradiated food to increased morbidity and mortality.

Exactly whose needs is the FDA serving, anyway?

*Gary Gibbs, D.O.
Kirksville, Mo.*

Practically no downside

I do recommend vitamin E and I do use it myself.

I don't know if it's any good, but I can think of almost no reason not to use it.

The dosage I use is 400 IU a day.

*James Bernstein, M.D.
Rockville Centre, N.Y.*

You can see the difference in red cells' appearance

I have been giving vitamin E to patients for 50 years. It doesn't seem that a dosage of 1,000 IU it has hurt anyone yet.

Patients seem to feel they have more energy and don't tire nearly as quickly.

And you can see the difference histologically by placing a drop of blood on a slide, putting the cover glass over it and making "live cell" observations.

My microscope has a turret below the stage which has a phase for each objective. I like this much better than the traditional dark field. All cells can be seen more distinctly and the outer membrane can be readily viewed. Red cells appear about a third larger than when they are embalmed.

My understanding is that red cells transport oxygen to peripheral tissues and the more oxygen delivered to a muscle, the better it works.

Before I start patients on vitamin E, some appear tired, and upon examination, their red cells appear a little pale and the membrane abnormally thin.

A couple of weeks after starting to take 1,000 IU of vitamin E daily, patients feel better and the cells look a lot different—more uniform in size, larger, with much less rouleaux formation and a thicker cell membrane.

Of course I realize that how a patient feels or thinks he or she feels is not “scientific medicine” if there is no accompanying lab test to show something. But after seeing the red-cell changes, no wonder patients imagine they feel better.

In current practice, we have to follow the government cook book. That book does not include live-cell examination.

*Marvin L. Ford, D.O.
Elmo, Mo.*

A report just the opposite: It causes severe fatigue

I find it interesting that everyone is so strongly in support of vitamin E, with no side effects mentioned.

I want to report that I've taken 400 units on multiple occasions and after 10 days or so begin suffering from severe fatigue in the evening, to the point where I have to go to bed at 9:30 or 10 when ordinarily I never go to bed before 11:30 at night.

So I've stopped vitamin E. As to whether it's psychological, I would emphasize that on multiple occasions, I'll stay off it for a month or two and then try again, and after 7–14 days the same thing happens.

It takes only two days for me to feel normal again.

Has anyone else ever described fatigue as a side effect of taking 400 IU of vitamin E a day?

*Dr. Edward Stilken
San Diego, Calif.*

From Pine Tree State

I'm in favor of vitamin E. I think it is very useful and helpful.

*Dr. George Fontaine
Bangor, Maine*

Why not tote up the cost for neglecting to use it?

I'm an internist and I'm just calling to say that I think it would be ingenious to print the whole history of vitamin E and how it was available back in the 1950s and what the cost in terms of morbidity and mortality was from selectively not disclosing its availability and potential.

So much for your objectivity. So much for your medicine.

Anonymous

Louisianan an advocate

In regard to vitamin E, I do use it daily and I support it.

*Dr. K. M. Size
Kenna, La.*

Disease risk minimized

I wish to express agreement on the use of vitamin E and vitamin C in minimizing heart risk, minimizing cancer risk and minimizing amyloid deposits.

*Dr. Stanley Levy
Berkeley, Mich.*

Take it and recommend it

Number 1, yes, I take 400 IU of vitamin E daily. Number 2, yes, I recommend it to some patients, and number 3, of course I think it's beneficial with regards to aging and heart disease.

*Frank Schmidt, D.O.
Saranac, Mich.*

The heart of a 50-year-old

I'm a GP, 75, and very much in favor of vitamin E therapy.

I've been in the Harvard University physicians' study from the beginning and take aspirin and beta carotene on alternate days. Along with that, I take vitamin E 800 IU and vitamin C 1 gm. And I have not been ill in years.

My cardiologist tells me I have the heart of a 50-year-old.

*Grayson B. Davis, M.D.
Lafayette, Ind.*

Oklahoman's testimonial

I've been on 400 IU of vitamin E for 23 years and on an aspirin a day for eight years. In addition, I take Myadec, a multivitamin, and I must say that I practically never miss a day's work

I don't think there's any question but that E is definitely beneficial and I would recommend it for anybody.

*Don Cooper, M.D.
Stillwater, Okla.*

Appendix C**(Chapter 2)**

Conversation Between Herbert Bailey and Muhammad Ali**Via Telephone—August 29, 1974**

(Conversation opens with Mrs. Belinda Ali)

BA: Hello . . . Who is it?

HB: This is Herbert Bailey—remember, the author of the Vitamin E book?

BA: Right!

HB: How've you been?

BA: Fine!

HB: That's good! You know, Ali was talking with Harold Salkin, and I told Harold to tell him about the increased dosage before the fight, of Vitamin E

BA: Uh-huh . . .

HB: I wanted to make sure that he got that straight. You know, he's been taking the two thousand units, he said . . .

BA: Uh-huh . . .

HB: Is that right? I mean, he's been taking the two thousand every—

BA: Just a minute—I'll find out.

HB: Okay.

(At this point, Ali comes to the phone:)

ALI: Hello . . .

HB: Hello . . .

ALI: Who is this?

HB: Harold asked me to call you. I am the author of the Vitamin E book. Herbert Bailey.

ALI: Right! Oh, you're my friend!

HB: Right. I wouldn't bother you, but—

ALI: Harold said you're writing a book on Vitamin E.

HB: I've already written it—you know, but I'm writing another one. He was telling me that you were taking the two thousand units a day, and it did help you in your last two fights.

ALI: Yes, it did.

HB: That's wonderful. Now, what I want you to do—you see, if you read the book, on page 43, you'll see what happens to the race horses. They win consistently when they get the Vitamin E.

ALI: Right.

HB: And now, what they do, is they give them double the dose, right before the race. About twenty-four hours before, they take a double dose. In other words, if the horse has been getting five thousand units, they give him ten thousand. It's an added thing, you know, that gives you that extra stamina. I just thought you ought to know about this, and I wish that you would take it. It won't hurt you—you know that. But it very likely will help you.

ALI: Well, I take it—I *do*. I take a double dose myself, twenty-four hours before the fight.

HB: Wonderful! Now *do* that, because I think it might make a difference. . . .

ALI: Okay!

HB: And I appreciate that, because I appreciate what you said about it, and I think we'll help the world with it. I know we will

ALI: Right!

HB: And so, good luck to you in everything else!

ALI: Right!

HB: And we'll see you, then . . .

ALI: Okay—thank you so much for calling!

HB: I do appreciate what you said about the book, and what you said—

ALI: It's *true*!

HB: I know it is, but I want the world to know it, more than just—you know, I've got ten thousand letters from people all over the world that have done this, and it has helped them.

ALI: Okay! Thank you!

HB: Okay—we'll see you, then, Ali. . . .

ALI: Bye-bye.

HB: Righto—bye!

Interview with Muhammad Ali Via Telephone—Early September 1974 (Before Foreman fight)

Interviewer: Harold Salkin (HS)

Associate of Herbert Bailey

(Opening remarks were about UFO's, one of Ali's favorite subjects.)

HS: Now, we want to ask you about the Vitamin E. Belinda [Ali's wife] says you are taking it regularly.

ALI: Yes, feeling good, too!

HS: Feeling good, huh?

ALI: Yes.

HS: Great! And you take the amount that you wrote down. Right?

ALI: Right!

HS: Two thousand . . .

ALI: Right!

HS: Two thousand a day . . .

ALI: We're gonna get him, Harold! (Talking about Foreman fight)

HS: You're gonna get him this time, eh?

ALI: We'll *shock the world!*

HS: You're gonna really shake everybody up!

ALI: Yep!

HS: Would you say that in the last two fights, you took the Vitamin E—do you think it helped you?

ALI I *know* it did! If I miss a day or two without it, I get real tired, and a *low* feeling.

HS: Is that right?

ALI: Real sleepy and tired, after a day's work, without those vitamins.

If it wasn't for *them*, I wouldn't make it, Harold!

HS: Oh, wonderful!

ALI: You *know* it!

HS: Well, now—Herb Bailey—you know, Bailey wrote the book—he's writing a new book on it, and he wants to mention you in the book. Is that okay?

ALI: It's *okay*, Harold!

HS: And he'll mention what you—

ALI: I've gotta go, Harold, but I'll contact you when I come in! Thank you for all the information!

HS: Okay, 'bye!

ALI: Good luck, now!

Interview with Muhammad Ali on Telephone—February 1, 1975 (After Foreman Fight)

Interviewer: Harold Salkin (HS)

Associate of Herbert Bailey

HS: Hi, Muhammad!

ALI: Yeah, Harold . . .

HS: How are you?

ALI: All right, I got about one minute. You're lucky—you caught me!

I'm on my way to Allentown airport.

HS: Oh, God! Well, the main thing is—will you be in New York again soon? We want to meet you. We have some good UFO slides.

ALI: Something new?

HS: Yeah!

ALI: Good!

HS: I hate to trust them in the mail

ALI: Right—you're right . . . Hold on, Harold, hold on—(Pause)

Okay—I'll be in New York on the tenth [Feb. 10].

HS: The tenth? Where will you be staying?

ALI: I don't know yet. I'm gonna call you—I'll meet with you. [Writing it down] H-A-R-O-L-D. Spell your last name.

HS: S-A-L-K-I-N.

ALI: Home number, Harold?

HS: Mu 5-4080.

ALI: Office?

HS: That's the same—day and night.

ALI: Okay.

HS: Let me ask you one thing. You see, we're rewriting the Vitamin E book, with Herbert Bailey.

(Much background noise, he was having difficulty hearing HS.)

ALI: Doing *what*, now?

HS: The Vitamin E book—remember, that we gave you, with Herb Bailey. And we want to—the way you said, we would talk about your using it in the fights, right?

ALI: Talk *what*?

HS: About how you used it—you've been taking it for all your recent fights.

ALI: Yeah! *Right—okay!*

HS: And how was it in your last fight—did you feel the difference?

ALI: Yes, sir! I felt *stronger* for it! I had *knockout* power—I usually don't *have* that!

HS: You had knockout power—*good!*

ALI: I took all his punches, and everything. I was stronger than him, and everything! He was the world's strongest fighter, till he met me, that night!

HS: Right!

ALI: It was Vitamin E, and this doctor's book—[referring to H. Bailey].

HS: Wonderful!

ALI: Right!

HS: Okay, so we can put this in the book?

ALI: It's okay—

HS: And we can use your picture on the jacket—

ALI: Okay!

HS: That's all right?

ALI: Well, Random House is doing a book, on me—

HS: Wait—*who* is?

ALI: Random House, but you can use a pic—I guess you can use a picture . . .

HS: We want to do a little chapter about your taking it, see?

ALI: Okay

HS: So we'll put all that in—

ALI: All right! Now Harold, I got to go! I'll call you on the tenth.

HS: On the tenth—fine.

ALI: Okay! 'Bye!

HS: Okay, Muhammad—'bye!

Appendix D

(Chapter 3)

Washington Post, March 4, 1972

The Latest Fad In Alphabet Nostrums: Vitamin E by Nancy L. Ross

“Though E has no champion as distinguished as two-time Nobel Prize winner, Dr. Pauling, it does have many advocates in print. One of these is Herbert Bailey, described in his book *Vitamin E: Your Key to a Healthy Heart* as a veteran medical reporter. After suffering a heart attack, Bailey doctored himself with Vitamin E. Six hundred milligrams a day.

“‘On the third day of Vitamin E therapy,’ wrote Bailey, ‘I felt a surge of energy, such a new awareness of physical well-being that I said to my wife, ‘I’ll be damned if I’ll stay in bed any longer.’ Nor did I.’

“Bailey devoted his book to revealing what he calls the ‘suppressed record of the curative values of this remarkable vitamin.’ Though a major portion of it deals with heart disease, the author also recommends Vitamin E for other ailments, including sterility and ‘stress and old age.’

“Originally published in 1964, it sold only 80,000 copies or so. Since being reissued in paperback in 1966, it has sold between 700,000 and 800,000 copies, with the bulk of them in the past 18 months. It sells particularly well in organic food stores and drugstores.”

The New York Times, February 13, 1972

Vitamin E New Fad. Only Lack Is Popularizer of Pauling's Eminence
by Douglas W. Cray

“Vitamin E is not lacking in print champions, however. Among them is Herbert Bailey's *Vitamin E: Your Key to a Healthy Heart*. Originally published in 1964, Mr. Bailey's effort has lately achieved shop-window display, its paperback cover including this billing: ‘The suppressed record of the curative values of this remarkable vitamin.’ ”

Chicago Sun Times, October 18, 1971

Kup's Column
by Irv Kupcinet

“Author Herb Bailey, a former Chicagoan, is celebrating the sale of the 1,000,000th paperback copy of his book, *Vitamin E: Your Key to a Healthy Heart*. (Vitamin E is the ‘in’ pill these days.)”

New York magazine, June 28, 1971 (page 44)

“Mr. California is reading Herbert Bailey's *The Vitamin Pioneers* and *Vitamin E: Your Key to a Healthy Heart*, talking about the Shute doctors in Canada and enjoying a wonderful sex life.”

Glamour magazine, March 1971

Vitamins Do They Really Cure Everything from Colds to Menstrual Cramps to Schizophrenia?
by Rivvy Berkman

“Whereas the mystique of vitamin C has been around long enough to make it a ‘blue-chip’ investment, the fastest rising glamour vitamin is E. One of the most important effects of E is that it helps the body conserve oxygen. It serves in the body as a natural anti-oxidant while also acting as a guard to prevent oxygen from ‘poisoning’ cell membranes. . . . Other animal studies have shown large dosages of E allow mice to withstand longer the poisoning effects of smog and pollution. And the vitamin’s oxygen properties are acknowledged by Dr. Evan Shute and Dr. Wilfred Shute of the non-profit Shute Foundation for Medical Research in Ontario, Canada, as being one of the main reasons E is helpful in treating heart disease where lack of oxygen is the trigger setting off heart pain. They have been researching the power of vitamin E to treat and prevent heart disease, whether coronary or rheumatic, and alone have successfully treated over twenty thousand heart patients with excellent results. People interested in the Shutes’ research and beliefs might read the book *Vitamin E: Your Key to a Healthy Heart* by Herbert Bailey.”

Health Monitor, Summer 1993

When It Comes to Vitamin E, Bailey Wrote the Book by Joanne G. Rochman

“Last month, every major American newspaper, radio station, and television news show reported the results of studies on vitamin E.

“Vitamin E Greatly Reduces Risk of Heart Disease” read the headline in the *New York Times*; “Vitamin E for a Healthy Heart,” was the lead for the article in *Newsweek*; “Vitamin E Appears to Cut Heart Disease Risk” reported *Science News*.

The only problem with all of this welcomed news was that it just *wasn’t* news. Newtown author and investigative reporter, Herbert Bailey, thoroughly researched studies on vitamin E and wrote the book *Vitamin E: Your Key To A Healthy Heart* in 1964.

The book sold 1½ million copies. So to a lot of people, word of the benefits of vitamin E came as no surprise. It certainly was no surprise to Mr. Bailey.

“I’m very happy to see all this publicity. It’s about time. Now the public will be able to avail itself of the benefits from vitamin E,” he said.

According to Mr. Bailey’s writings, those benefits include: lengthening the lifespan of your red blood cells, slowing down your “biological clock,” strengthening your immune system against infectious and toxic diseases, and heightening energy and endurance. Vitamin E also reduces stress, inhibits the

deteriorating effects of smog and cigarette smoke, helps heal burns and many other skin conditions and helps increase male and female fertility.

Regarding its effect on heart disease, Mr. Bailey said that he got involved with the study of vitamin E when he suffered a heart attack many years ago. He investigated ways to improve his health and to protect against another heart attack. That investigation led him to the research conducted at that time on vitamin E.

"The recent publicity proves that the most orthodox of researchers have finally accepted the research that I was investigating and reporting years ago," said Mr. Bailey. "What amazes me is that some of these people [researchers] act as if their research was the only research ever conducted on the subject, when in actuality most of their 'news' findings had already been reported years ago!" he said, producing a folder filled with articles and news releases that substantiated his claim. . . ." [Note: This recent article in *HEALTH MONITOR* (July 1993) substantiates the previously published articles (1971-72) about my first book's prime role in the widespread acceptance of Vitamin E.]

Appendix E

(Chapter 4)

OCHSNER CLINIC

1514 JEFFERSON HIGHWAY

NEW ORLEANS, LA. 70121

CABLE ADDRESS: OCHSCLINIC

TELEPHONE: 834:7070

AREA CODE: 504

Dear Mr. Bailey:

I apologize for not having answered your letter of November 12. It arrived when I was out of the city and this accounts for my delay in answering it.

My interest in alpha-tocopherol began about 30 years ago at the time I brought a young surgeon who had trained at Yale to Tulane when I was Professor of Surgery at the latter institution. Dr. Jack Kay was tremendously interested in blood clotting and he did a great deal of investigative work. He found that alpha-tocopherol in vitro in the presence of calcium interfered with clotting because it acted as an

antithrombic agent. We began using this then in the treatment of individuals who had been operated upon and who were candidates for venous thrombosis. In the beginning we gave alpha-tocopherol and calcium but then later found out that usually the calcium content of the blood in the patient was sufficient so that it was not necessary to give calcium. Since that time, over twenty-five years, I have had one case of postoperative pulmonary embolism which I thought was probably the exception that proved the rule until autopsy showed that the thrombus had originated not in the veins of the lower extremity, which is the usual site, but it originated from the auricular appendage of the heart.

I have continued to use alpha-tocopherol in the prevention of venous thrombosis in individuals who are so-called chronic thrombosers and it has been very efficacious. It has the principal advantage of not producing a bleeding tendency in contradistinction to other agents used to prevent clotting.

I am sure that vitamin E has been used much more extensively than it should be, although it is apparently perfectly safe. I think it should be emphasized that only the the alpha fraction of vitamin E is the one that acts as an antithrombic agent and unless the potency of the alpha fraction is known the product may not be efficacious.

I have had no experience with the use of alpha tocopherol in arterial clotting and I do not know whether it would be of any value or not. I do know it is of value in venous clotting.

I sincerely hope that this information may be of value to you.

Yours very sincerely,

Alton Ochsner, M.D.

APPENDIX F

(Chapter 7)

From: *Age*, Vol. 1, Pages 145-152, October, 1978

[Note: This article is the first complete (nutritional) presentation of Dr. Harman's Theory of Aging. H.B.]

Free Radical Theory Of Aging: Nutritional Implications*

Denham Harman

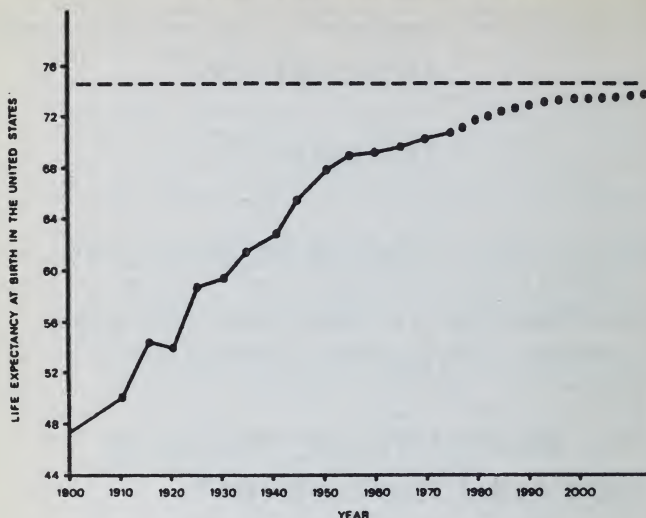
Departments of Medicine and Biochemistry
University of Nebraska College of Medicine
Omaha, Nebraska 68105

Abstract

Life expectancy at birth, now about 72 years, is slowly approaching a plateau value of 74-75 years. Further increases in life expectancy may be achieved by a nutritional approach based on the possibility that the free radical reactions going on continuously in biological systems contribute to the degradation of these systems. Growing evidence indicates that random free radical reactions are a major factor in the break-

*This is the second paper of the mini-symposium on nutrition and aging organized by Charles H. Barrows, Jr., and presented on Thursday, September 29, 1977 as part of the 7th Annual Meeting of the American Aging Association in New York City.

FIGURE 1
LIFE EXPECTANCY AT BIRTH IN THE UNITED STATES SINCE 1900



down of such systems, suggesting that the aging process may be simply the sum of these everpresent deleterious reactions throughout the cells and tissues and that longevity is a reflection of the ability to cope with such reactions. Dietary changes based on this possibility have been shown to: 1) increase the life span of mice, rats, fruit flies and nematodes, 2) inhibit the development of cancer; at least of some forms, 3) slow formation of amyloid, and 4) enhance immune responses. The available data strongly supports the possibility that such dietary changes also have a beneficial effect on the central nervous and cardiovascular systems.

Formulation of human diets, adequate in essential nutrients but designed to minimize random free radical reactions in the body, may reasonably be expected to add five to ten or more years of healthy, productive life to the life span of the average person. [Recently (1993) Dr. Harman advocated supplementation with Vitamin E and other antioxidants.]

Introduction

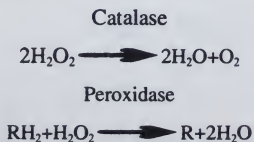
Life expectancy at birth in the United States rose rapidly from a value of 47.2 years in 1900 to 67.2 years in 1954–1955, and then more slowly to the present value of about 72 years on an advance towards a plateau value of 74–75 years (1) (Figure 1). Today the average life span—a rough measure of the period of healthy productive life—is the highest in history in the United States and other developed countries; it is about ten years less than the maximum average life expectancy at birth (2) and around 30 years less than the maximum life span.

Improvements in nutrition contributed to past increases in life expectancy, further improvements may do so in the future. One promising approach is based on the possibility that free radical reactions (3,4) contribute to the degradation of biological systems (5,6,7,8).

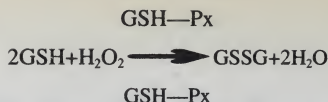
Free radical reactions, initiated by both enzymatic and nonenzymatic means, go on continuously throughout the cells and tissues. Because of the high chemical reactivity of the intermediates, free radicals, formed in free radical reactions, it would be expected that all components of the body would be constantly subject to some degree of chemical change in a more-or-less random manner. These expected changes include: a) accumulative oxidative alterations in the long-lived molecules collagen (9), elastin (10,11) and chromosomal material; b) breakdown of mucopolysaccharides through oxidative degradation (12); c) accumulation of metabolically inert material such as ceroid and age pigment through oxidative-polymerization reactions involving lipids, particularly polyunsaturated lipids, and proteins (13,14); d) changes in membrane characteristics of such elements as mitochondria and lysosomes, due to lipid peroxidation (15,16) and e) arteriolocapillary fibrosis due to peroxidation products of serum and vessel-wall irritants (17,18).

Enzymes have evolved which help limit the rate of production of free radical reaction damage to “tolerable” levels. They include:

a) heme-containing peroxidases—catalase is an important member of this group—which serve to “safely” lower the steady state concentration of hydrogen peroxide and hydroperoxides (19):



b) the selenoenzyme, glutathione peroxidase (20), (GSH-Px), which catalyses the reactions:



where ROOH is an organic hydroperoxide

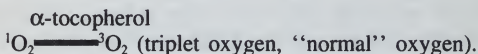
c) superoxide dismutases (SD) c21), metalloproteins which catalyses the reaction:



The highly reactive form of oxygen, singlet oxygen ($^1\text{O}_2$) can be formed *in vivo* (22). For example,



Compounds such as β -carotene and α -tocopherol are good quenchers for singlet oxygen.



It should be possible to further reduce the rate of production of damage by free radical reactions through at least three dietary changes: 1) caloric reduction—this will lower the level of free radical reactions arising in the course of normal metabolism; 2) minimize dietary components—such as copper and polyunsaturates—which tend to increase the level of free radical reactions; and 3) add to the diet one or more radical reaction inhibitors such as 2-mercaptoethylamine (2-MEA), α -tocopherol, butylated hydroxytoluene (BHT), and 1,2-dihydro-6-ethoxy-2,2,4-trimethylquinoline (Santoquin, Ethoxyquin).

If free radical reactions contribute to the degradation of biological systems and if the level of deleterious reactions can be lowered significantly by diet as suggested, then such dietary alterations should have a beneficial effect. A number of experiments have been reported which support this possibility. Some of these studies are briefly presented below.

Effect of Diet on Life Span

Caloric reduction

Humans of less than average weight have a relatively long life span (23). Results of rat experiments parallel experience in man. For example, with Sprague-Dawley rats, “when food intake was restricted by

33 or 46 percent (starting at weaning), levels which prevented fat accumulation and had little retarding effect on skeletal growth, longevity was extended and onset of disease was delayed (24).”

Dietary fat

Increasing the amount and/or degree of unsaturation of dietary fat would be expected to increase the level of more-or-less random free radical reactions. The results of a study (25) in which mice and rats were fed a semi-synthetic diet containing 5, 10, or 20%w (percent by weight) of either lard, olive oil, corn oil or safflower oil as the sole source of lipid are summarized below; in this, as well as other mouse and rat experiments to be described, the diet was started shortly after weaning:

C3H/HeJ female mice: increasing the amount and/or degree of unsaturation of the dietary fat shortened the average life span.

Swiss male mice: no apparent relationship between dietary fat and life span.

Sprague-Dawley male rats: life spans of the 5 and 10%w groups not related to dietary fat. The life spans of the 20%w groups tended to become shorter as the degree of unsaturation was increased.

Dietary protein

Increasing the dietary content of easily oxidized amino acids might increase the level of free-radical reactions and thereby decrease life expectancy. This proved to be the case (Harman, D: Free radical theory of aging: effect of dietary protein on mortality rate. Proceedings, 8th Intern. Congr. Gerontol., Washington, D.C., II 11, 1969) when either 1%w histidine or 1%w lysine was added to a semi-synthetic diet containing 20%w of casein as the sole source of protein. Conversely, average life expectancy was increased when the casein was replaced by a soybean protein which contained a lesser amount of easily oxidized amino acids. (Table 1).

Antioxidants

A number of studies have shown that the addition of antioxidants, such as 2-MEA and BHT, to the diet can increase the life span. Some of the data for mice (26,27,28) and rats (29), drosophila (Miquel, J., and Johnson, J. E., Jr.: Effects of various antioxidants and radiation protectants on the life span and lipofuscin of drosophila and C57BL/6J mice. The Gerontologist, 15: II, 25, 1975) and nematodes (30), are presented in Tables 2 (mice), 3 (drosophila) and 4 (nematodes).

Table 1.Dietary Protein^a: Effect on Life Span of Male LAF₁ Mice

	Average Life Span, Months	Percent Difference from Control
Control	14.5±3.7 ^b	—
" +1%w tyrosine	14.6±3.7	0.7
" +1%w histidine	13.8±4.1	-4.8
" +1%w lysine	13.6±3.4	-6.2
Soybean protein in place of casein	16.4±5.6	13.1

^aBase diet: semi-synthetic with 20%w casein as the sole source of protein.

^bStandard deviation.

Effect of Diet on Disease Processes

Lowering the level of free radical reactions might be expected to decrease the incidence at any given age of those specific diseases in which such reactions play an etiologic role. Diseases for which accumulating evidence implicates free radical reactions in pathogenesis include cancer, atherosclerosis, hypertension, amyloidosis, and senile dementia. Some of this evidence is briefly presented below:

Cancer

From radiation studies, it would be anticipated that an increase in the level of free radical reactions would be associated with an increased tumor incidence. Data supporting this possibility include: a) the mammary tumor incidence of C3H/HeJ female mice increased as the amount and/or degree of unsaturation of the dietary fat increased (25); b) the incidence of breast cancer in women increases as the amount of fat in the diet increases (31); c) the tumor incidence of Sprague-Dawley female rats fed a semi-synthetic diet, with safflower oil as the sole source of lipid, was decreased by adding α -tocopherol acetate to the diet at a level of 20 mg/100 grams of diet (Harman, D.: Free radical theory of aging: effect of Vitamin E on tumor incidence. Abstract, 25th annual meeting of the Gerontology Soc., San Juan, Puerto Rico, 12/20/72); d) butylated hydroxyanisole (BHA) and Santoquin inhibited the carcinogenicity of benz(α)pyrene (BP), and 7,12-dimethylbenz(α)-anthracene (DMBA) on the foresto-

Table 2.

Antioxidants: Effect on Life Span of Mice.

Antioxidant	Percent Increase in Mean Life Span Mouse Strain			
	LAF ¹	C3H	SHK	
	Male	Male	Female	Female
2-MEA ^a				
0.5%w	12.8			
1.0%w	29.2			
Santoquin ^b , 0.5%w		18.1	20.0	
2—Ethyl-6-methyl-3-hydroxypyridine HC1 ^c				23.1

^aReference 26.^bReference 27.^cReference 28.**Table 3.**Antioxidants^a: Effect on Life Span of *Drosophila Melanogaster*.

	Percent Increase in Mean and Maximum Life Span	
	Mean	Maximum
Tocopherol-p-chlorophen- oxyacetate	13	13
Nordihydroquaiaretic acid (NDGA)	20	20
Magnesium thiazolidin carboxylate	20	20

^aThe media contained 0.1%w of the free radical reaction inhibitor.**Table 4.**Antioxidants^a. Effect on Life Span of the Nematode, *Caenorhabditis Briggsae*.

	Percent Increase In	
	50% Survival Time	Maximum Life Span
α-tocophertol or α-tocophertolquinone	31.4	23.2

^aThe media contained 400 μg of antioxidant/ml.

mach of the mouse. The antioxidants also inhibited mammary tumor formation produced by oral administration of DMBA to female Sprague-Dawley rats (32); e) dietary BHA protected against pulmonary neoplasia from acute exposure to DMBA, BP, urethan or uracil mustard (33), f) BHA and ethoxyguin, given intraperitoneally, inhibited the carcinogenic effects of diethylnitrosamine and 4-nitroquinoline-N-oxide on the mouse lung (Wattenberg, L. W.: Inhibition of carcinogenic effects of diethylnitrosamine (DEN) and 4-nitroquinoline-N-oxide (NOO) by antioxidants. *Fed. Proc.*, 31: 633 abs., 1972); g) dietary antioxidants were effective in reducing the number and severity of ultraviolet light-induced squamous cell carcinomas in the skin of hairless mice (34); and h) human gastric cancer (35). "The declining American death rate from gastric carcinoma appears to be associated with the introduction of breakfast cereals, particularly wheat cereals rich in tocopherols." "It is likely that the decline of mortality from gastric carcinoma in 1947 is associated with the introduction of the antioxidant preservatives BHT and BHA at that time."

Cardiovascular Disease

About 60 percent of all deaths are due to cardiovascular disease; atherosclerosis and hypertension are the two major underlying causes.

Atherosclerosis—Arterial injury predisposes to the development of atherosclerotic lesions (36). Peroxides and other compounds formed by the peroxidation of serum and vessel wall lipids and proteins may serve as a constant direct, or indirect, irritant contributing to the development of atherosclerosis (37). Data supportive of this possibility include: a) free radical reactions are going on continuously in the serum (18); b) polyunsaturated fatty acids comprise about 30% of the total fatty acids in the lipids of both serum (38) and atherosclerotic plaques (39); c) ceroid is present in plaques (40,41,42); d) in mammalian species, the higher the arachidonate content of serum cholesterol esters—suggesting a lower lipid peroxidation rate, the lower the incidence of atherosclerosis (43); e) the rise in the incidence of cardiovascular disease in the last 50 years was accompanied by a 7% increase in consumption of saturated fats, but rise in polyunsaturated fats was 37% (44); f) men with a past history of myocardial infarction have higher serum levels of copper, as well as of cholesterol, than men with a negative history (Harman, D.: Role of serum copper in coronary atherosclerosis. *Circulation*, 28 II: 658, 1963); g) areas where the drinking water is hard have a relatively low incidence of cardiovascular disease (Harman, D.: Atherosclerosis: possible role of drinking water copper. *Clin. Res.*, 13: 91, 1966) (45)—the copper content of hard water is relatively low; and h) lipid peroxides inhibit the prostacyclin synthetase

found in the arterial wall, thus predisposing to thrombus formation (46,47,48).

The large majority of deaths from ischemic heart disease are associated with extensive atherosclerosis of the coronary arteries. The thrombosis frequently found in these cases are apparently generally secondary to the fatal event (49), rather than the cause. The latter observation raises the possibility (45) that the same factors involved in coronary atherogenesis also produce in parallel more subtle, free radical-induced, degradative changes in the fixed post-mitotic cells of the myocardium which set the stage for fatal myocardial malfunction to occur under some circumstances.

Hypertension—The etiology of most cases of hypertension is unknown, i.e., essential hypertension. The single most important factor in the production of hypertension is an increase in peripheral resistance (50) in the small blood vessels, mainly arteriolar.

The free radical reactions postulated to contribute to atherosclerosis may also play an etiologic role in hypertension. Studies in support of this possibility include: a) by analogy with the effects of ionizing radiation (17), the free radical reactions in serum and vessel walls may contribute to the development of arteriolocapillary fibrosis (18); b) the hypertension of the toxemia of pregnancy is associated with elevated serum levels of copper (51,52) and lipid (53) and depressed levels of ascorbic acid (54); c) increased incidence of hypertension (55) in women on "the pill"—associated with increased serum copper (56) and lipid levels (57); and d) lipid peroxides inhibit production of prostacyclin, a compound that tends to decrease vessel tone (58).

Amyloidosis

Amyloid is associated with age and a large number of seemingly unrelated diseases, both in man and in many species of animals (59). The involvement of a free radical reaction(s) in the formation of amyloid was first noted (26) in a study with LAF₁ male mice; spontaneous amyloidosis was markedly inhibited by adding BHT or Santoquin to the diet. Subsequently, antioxidants were found to inhibit casein-induced amyloidosis in both C3HeB/Fe and CBA/J mice (60)(Cathcart, E. S., Yonkosky, D., and Harman, D.: Santoquin therapy in casein-induced experimental amyloidosis. 14th Intern. Congr. Rheumatology, San Francisco, 6/26/77-7/1/77). Consideration of the current knowledge of amyloid, in the light of the antioxidant studies, prompted the following hypothesis (60). Amyloidosis is largely the result of an enhanced rate of oxidative degradation of a connective tissue glycoprotein(s) coupled with oxidative/enzymatic changes in the plasma, both of the tissue-derived substances and of immunoglobulins, to form the amyloid fibril

protein subunits which subsequently aggregate to form the amyloid fibrils.

Senile Dementia: Aging of the Central Nervous System

The central nervous system (CNS) may be particularly susceptible to degradation by free radical reactions because the neurons are fixed post-mitotic cells with unique connections between them and a predilection for the highly unsaturated fatty acid, docosahexanoic acid (22:6 ω 3).

Docosahexanoic acid, either free or in triglyceride form, as well as the precursor linolenic acid (18:3 ω 3), is avidly removed from the diet and incorporated into brain 22:6 ω 3 phospholipids (61). Brain 22:6 ω 3 is tenaciously retained.

To evaluate the possibility that increases, in the level of more-or-less random free radical reactions secondary to increased lipid peroxidation might have a selective adverse effect on the CNS, two experiments (62) were conducted in which the relative efficiency in a learning situation was used as a measure of CNS deterioration. In the first study, Sprague-Dawley male rats, fed semi-synthetic diets which did not significantly alter mortality rates, made more errors in the Hebb-Williams maze test of relative intelligence as the amount and/or degree of unsaturation of the dietary fat was increased. The second experiment evaluated operative behavior using a discrimination learning paradigm; rats fed a safflower oil diet supplemented with 20 mg of δ -tocopherol acetate per 100 grams of diet did better than rats fed the unsupplemented diet.

These studies indicate that variations in the amount or degree of unsaturation of dietary fat and of factors (e.g., Vitamin E) that can modify lipid peroxidation rates, may contribute to the variability in age of onset of evident degradative CNS changes such as senility, above and beyond the variations expected from differences in mortality rates.

Effect of Diet on the Immune System

The immune system declines with age (63). To determine if free radical reaction inhibitors could delay the decline, C3HeB/FeJ female mice had 0.25%w of either Santoquin or Vitamin E added to their diet starting at age 40 days (64). Humoral and cell-mediated responses were measured to age 22 months. Whether or not the antioxidants slowed the decline of the immune system with time could not be determined from the data. However, the compounds enhanced the immune response throughout the study period suggesting that endogenous free radical reactions had a deleterious effect on the immune response. Prompted by the foregoing, a number of other free radical reaction inhibitors

were studied and found to have a beneficial effect on the humoral response; BHT and 2-MEA were as effective as levamisole.

These studies suggest that at least part of the decline of the immune response with age may be related to an increasing level of random free radical reactions with age; in the serum, copper—an oxidation catalyst—increases with age (65) while the concentration of mercaptans—oxidation inhibitors—decreases (66,67). Likewise, the depression of the immune system frequently noted in cancer patients may be a reflection of their elevated serum copper levels (68).

Comment

These data demonstrate that free radical reactions shorten life span, contribute to the development of cancer (at least of some forms) and amyloidosis, dampen immune responses, and most likely play a significant role in the degradation of the central nervous and cardiovascular systems. The data further demonstrate that the natural defenses of living things against deleterious free radical reactions can be significantly influenced by diet.

As knowledge of the role of free radical reactions in biological systems accumulates, evidence mounts that random free radical reactions are a major factor in the degradation of such systems, suggesting that the aging process may be simply the sum of these deleterious reactions going on continuously throughout the cells and tissues. It would appear that living things not only selected free radical reactions to play major metabolic roles but also utilized the side reactions that naturally arise from them to provide for mutation and death, thereby assuring evolution. From this point of view, life span—a species characteristic, has evolved in parallel with the ability of organisms to cope with random free radical reactions.

Formulation of human diets, adequate in essential nutrients but designed to minimize random free radical reactions in the body (Vitamin E), will almost certainly increase the healthy life span by slowing accumulation of nonspecific deleterious changes and the development of specific diseases. With such diets, it is not unreasonable to expect that the average person will enjoy five to ten or more years of healthy, productive life.

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Vitamin E Abstracts by Dr. M. K. Horwitt

The clinical reports on vitamin E therapy have been of special high quality and significance.

The Effect of Tocopherol on High-Density Lipoprotein Cholesterol. W. J. Hermann, K. Ward and J. Faucett. *Am. J. Clin. Path.*, 72:848, 1979 (From Memorial City General Hospital, Department of Laboratory Medicine, Houston, Texas.)

The cholesterol in the blood is transported by several lipoproteins. Changes in the proportions of one of these lipoproteins, termed high density lipoprotein (HDL), have been correlated with the incidence of heart disease; the lower the proportion of HDL, the higher the incidence of cardiac failure. The senior author (W.J.H.) had noted that when he took 600 I.U. of d-alpha-tocopheryl acetate for 30 days that the proportion of his HDL cholesterol increased dramatically from 9% to 40%. To check further he repeated the experiments on 10 volunteers, five who had normal blood lipid profiles and five who had low HDL cholesterols. After about one month of supplementation with 600 I.U. of vitamin E, the normal group had an increase in the proportion of HDL cholesterol of 50 percent and the other group had an increase of more than 200 percent. There was no significant change in total plasma cholesterol. This work is now being repeated. *If it is substantiated, it could be one of the most important developments in the use of vitamin E of this decade.*

Evidence That Vitamin E Actually Prevents Heart Attacks*

Richard A. Passwater

Research explaining how tocopherol prevents heart disease is important, but the final proof is in testing a large number of people taking tocopherol to see if their hearts are healthier than the hearts of those not taking it. This is what I did in 1974 and 1975. The results were published in *Prevention* in the January to May and July to September 1976 issues.

I studied reports from 17,884 people between the ages of 50 and 98 years (nearly equally balanced between sexes) of whom 3,725 had heart disease. I discovered two important facts.

First of all, more than 80 percent of those with heart disease reported that tocopherol helped the condition measurably. People requiring 18 or more nitroglycerin pills for their angina reported they threw their nitroglycerin pills away within a month or two after starting tocopherol. Many reported that angina pains would return in a few weeks or months if they discontinued tocopherol, but would disappear once again soon after they resumed taking tocopherol.

It is possible that psychological factors could account for the improvement, but results of studies show that any placebo effect is small. And it is doubtful that the prevention of a heart attack can be greatly influenced by the placebo effect.

The second major finding of the survey was that tocopherol apparently prevents heart disease. It was notable that the amount of heart disease in any age group decreased proportionally with the length of time that tocopherol had been taken. In fact, the length of time was more important than dosage after a minimum level of 400 IU daily was reached.

Two Groups Show Lower Heart-Disease Incidence with Vitamin E

Two groups were especially interesting. One group consisted of those who had taken 400 IU or more of tocopherol daily for 10 years or

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more. The survey included 2,508 such people between the ages of 50 and 98. Normally, based on the U.S. Dept. of Health, Education, and Welfare figures (HRS 74-1222), you would expect to find 836 of the 2,508 having heart disease. Instead there were only 4. This is less than one-half of 1 percent of the expected number.

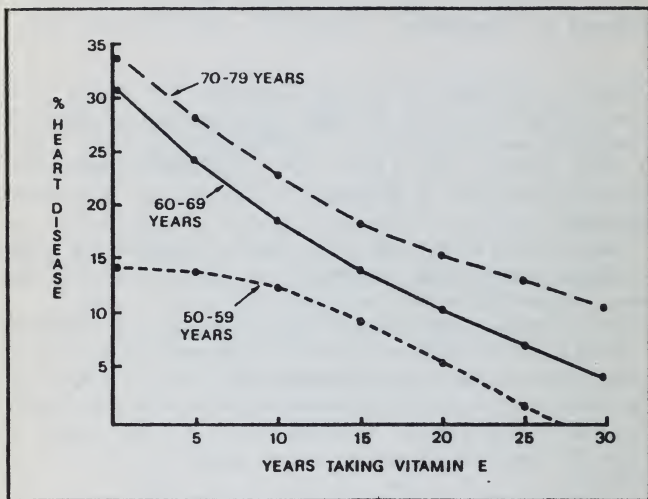


Figure 10.1 There is a strong correlation between length of time taking vitamin E and freedom from heart disease in all age groups responding to survey.

A second group of 1,038 had taken 1,200 IU or more of tocopherol daily for 4 years or more. Among this group, you would expect to find 323 suffering from heart disease. But the actual number was 7.

A survey does not prove cause and effect. It only shows association, yet tocopherol was shown to be strongly associated with reduced incidence of heart disease.

The conclusions and projections that I have determined from the study are shown in tables 10.1, 10.2 and 10.3.

Still, few physicians will be convinced by either the findings of the survey or the early evidence from the 1940s and 1950s. However, the general population has found out for itself that tocopherol is effective.

The use of tocopherol tripled from 1970 to 1971, correlating well with the reports that heart-disease deaths in two age groups (45-54 and 55-64) have decreased by as much as 15 percent since that time through 1976.

Table 10.1

Heart Disease Incidence among Those Taking 400 IU of Vitamin E or More for 10 Years or More

Age Group	Number in Group	Expected Number with Heart Disease (National Rate from HEW Figures)	Actual Reported Number with Heart Disease (Possible Undetected Cases in Parentheses)	Percentage of Expected Number with Heart Disease
50-59	569	114	0 (3)	0
60-69	871	279	3 (1)	1.1
70-79	806	323	1 (2)	0.3
80-89	245	111	0 (1)	0
90-98	17	9	0 (0)	0
total	2,508	836	4 (7)	0.5

400 IU or more of vitamin E taken for 10 years or more is strongly associated with reduction of the incidence of heart disease prior to 80 years of age to less than 10 percent of the present rate (32 per 100 people expected to have heart disease reduced to 3 per 100 people who actually do have heart disease). Table 10.1 gives the data for this finding.

Table 10.2
Heart Disease Incidence among Those Taking 1,200 IU of Vitamin E or More for 4 Years or More

Age Group	Number in Group	Expected Number with Heart Disease (National Rate from HEW Figures)	Actual Reported Number with Heart Disease (Possible Undetected Cases in Parentheses)	Percentage of Expected Number with Heart Disease
90-98	1	0	0 (0)	—
80-89	64	28	0 (0)	0
70-79	284	114	2 (2)	1.8
60-69	356	114	4 (3)	3.5
50-59	333	67	1 (2)	1.5
	1,038	323	7 (7)	2.2

1200 IU or more of vitamin E taken for 4 years or more is strongly associated with reducing the incidence of heart disease prior to 80 years of age from 32 per 100 people to 10 per 100 people. Table 10.2 gives the data for this finding.

Table 10.3

Correlation Between Length of Taking Vitamin E and Freedom from Heart Disease

Years of Use of Vitamin E	Percentage Free from Heart Disease								
	0	Less than 1	1-3	4-5	6-9	10-19	20-29	30+	AVG.
Age Group									
50-59	85.7	86.6	86.1	86.2	86.7	89.2	98.3	100	86.8
60-69	68.4	69.6	73.0	74.9	78.7	85.5	89.1	100	76.1
70-79	67.8	64.6	69.1	71.0	70.7	82.1	81.8	88.5	73.6
80-89	65.2	71.8	69.3	67.5	66.9	81.2	85.6	100	73.3
90-98	87.5	100	94.8	77.8	91.7	96.8	—	100	93.4

The association between length of time taking vitamin E and freedom of heart disease is extremely strong statistically. The association (Spearman-Rho) is, in fact, stronger for vitamin E than any other tested variables, which included exercise, diet, and cholesterol intake. The association varies slightly with age, but typically shows a 0.96 association where 1.00 is a perfect association. This correlation is significantly different from chance occurrence at the 0.01 level. Table 10.3 gives the data for this finding, and further data are given in Appendix 5.

Eighty percent of those with angina pectoris, fibrillation, or tachycardia are substantially helped by tocopherol within the first year of use. Most patients find that they can gradually reduce other medication with increasing time and dosage of vitamin E.

An Indication that Vitamin E Prevents Heart Disease

A survey I conducted of nearly twenty thousand vitamin E users produced an unexpected result. It was found that the amount of heart disease in any age group decreased proportionally with the length of time of taking vitamin E. A summary of this statistical survey and its interpretation is given in Chapter 10. The data on which those conclusions are based are presented here. The questionnaire used in the survey was published in the August 1974 issue of *Prevention* magazine, and detailed reports were published in the January through August 1976 issues of *Prevention*. *Prevention's* cooperation in conducting this survey is greatly appreciated.

Table A5.1
Survey Respondents

Age Group	Number in Group	Number with Heart Disease
90-99	106	7
80-89	1,064	284
70-79	4,060	1,073
60-69	6,459	1,543
50-59	6,205	818
	17,894	3,725

Table A5.2
Heart Disease Incidence in Vitamin E Users Aged 50-59

Years Taking Vitamin E	Number in Group	Survey Participants with Heart Disease Developed Prior to Taking Vitamin E			Survey Participants with Insufficient Data to Classify Heart Disease as Prior or Not			Survey Participants with Heart Disease Developed While Taking Vitamin E			Survey Participants Free of Heart Disease		
		Better #	Same #	Worse %	Better #	Same #	Worse %	Better #	Same #	Worse %	#	%	%
30+	24	0	0	0	0	0	0	0	0	0	24	100	
25-29	18	0	0	0	0	0	0	0	0	0	18	100	
20-24	116	4	3.4	0	0	0	0	0	0	0	112	96.6	
15-19	228	14	6.1	2.09	0	4	1.8	2	0.9	0	206	90.4	
10-14	405	36	8.9	2.05	0	5	1.2	1	0.2	0	360	88.9	
6-9	690	73	10.6	6	0.9	1	0.1	1	0.1	1	598	86.7	
4-5	1407	141	10.0	28	2.0	7	0.5	9	0.6	1	1213	86.2	
1-3	2887	276	9.6	62	2.1	11	0.4	22	0.8	4	2485	86.1	
less than 1	269	25	9.3	3	1.1	8	3.0	0	0	0	233	86.6	
Subtotal	6044										5249	86.8	
0	161										138	85.7	
Total	6205										5387	86.8	

Spearman-Rho Stat: Rho is 0.92

The percentages listed in the chart refer to the members of each group of "years taking vitamin E" applying to each category listed above. In Table A5.2, as an example, 3.4 percent of those taking vitamin E for 20-24 years and had prior heart disease report their heart disease better; this indicates that of the 116 respondents, 4 had prior heart disease (3.4 percent of the group) and all 4 were helped appreciably by vitamin E. Thus of the 3.4 percent of the responding group with prior heart disease, 100 percent were helped by vitamin E.

Heart Disease Incidence in Vitamin E Users Aged 60-69

[illegible]

Spearman-Rho Stat: Rho is 0.94

The percentages listed in the chart refer to the members of each group of "years taking vitamin E" applying to each category listed above in Table A5.3, as an example, 6.1 percent of those taking vitamin E for 25-29 years report their heart disease better; this indicates that of the 33 respondents, 3 had prior heart disease (9.1 percent of the group) and 2 were helped appreciably by vitamin E. Thus of the 9.1 percent of the responding group with prior heart disease, 66 percent were helped by vitamin E.

Heart Disease Incidence in Vitamin E Users Aged 70-79

Years Taking Vitamin E	Number in Group	Survey Participants with Heart Disease Developed Prior to Taking Vitamin E			Survey Participants with Insufficient Data to Classify Heart Disease as Prior or Not			Survey Participants with Heart Disease Developed While Taking Vitamin E			Survey Participants Free of Heart Disease										
		Better		Worse	Better		Same	Worse	Better		Same	Worse	Better		Same	Worse					
		#	%	#	%	#	%	#	%	#	%	#	%	#	%	#	%				
30+	96	6	6.3	0	0	0	0	1	1.0	0	0	4	4.2	0	0	0	85	88.5			
25-29	268	38	14.2	2	0.7	0	0	6	2.2	2	0.7	0	0	1	0.4	0	2198	81.1			
10-19	862	132	15.3	11	1.3	2	0.2	4	0.5	2	0.2	1	0.1	2	0.2	0	708	82.1			
6-9	567	138	24.3	24	4.2	2	0.4	0	0	0	0	0	0	2	0.4	0	401	70.7			
4-5	713	172	24.1	28	3.9	3	0.4	0	0	0	0	0	0	3	0.4	0	506	71.0			
1-3	1340	337	25.1	62	4.2	14	1.0	0	0	0	0	0	0	1	0.1	0	926	69.1			
less than 1	96	22	22.9	4	4.2	8	8.3	0	0	0	0	0	0	0	0	0	62	64.6			
Subtotal	3942	845	21.4	131	3.3	29	0.7	11	0.3	4	0.1	1	0.0	11	0.3	2	0.05	1	0.0	2907	73.7
0	118																			80	67.8
Total	4060																			2987	73.6

Spearman-Rho Stat: Rho is 0.64

The percentages listed in the chart refer to the members of each group of "years taking vitamin E" applying to each category listed above. In Table A5.4, as an example, 6.3 percent of those taking vitamin E for 30 years or more had prior heart disease, and all report their heart disease better; this indicates that of the 96 respondents, 6 had prior heart disease (6.3 percent of the group) and all 6 were helped appreciably by vitamin E. Thus of the 6.3 percent of the responding group with prior heart disease, 100 percent were helped by Vitamin E.

Heart Disease Incidence in Vitamin E Users Aged 80-89

Years Taking Vitamin E	Number in Group	Survey Participants Heart Disease Developed Prior to Taking Vitamin E				Survey Participants with Heart Disease Developed While Taking Vitamin E				Survey Participants Free of Heart Disease	
		Better		Same		Worse				#	%
		#	%	#	%	#	%	#	%	#	%
30+	6*	0	0	0	0	0	0	0	0	6	100
20-29	104	15	14.4	0	0	0	0	0**	0**	89	85.6
10-19	266	48	18.0	0	0	0	0	2	0.8	216	81.2
6-9	136	43	31.6	1	0.7	0	0	(1)**	(0.7)	91	66.9
4-5	200	52	26	8	4.0	4	2.0	(1)**	(0.5)	135	67.5
1-3	274	67	24.5	11	4.0	6	2.2	0	0	190	69.3
less than 1	32*	6	18.8	2	6.3	1	3.1	0	0	23	71.8
Subtotal	1018	231	22.7	22	2.1	11	1.1	(4)*	(0.4)	750	73.7
0	46*	—	—	—	—	—	—	—	—	30	65.2***
Total	1064	—	—	—	—	—	—	—	—	780	73.3

*Sample group size insignificant

**No confirmed cases

***Higher than normal due to proper diet and multivitamins

Spearman-Rho Stat: Rho is 0.92

The percentages listed in the chart refer to the members of each group of "years taking vitamin E" applying to each category listed above. In Table A5.5, as an example, 14.4 percent of those taking vitamin E for 20-29 years report their heart disease better; this indicates that of the 104 respondents, 15 had prior heart disease (14.4 percent of the group) and all 15 were helped appreciably by vitamin E. Thus of the 14.4 percent of the responding group with prior heart disease, 100 percent were helped by vitamin E.

Heart Disease Incidence in Vitamin E Users Aged 90-99

Years Taking Vitamin E	Number in Group	Survey Participants Heart Disease Developed Prior to Taking Vitamin E			Survey Participants with Heart Disease Developed While Taking Vitamin E			Survey Participants Free of Heart Disease		
		Better		Same		Worse				
		#	%	#	%	#	%	#	%	%
30+	3	0	0	0	0	0	0	3	100	
20-29	0	0	0	0	0	0	0	0	0	
10-19	31	1	3.2	0	0	0	0	29	96.8	
6-9	12	1	8.3	0	0	0	0	11	91.7	
4-5	9	1	11.1	0	0	0	0	7	77.8	
1-3	38	1	2.6	1	2.6	0	0	36	94.8	
less than 1	5	0	0	0	0	0	0	5	100	
Subtotal	98	4	4.1	1	1.0	0	0	92	93.9	
O	8	—	—	1	12.5	—	—	7	87.5	
Total	106	4	3.8	2	1.9	0	0	99	93.4	

Spearman-Rho Stat: not applied because of small sample size

The percentages listed in the chart refer to the members of each group “years taking vitamin E” applying to each category listed above. In Table A5.6, as an example, percent of those taking vitamin E for 10-19 years report their heart disease better; this indicates that of the 31 respondents, 1 had prior heart disease (3.2 percent of the group) and that individual was helped appreciably by vitamin E. Thus of the 3.2 percent of the responding group with prior heart disease 100 percent were helped by vitamin E.

From: *Cutis*, Vol. 21, March 1978.

Abstract

Is Vitamin E Involved in the Autoimmune Mechanism?

**Samuel Ayres, Jr., M.D., F.A.C.P., Los Angeles,
California**

Richard Mihan, M.D., Los Angeles, California

Autoimmune diseases are characterized by an alteration of the body's defense mechanism, designed for protection against infections and toxic injuries, which for unknown reasons attacks and destroys normal tissue.

Some evidence strongly suggests that such diseases are the result of hydrolytic enzymes that escape from lysosomes whose membranes have been damaged by lipid peroxidation or other causes and that combine with and denature normal tissue proteins—in effect converting them into foreign proteins—to which the body then reacts by producing antibodies.

During the past ten years, in a private dermatologic practice, we have conducted clinical investigations on the possible therapeutic value of vitamin E in the management of a number of disabling skin diseases of unknown etiology as well as several muscular disorders.

Among the diseases that were successfully controlled were a number in the autoimmune category, including scleroderma, discoid lupus erythematosus, porphyria cutanea tarda, several types of vasculitis, and polymyositis.

Since vitamin E is a physiologic stabilizer of cellular and lysosomal membranes, and since some autoimmune diseases respond to vitamin E, we suggest that a relative deficiency of vitamin E damages lysosomal membranes, thus initiating the autoimmune process.

Dr. Ayres is Emeritus Clinical Professor of Medicine (Dermatology), University of California at Los Angeles, The Center for the Health Sciences; Dr. Mihan is Associate Clinical Professor of Medicine (Dermatology and Syphilology), University of Southern California School of Medicine.

Vitamin E and Prostaglandins (PG)

Lawrence J. Machlin, Ph.D.
Department of Biochemical Nutrition
Roche Research Center
Nutley, New Jersey 07110, U.S.A.

Summary

In rats maintained on diets containing different concentrations of vitamin E, serum concentrations of PGE_2 and PGF_{2a} were inversely related to serum concentrations of α -tocopherol. Since platelets are the only source of serum prostaglandin, these data suggest that, *in vivo*, α -tocopherol inhibits the endogenous conversion of arachidonic acid (AA) into PG by platelets. The increased platelet aggregation observed in vitamin E-deficient rats could be related to increased PG synthesis. Increased synthesis of PG in testis from vitamin E-deficient animals has also been observed by Carpenter et al.

It is possible that excessive PG biosynthesis could play a role in the pathology observed in vitamin E deficiency and therefore, that PG inhibitors could have a vitamin E-like effect. We have found that aspirin, a PG inhibitor, prevented the anemia, and the thrombocythemia observed in rats maintained on diets low in vitamin E for 16 weeks. However, aspirin exacerbated testis degeneration, increased the vitamin E requirement for the amelioration of myopathy and did not prevent fetal resorption.

[Dr. Machlin is also president of Nutrition Research and Information, Inc., Livingston, N.J.]

Appendix G

(Chapter 8)

From: *Family Health* September 1980.

[Note: This is the first large-scale revolt against the cholesterol theory that only cholesterol was the major cause of heart disease. As we have noted, there are still lingering traces of this myth in 1993. H.B.]

The Cholesterol Controversy

For nearly 20 years, the American Heart Association and numerous Federal agencies, as well as doctors and nutritionists, have urged consumers to cut back on cholesterol-rich foods such as red meats, eggs and dairy products in order to reduce the danger of heart disease. But recently, the Food and Nutrition Board of the National Research Council, which is an arm of the prestigious National Academy of Sciences, has challenged the long-standing belief that dietary cholesterol is a major cause of heart disease.

The Food and Nutrition Board, which conducted an exhaustive two-year study, concluded that the dietary-cholesterol theory lacks scientific foundation. Declares the board: "Intervention trials in which diet modification was employed to alter the incidents of coronary artery disease and mortality in middle-aged men have been generally negative. Seven large scale studies were carried out in London, Oslo, Helsinki, New York City, New Jersey, and Los Angeles. In these studies, comprising about twenty thousand man-years of observation, [and] in which decreases in serum cholesterol concentrations of seven to sixteen percent occurred, there was a marginal decrease in coronary incidents—but no effect on overall mortality."

The board goes on to say that, nonetheless, people at specially high risk for coronary artery disease might profit by low-cholesterol diets, and that individuals should consult their own doctors. "We are not saying that people with angina pectoris should not see their doctors and follow their advice with respect to risk factors," says Robert E.

Olson, M.D., of the St. Louis University School of Medicine, who chaired the research project. "[But] I think there has been a food fad created around the bogeyman of dietary cholesterol. Good food should not be regarded as a poison, a medicine, or a talisman. It should be eaten and enjoyed."

Those, of course, are fighting words. "It's true that not all the facts are in," retorts Robert Levy, M.D., director of the National Heart, Lung and Blood Institute. "But to recommend doing nothing in the meantime is inappropriate. The existing information indicates that Americans should hedge their bets and seek a diet lower in saturated fats and cholesterol, at least until more evidence is available."

The American Heart Association and a variety of other experts have also disagreed sharply with the board's conclusions. Some authorities have pointed out that although the board relied heavily on epidemiological (population) studies, no board members are epidemiologists, public health experts, or, for that matter, cardiologists. Most members are nutritionists and biochemists, and a few have served as paid consultants to food-industry organizations such as the American Egg Board.

To sum up: *The last word still isn't in*; so when in doubt, consult your own physician.

William Barnhill

[*The last word is almost in. See Chapter VIII, Cholesterol Theory Examined. H.B.*]

Appendix H

Chapter 10

From *JAMA*, September 5, 1981—Vol. 244 No. 10
MEDICAL NEWS 1077-1078

[Note: One of the first reports of Vitamin E and Cystic Breast Disease. Many other reports were to follow up to the present time. H.B.]

Vitamin E Relieves Most Cystic Breast Disease; May Alter Lipids, Hormones

Noncancerous lumpy breast tissue plagues up to 20% of American women. This disorder, or set of disorders, is often diagnosed as fibrocystic breast disease, mammary dysplasia, or fibrous mastopathy—general terms that subsume every kind of benign histological alteration and are rejected by some investigators as “catchall” (*N. Engl. J. Med.* 229:381–385, 1978).

Women with at least some types of fibrocystic breast disease are thought to be at a twofold to eightfold greater risk of development of breast cancer, and even those whose lumps remain benign often experience extreme discomfort. Long before scar tissue develops around the cysts and sclerosing adenosis occurs, a woman may have such severe premenstrual aching and tenderness that she cannot tolerate breast stimulation or sleep without wearing a bra.

“Some of our patients fly in from other parts of the country because they’re so symptomatic,” says Robert S. London, MD, director of reproductive endocrinology at Baltimore’s Sinai Hospital and assistant professor of obstetrics and gynecology at The Johns Hopkins University School of Medicine. “They’ve seen local physicians who have tried everything and have run out of remedies.”

Several years ago, London and associates began to prescribe exogenous vitamin E supplementation for these patients. They reported at the recent meeting of the Endocrine Society in Washington, D.C., that the

vitamin relieves symptoms in a majority of women and causes regression of the disease in some.

The vitamin's mechanism of action is unknown, but the investigators have learned that it alters blood levels of various hormones, including the adrenal androgens and the gonadotropins. Moreover, it causes a statistically significant increase in the level of free cholesterol in low-density and very-low-density lipoproteins (not the esterified form that is found in plaques) in both patients and control subjects. While the group urges utmost caution in the interpretation of this last finding, they acknowledge that it could conceivably have implications for the prophylaxis of atherosclerosis and breast cancer, both of which are thought to involve, in part, altered lipid levels.

London and co-workers began their research with vitamin E after learning that Archie A. Abrams, MD, of the Boston University School of Medicine, had achieved moderate to complete relief of premenstrual symptoms in 16 patients with fibrocystic breast disease, and palpable softening of the breasts, with reduction in cyst size, in 13 others by administering vitamin E (*N. Engl. J. Med.* 272:1080-1081, 1965). Subsequently the Baltimore group reported that vitamin E relieved breast tenderness, caused cyst regression, and altered urinary adrenal steroid excretion in 20 women (*Breast* 4:19, 1978).

In their most recent study, these investigators treated 26 patients and eight control subjects with a placebo for four weeks followed by 600 IU of vitamin E per day for eight weeks in a double-blind study funded by the American Cancer Society. Hoping to learn whether the vitamin's effect is mediated through steroid synthesis or alteration of lipoproteins, they assayed serum leuteinizing hormone, follicle-stimulating hormone, prolactin, estrone, estradiol, progesterone, testosterone, dehydroepiandrosterone (DHEA), and lipid levels in blood drawn between 8:30 and 9:30 A.M. on day 21 of the menstrual cycle, both before and after therapy. All women were between the ages of 23 and 40 years, with an average age of 34 years, and all had diagnoses of middle-stage mammary dysplasia, with the diagnosis corroborated by mammography and biopsy specimens when necessary.

London told *JAMA MEDICAL NEWS*: "Ten patients were good responders, 12 were fair responders, and 4 were nonresponders. By 'good response,' we mean regression of the disease process. The lumps went away and the patients noticed tremendous clinical improvement."

Laboratory Findings

More startling than the positive responses to vitamin E, which the group had expected, were some of the laboratory findings. In contrast to previous studies showing alterations of various hormones and hormone combinations in patients with mammary dysplasia, London and col-

leagues found no significant pretreatment differences between patients and control subjects except for higher amounts of the weak adrenal androgen DHEA in patients. These high levels were normalized in the course of vitamin E treatment. For unknown reasons, the decrease in DHEA levels was most striking in women who did *not* respond to therapy with improvement of their fibrocystic disease.

In both patients and control subjects, some changes in serum gonadotropin levels occurred—probably, London speculates, because of “a direct effect of the vitamin E on the CNS or on one of the mediators at the releasing level in the hypothalamus.”

As for serum lipid levels, they were comparable in patients and control subjects before treatment; afterward, however, levels of free low-density cholesterol increased from 36.60 ± 4.97 mg/dL to 57.60 ± 5.65 mg/dL in control subjects and from 41.50 ± 2.09 mg/dL to 49.28 ± 3.03 mg/dL in patients who responded to vitamin E. A much smaller increase occurred in nonresponding patients.

None of these chemical changes appeared to have clinical manifestations, London says. All patients were ovulatory both before and after treatment, as indicated by normal progesterone levels, and all reported that their libidos were unchanged during the course of therapy.

London has no clear explanation for the laboratory results. He comments, “I can give you five pages of theory, but I’m really not sure. I think the important thing is that vitamin E has profound effects on the homeostatic mechanism both in normal women and in women with mammary dysplasia.

“Right now you can go to the drugstore and buy thousands of grams of vitamin E and take as much as you want, not knowing that it may affect you profoundly. It may change your lipids; it may, and probably will, alter some of your steroid hormones. At least, at the dosage levels we prescribed, which weren’t all that high, vitamin E is not a benign vitamin that you can take like vitamin C if you think you’re getting a cold. It is—and we need to stress this—a pharmacologic agent.”

London also stresses that he does not know why certain women respond to the vitamin and others do not. He postulates that differences in response may be dose-related, with some women, perhaps, needing more vitamin E than the standardized dosage given during the study. “Alternatively,” he suggests, “perhaps there are intrinsic differences among patients in the metabolism of vitamin E. Or perhaps the patient who will progress to carcinoma is the one who doesn’t respond, whereas the patients who do respond to vitamin E are not the ones who progress to cancer, or vice versa.”

He adds, “Women with breast cancer are thought to have certain abnormal parameters, such as very high estrogen levels, abnormal estrogen-progesterone ratios and lipoprotein abnormalities, but we did not confirm any of this in women with mammary dysplasia.”

Among mammary dysplasia patients who progress to carcinoma, he proposes, such abnormalities may develop only with the onset of malignancy.

London and associates now will attempt to identify specific vitamin E receptors in breast tissue and determine whether the vitamin has a direct effect on normal or abnormal breast tissue. "I'm not continuing the present study because I don't have any more money," he says.

But on the basis of his current findings, *London recommends that clinicians in general try prescribing vitamin E for their patients with cystic breast disease.*

"I think our work highly suggests that vitamin E is effective in these patients," he says. "We found absolutely no side effects in terms of clinical derangements, and it worked in a high percentage of patients. If the clinicians can get symptomatic relief in patients with something as benign as vitamin E, I think it's a reasonable therapy. The other therapies are all more dangerous or have more side effects. Danazol, for instance, which is awfully expensive, has virilizing and androgenizing side effects."

London's co-workers in this research are G. S. Sundaram, PhD, Michael Schultz, M.D., P. P. Naier, Ph.D., and Philip Goldstein, M.D.

Elizabeth Rasche Gonzalez

Appendix I

(Chapter 11)

[Note: This report is so apropos to the present, it is a good example of our thesis in Chapter 11. H.B.]

From: *New York Daily News*, March 19, 1982

Heart of the matter

Better care didn't cut deaths: study

By EDWARD EDELSON
Science Editor

Costly new medical treatments for heart attacks apparently have not improved the survival rate for coronary patients, according to results of a Health Insurance Plan of New York study released yesterday.

Although the death rate from heart attacks dropped by more than 20% from the mid-1960s to the mid-1970s, the HIP study contradicts those medical experts who credited improved care for at least part of the drop. The study indicates instead that the decline in the death rate may be due to the decline in the number of heart attacks.

The study compared a group of patients who suffered heart attacks in the 1960s and another group who had coronaries in the 1970s. The decade between the studies saw the introduction of many new treatments, including more widespread use of coronary care units in hospitals.

Yet there was "no difference between the prognosis between patients in the two decades," a team headed by statistician Eve Weinblatt reported in the *Journal of the American Medical Association*.

The HIP researchers compared the survival rates for 436 men who suffered heart attacks in the early 1960s and 697 men who had coronaries in the early 1970s. In the five years after the heart attacks, the death rates for both groups were almost identical, the researchers found.

In an editorial comment, Dr. Richard Havlik of the National Health, Blood and Lung Institute noted that a previous study of heart attack survival done at John Hopkins University reached the same conclusion.

Appendix J

(Chapter 15)

[Note: One of the first reports showing the importance of selenium in heart disease and cancer. H.B.]

Selenium/Heart Attack Relationship

In a series of papers reported at the 60th annual meeting of the Federation of American Societies for Experimental Biology, new epidemiological evidence was presented linking low environmental and dietary levels of selenium and copper to hypertension-related diseases and cancer.

Americans who live in areas where selenium levels are low are up to three times more likely to die from heart attack, strokes, and other diseases related to high blood pressure than their neighbors living in high selenium areas, according to a survey carried out by Raymond J. Shamberger and Dr. Charles E. Willis of the Cleveland Clinic. Selenium is a trace metal found in soil, plants, and water and is required in small amounts for proper body functioning.

After comparing the selenium concentration of each state's forage crop with the death rate from high blood pressure-related diseases in the 55- to 64-year-old group, investigators found that in the selenium-rich states such as Texas, Oklahoma, Arizona, Colorado, Louisiana, Utah, Alabama, Nebraska, and Kansas the death rate is far below the national average (67% below in Colorado Springs and 53% below in Austin, TX). In selenium-poor states such as Connecticut, Ohio, Illinois, Oregon, Massachusetts, New York, Pennsylvania, Indiana, and Delaware the death rate due to heart disease is as much as 300% higher than it is in the high selenium states.

The Cleveland Clinic study was an outgrowth of earlier studies which

showed that animals raised on low selenium diets developed heart abnormalities while other work indicates that selenium in the proper amounts reduced angina. It has been known that people living in low selenium states have lower levels of selenium in their blood than the residents of the selenium-rich states.

Washington State University
November, 1976

WHAT WOULD YOUR BODY PREFER?

NATURAL VITAMIN E or SYNTHETIC VITAMIN E

New research shows Natural Vitamin E is preferred up to 5 times more than Synthetic Vitamin E by the body.

Dr. Graham Burton, associated with the National Research Council of Canada, recently presented the results of a study evaluating the body's response to Natural Vitamin E compared to Synthetic Vitamin E. This published study measured the proportion of Natural and Synthetic Vitamin E in various organs, tissues and fluids of experimental animals given equal doses of both types of Vitamin E. The preference was then confirmed by measurements of Vitamin E in blood and plasma from human subjects.

Data from this study strongly indicates the body prefers Natural Vitamin E to Synthetic Vitamin E. Here are some of the results:

5.3 times more Natural Vitamin E was found in the Brain than Synthetic Vitamin E, and:

- 3.6 times more in Red Blood Cells
- 2.6 times more in the Lungs
- 2.4 times more in the Plasma
- 1.9 times more in the Heart
- 1.7 times more in the Muscles

Your body knows the difference between Natural and Synthetic Vitamin E.

Appendix K

(Chapter 16)

Man's Frontiers Foundation, Inc.

**A personal message from Herbert Bailey,
Executive Director of MAN'S FRONTIERS**

For many years I have been intensely interested in founding an organization which would *in essence* be totally free of preconceived ideas about man, his present-day concepts of what he is, and his orthodox-derived knowledge. In other words, an organization which would be *truly* seeking knowledge no matter where that knowledge may be found, or where it might eventually lead us.

Such an organization would necessarily have to be constructed on an extremely broad concept. I believe in man's evolvement, physically, mentally, and spiritually. I think such a role for man *can be proved*—hence MAN'S FRONTIERS. The road ahead will be rocky, as it always is for a pioneering group. Yet by intelligent, courageous investigations, conducted by persons (scientists, scholars, and writers) with no axes to be sharpened by either governmental or quasi-official sources, many truths may be ascertained.

Man's evolution on this planet has in the past been most painful, and in the main produced by individual geniuses, usually working against the (then) presently accepted notions of what man and the Cosmos are. MAN'S FRONTIERS has no intention of regimenting anything—it seems to us that *individual* effort and genius are absolutely necessary and desirable, but that often a team of geniuses is necessary to solve the multi-faceted problems now facing mankind. However, many problems can be solved *only* by individual genius, and thus each problem must be evaluated in terms of what is needed.

MAN'S FRONTIERS believes in the destiny of mankind as being a sort of *evolving* process which will enable him to understand some of

the workings of the Cosmos as well as his own present-day operations. If it were possible—and it might be soon—MAN'S FRONTIERS would utilize scientists, scholars, researchers from *other* civilized worlds to study man so as to arrive at some tentative, at least, partially valid, objective conclusions concerning our roles in the Universe. Meanwhile, MAN'S FRONTIERS will utilize what it possesses: *true* and brilliant scientists, scholars, authors, reporters who will investigate and describe their findings regardless of the consequences.

It is also MAN'S FRONTIERS' belief that modern man with all the technological, medical, psychological, sociological, literary, and artistic advances thrust upon him within the last few thousand years, no longer has adequate opportunity to communicate with his fellows. Each so-called scientific and/or artistic discipline has developed its own jargon, which is mainly undecipherable except to its specialists, and even then, the geometrical progression of information renders it impossible for anyone to "keep up"—even in his specialty no matter if he spends almost all his time studying.

For instance, even in the highly specialized field of neurosurgery, a brain surgeon has almost no chance of absorbing all the recent developments in his science-art-craft. He can barely communicate with his confreres. He cannot even read all the recent developments associated with brain surgery. How can we ask him to communicate with *us*, even the "educated outside" world? Multiply this problem by about four billion persons, and you realize what mankind faces: a hopeless breakdown in communications. When the total breakdown in communications occurs, what can mankind expect except total chaos and disaster?

It is MAN'S FRONTIERS' desire to prevent such a breakdown and the complete anarchy which would automatically follow.

How can such a tragic situation (for man) be prevented? MAN'S FRONTIERS does not know the final answers, but it does have some important clues. It is one of its most compelling projects. Without going into details, we can mention that a process which MAN'S FRONTIERS terms "instantaneous learning" is not only possible, but probable, provided the proper approaches are made.

There are dozens of vital projects which MAN'S FRONTIERS is exploring. Those who know my previous and present work will understand its calibre—its research in depth which I have made for every one of my publications. Not one has been seriously challenged under any circumstances, because what I have written is demonstrable truth. Under my direction, MAN'S FRONTIERS will also investigate and publish demonstrable truths.

As some of you know, I have devoted a good part of my life in investigating and writing the facts about Krebiozen, Vitamin E, atomic energy in medicine, ultra-sound, mental health, and many other worthwhile subjects. We have conducted investigations in the fields of the

heart, lungs, liver, kidneys—and, in fact, every physical aspect of man as related to his environment and heredity.

But MAN'S FRONTIERS is also deeply concerned about man's mental and spiritual development. We have been engaged in several startling, exciting projects, but for lack of funds, we have not been able to set up further procedures which are essential to continuing this work.

If you believe in my work and in those who are associated with MAN'S FRONTIERS, and if you believe us when we say there are infinitely more worlds to explore, then we would welcome your support for MAN'S FRONTIERS.

MAN'S FRONTIERS, a non-profit organization, was incorporated in the State of New York to become a tax-exempt organization. Tax-exempt status will be applied for as soon as proper capitalization is achieved. We intend sending to donors a Progress Report, probably in the form of a newsletter, so that all supporters of MAN'S FRONTIERS will be informed of the activities of the organization.

We are not prepared at the present to accept members, *per se*; therefore, do *not* send a contribution unless you can well afford it and thoroughly understand the cost of activating MAN'S FRONTIERS along with its various projects. I specify I do not wish contributions from individuals, but only grants from foundations who can well afford it. So far, I have received no grants, although almost every day I read where the ultrarich, who naturally have ultracharging doctors, die with the same regularity—both patients and doctors—from diseases from which they could have been saved had the doctors had the time or inclination to read the literature on Vitamin E and/or other supplements.

Cordially,

HERBERT BAILEY

Executive Director

MAN'S FRONTIERS

P.O. Box 636

Sandy Hook, Ct. 06482

Copy

CERTIFICATE OF INCORPORATION

of

MAN'S FRONTIERS FOUNDATION, INC.

Pursuant to the Membership Corporations Law

1. The name of the proposed Corporation is MAN'S FRONTIERS FOUNDATION, INC.

2. The purpose for which the Corporation is to be formed are:

(a) To assimilate and disseminate information pertinent to man's physical and mental progress and well-being.

(b) To appraise and pay especial attention to the new, original, unexplored and the unorthodox.

(c) To sponsor, underwrite and encourage any promising humane and progressive endeavor in the fields of art, science and philosophy.

(d) To publish the unbiased results of any or all research projects deemed important to man's progress, whether sponsored by the Foundation or not.

(e) To challenge and resist any attempt to impede, abridge or censor any research project or the results thereof, by bringing all of the facts to the public.

(f) To promote in any way possible man's knowledge and wisdom.

As a means of accomplishing the foregoing purposes, the Corporation shall have the following powers in addition to those otherwise specified by law:

To accept, acquire, receive, take or hold by bequest, devise, grant, gift, purchase, exchange, lease, transfer, judicial order or decree or otherwise, for any of its objects and purposes, any property, both real and personal of whatever kind and wherever situated.

No part of the earnings of the Corporation shall inure to the benefit of any members of any other private individual, nor shall any part of the activities of the Corporation consist of carrying on propaganda, or otherwise attempting to influence legislation or participating in, or intervening in (including the publishing or distributing of statements), any political campaign on behalf of any candidate for public office.

Upon the dissolution of the Corporation no member shall be entitled to any distribution or division of its remaining property or its proceeds, and the balance of all money and other property received by the Corporation from any source, including its operation, after the payment of all debts and obligations of the Corporation of whatsoever kind and nature, shall be used or distributed, subject to the order of the Supreme Court of the State of New York, as provided by law, exclusively for purposes within those set forth in this Article of the Certificate and within the intentment of Section 501 (b) (3) of the Internal Revenue Code of 1954, as the same may be amended from time to time.

3. The territory in which the operations of the Corporation will principally be conducted is New York State.

4. The principal office of the Corporation is to be located in the City of New York, County of New York.

5. The number of its directors shall not be less than three nor more than fifteen.

STATE OF NEW YORK
COUNTY OF KINGS } SS.:

I HEREBY approve the foregoing Certificate of Incorporation and the filing thereof.

August 22, 1961.

Sidney A. Fine
Justice of the Supreme Court
of the State of New York

The undersigned has no objection to judicial approval being granted to the within certificate of incorporation of Man's Frontiers Foundation.
August 21, 1961.

Louis J. Lefkowitz

Attorney General, State of New York

By: Percy E. Schuberth, Jr.

Assistant Attorney General

(Approved and charter issued October, 1961, State of New York)

About the Author

Herbert Bailey, former newspaperman, is one of the most successful writers and reporters in the health/sciences field. His books include *A Matter of Life or Death* (Putnam); *The Vitamin Pioneers* (Rodale); *Food Facts and Fallacies* with Carlton Fredericks (Crown); and his bestseller, *Vitamin E: Your Key to a Healthy Heart* (Chilton/Arco) which was largely responsible for Vitamin E's general and medical acceptance. He is also the author of *E: The Essential Vitamin* (Bantam) and *GH3: Will It Keep You Young Longer?* (Bantam).

He is credited with bringing national attention to the first large-scale successful and practical use of atomic energy in medicine—a force which has saved or prolonged countless thousands of lives. Also, Mr. Bailey wrote about the first clinical trials of hydrocortisone in the treatment of arthritis and other diseases. He first acquainted the American public with ultrasound, which had been used for many years in Europe and elsewhere for treating various diseases.

He directed the world's attention to one of our greatest medical heroes—Dr. Emil Grubbé, who first used X-ray in the treatment of cancer on January 29, 1896. As a result of the author's efforts, Dr. Grubbé was finally accorded the recognition due him by the medical profession.

Mr. Bailey is the founder of *Man's Frontiers*, an organization devoted to explorations of the unknown and the dissemination of information on generally suppressed subjects.

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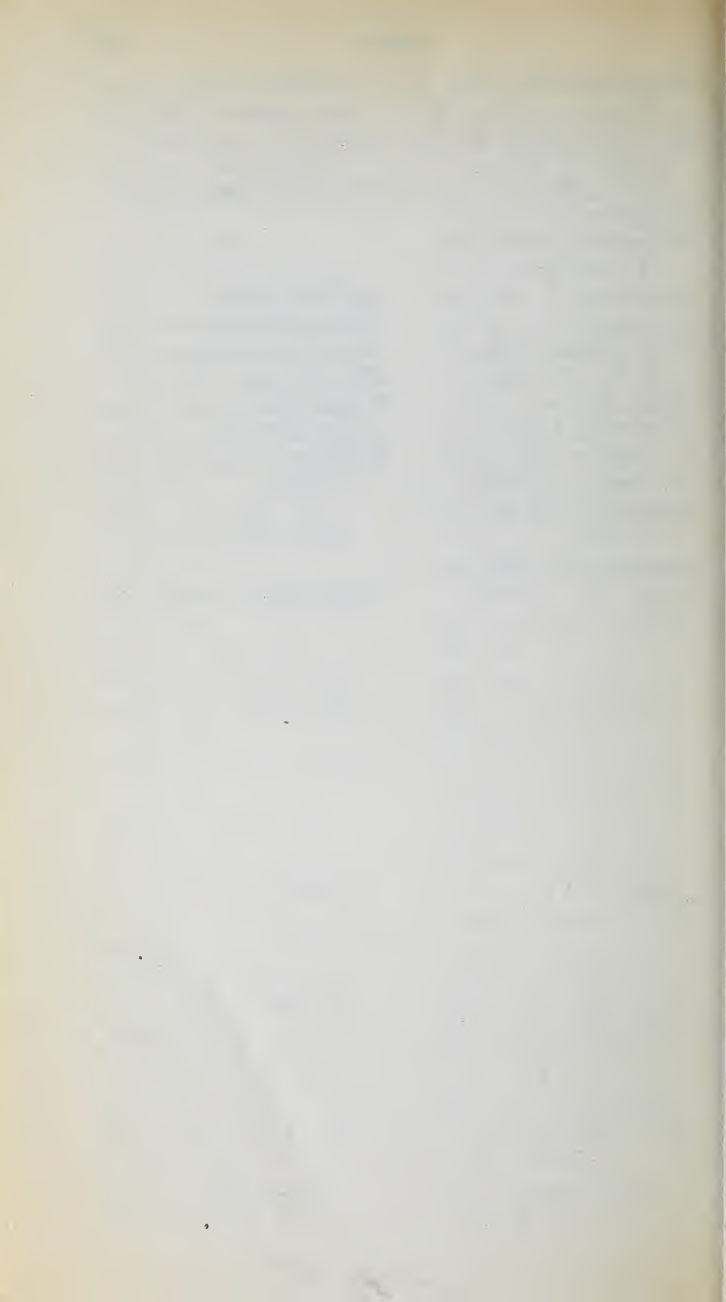
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